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(54) **EXTRACTION AND CLASSIFICATION OF AUDIO EVENTS IN GAMING SYSTEMS**

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H04R 5/04 (2006.01)
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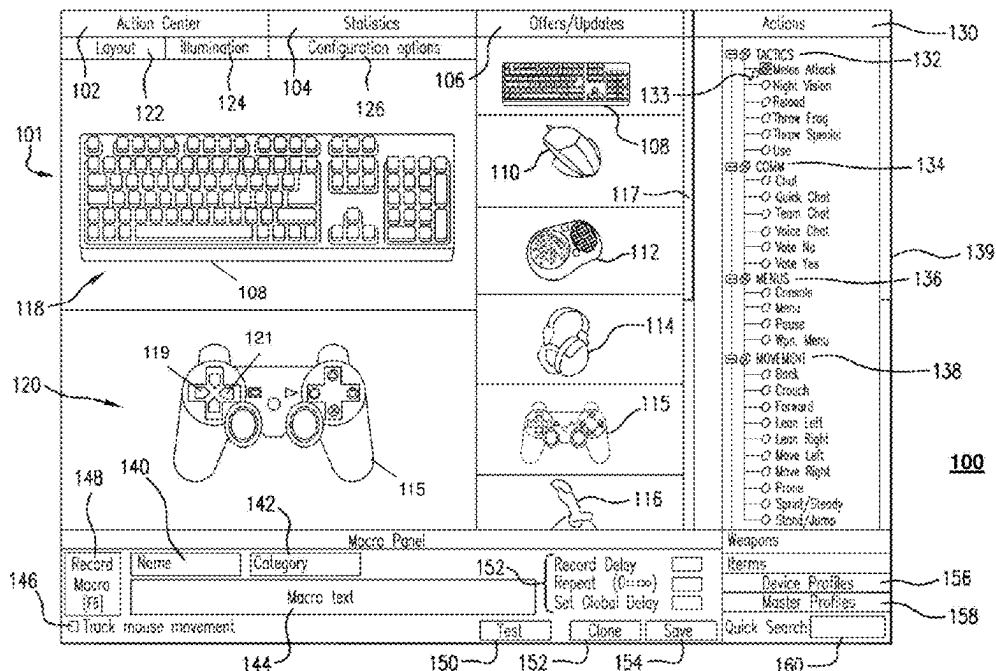
(52) **U.S. Cl.**

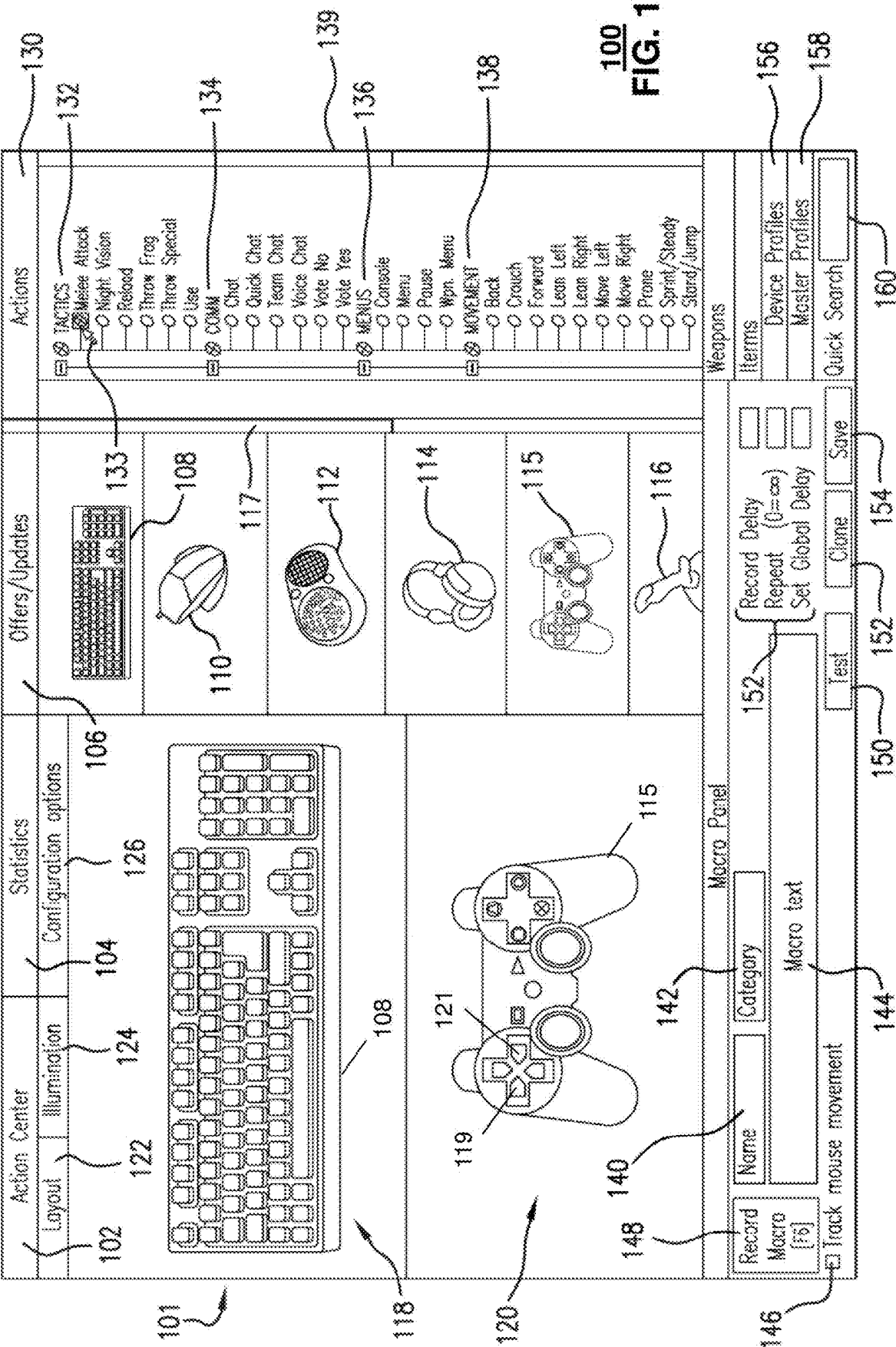
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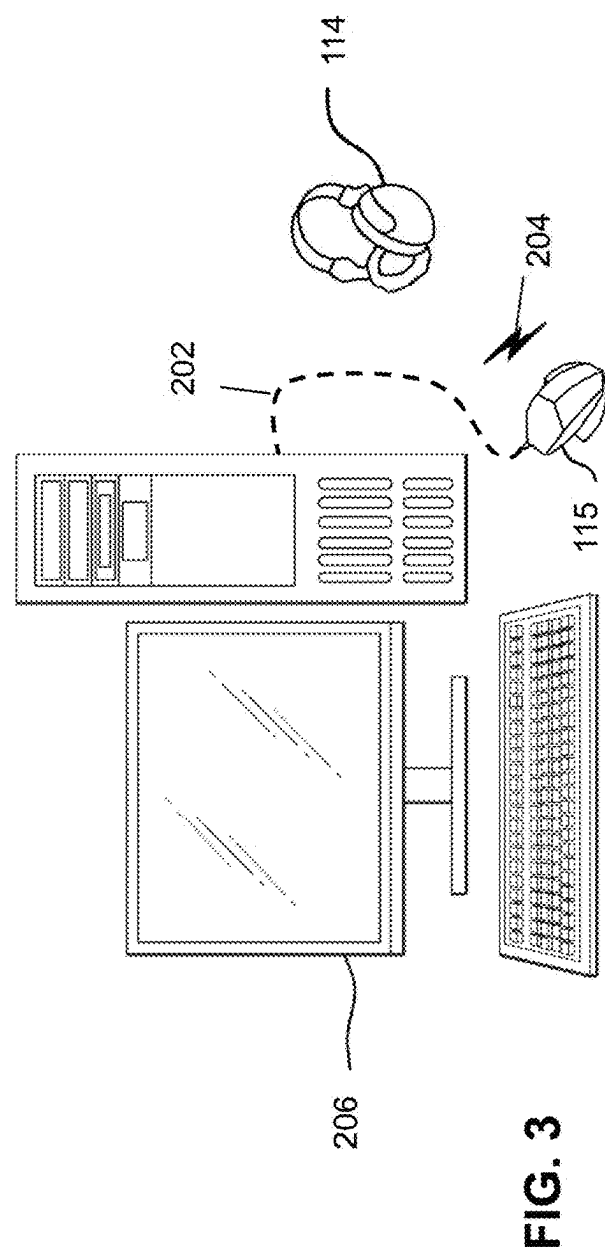
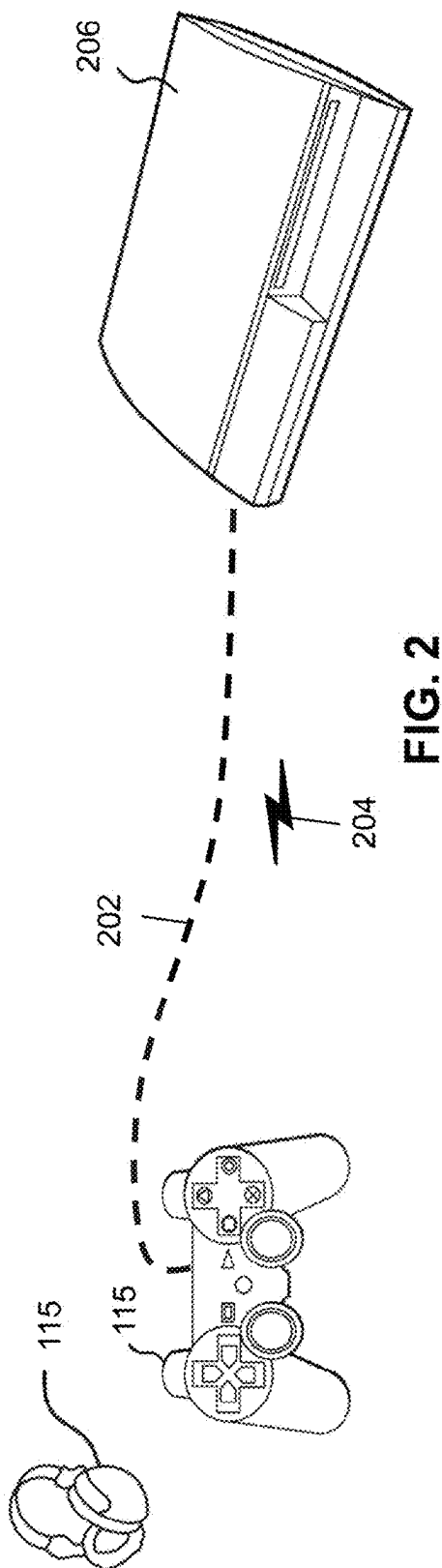
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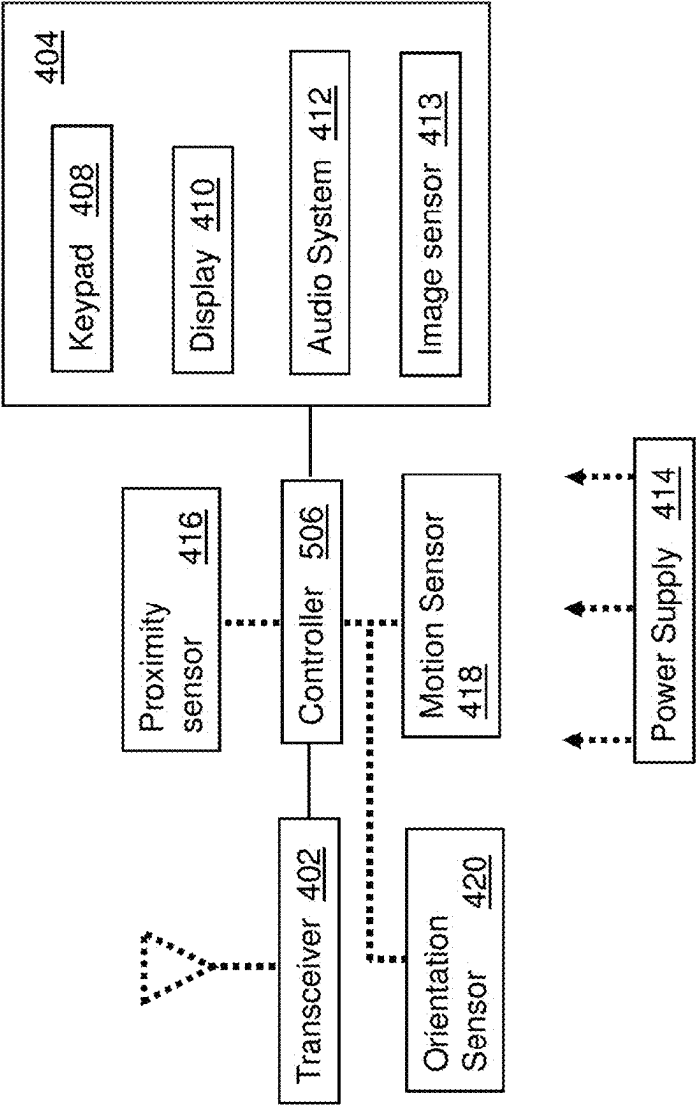
ABSTRACT

A system that incorporates the subject disclosure may include, for example, receiving an input audio stream from a gaming system, the input audio stream including gaming audio of a video game played by a game player, the input audio stream including a plurality of classes of sounds, providing the input audio stream to a neural network, extracting, by the neural network, sounds of a selected class of sounds of the plurality of classes of sounds, and providing a plurality of output audio streams including providing a first audio stream including the sounds of the selected class of sounds of the input audio stream and a second audio stream including remaining sounds of the input audio stream. Additional embodiments are disclosed.

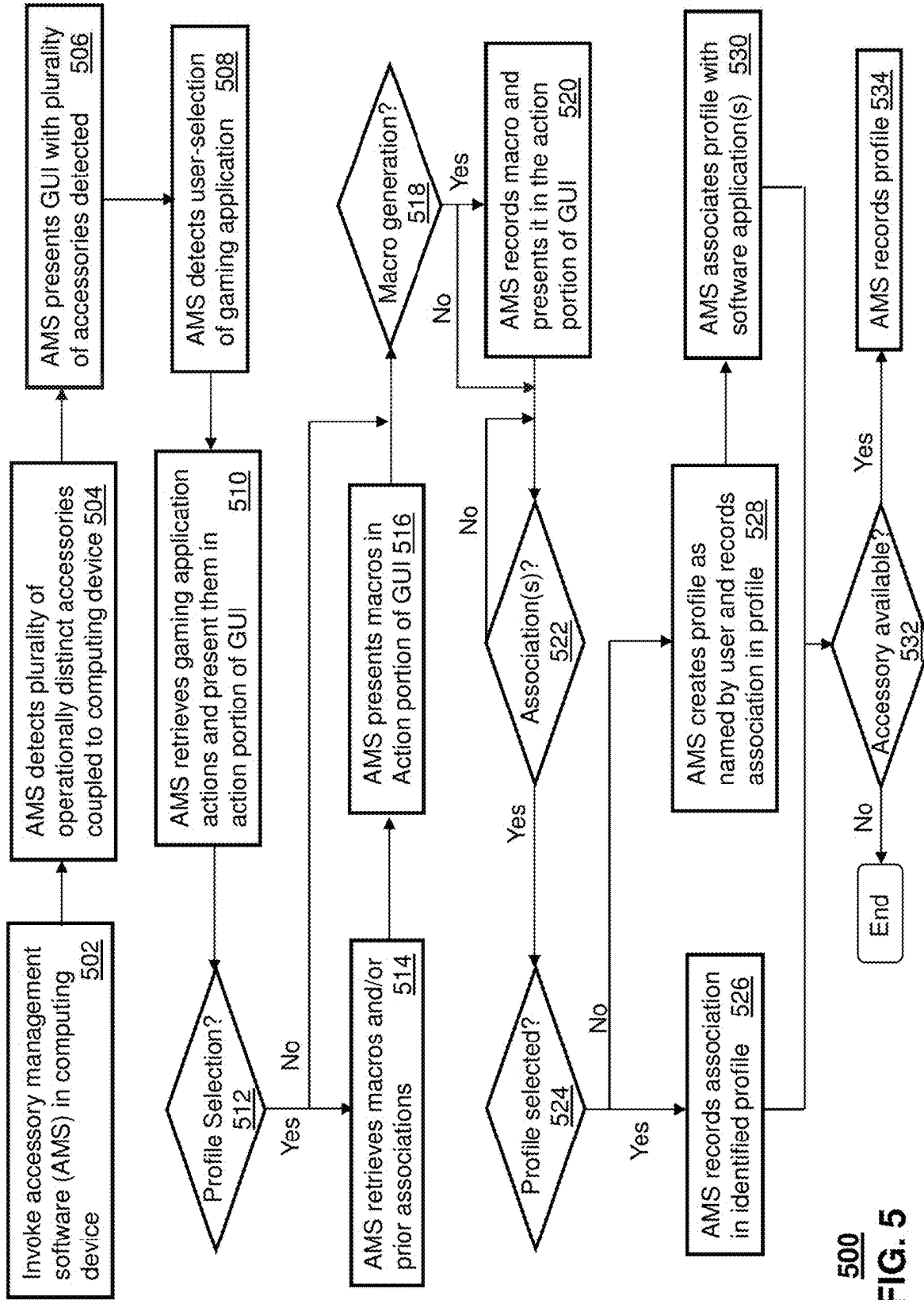


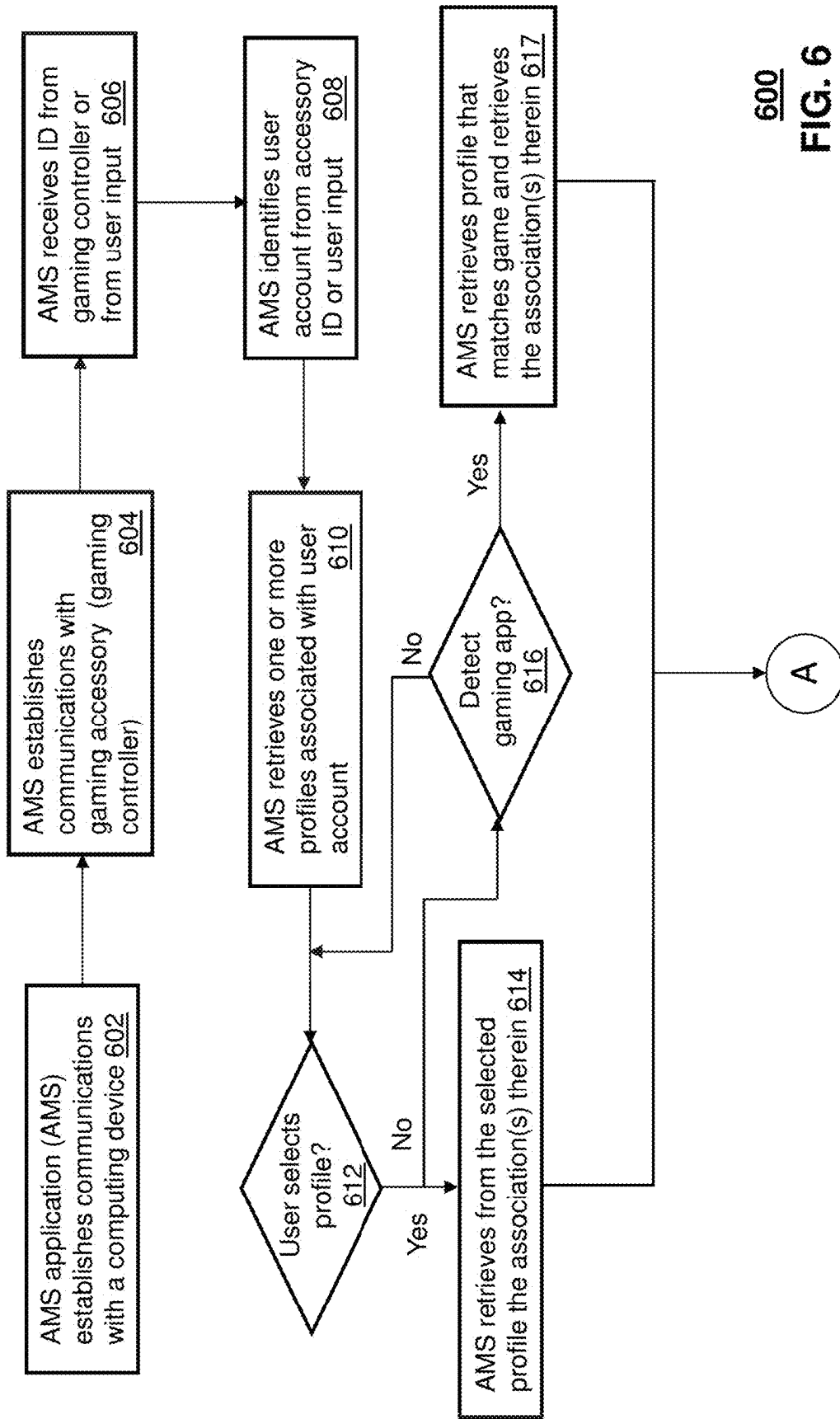






400
FIG. 4





600
FIG. 6

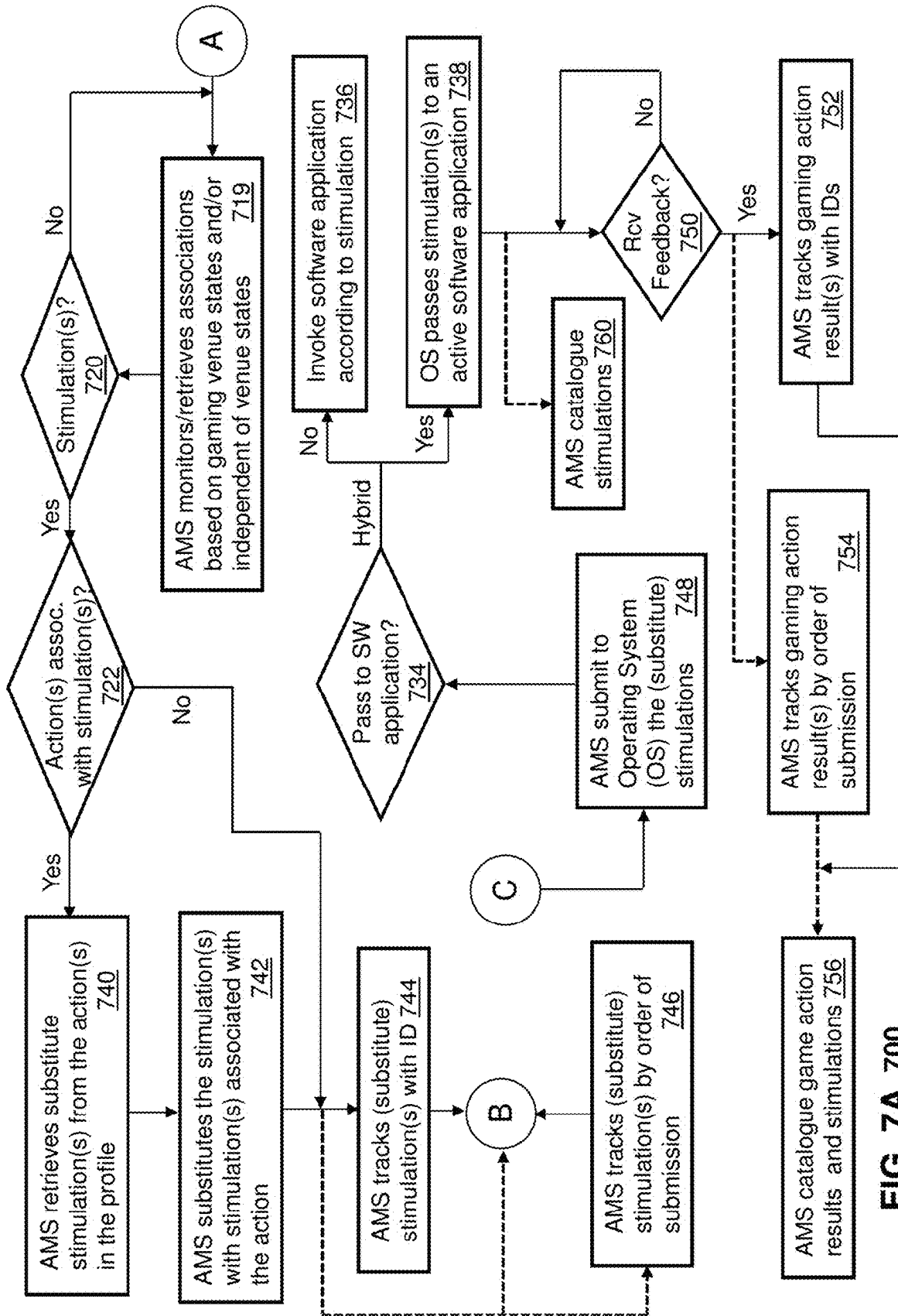


FIG. 7A 700

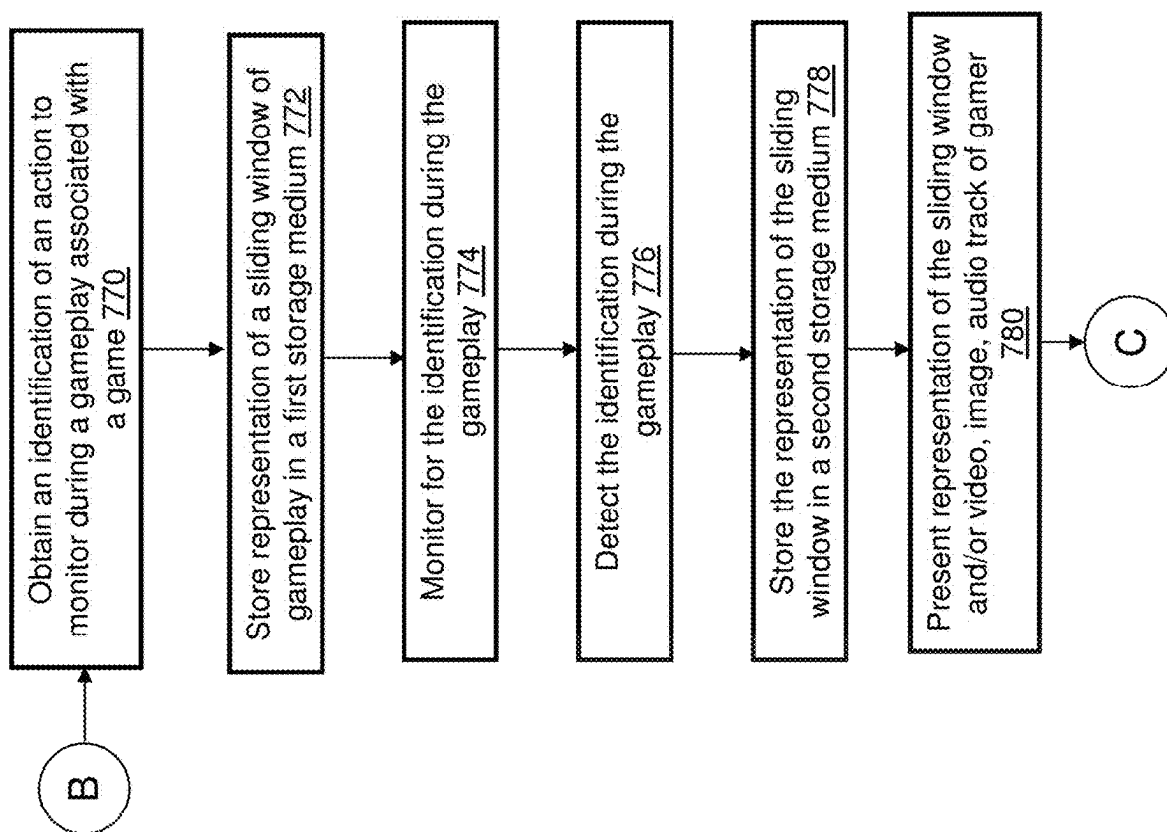


FIG. 7B

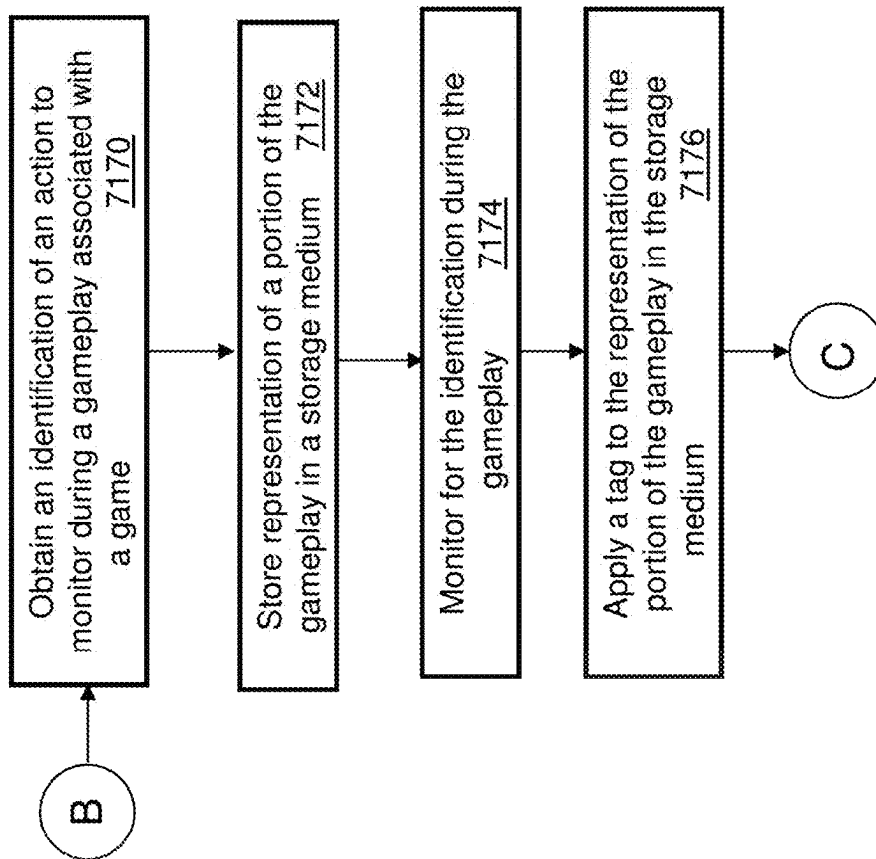


FIG. 7C

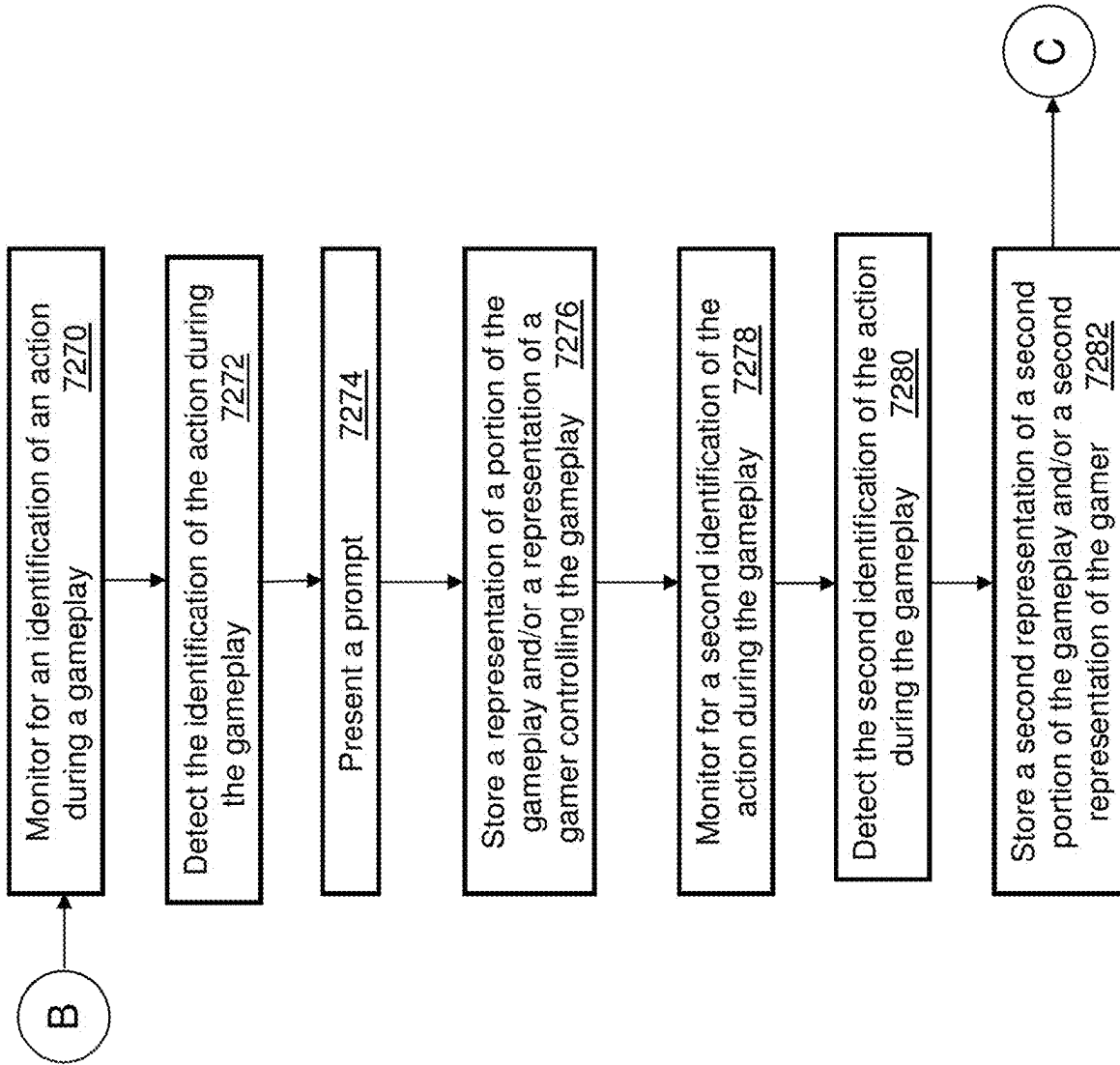
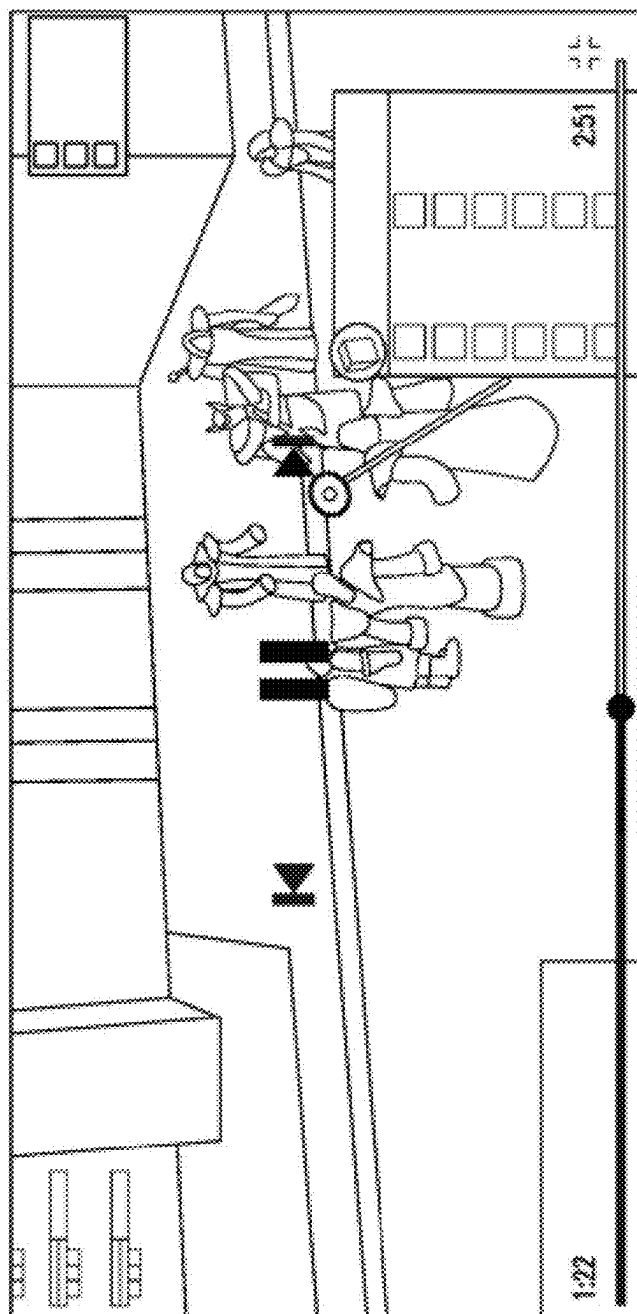


FIG. 7D



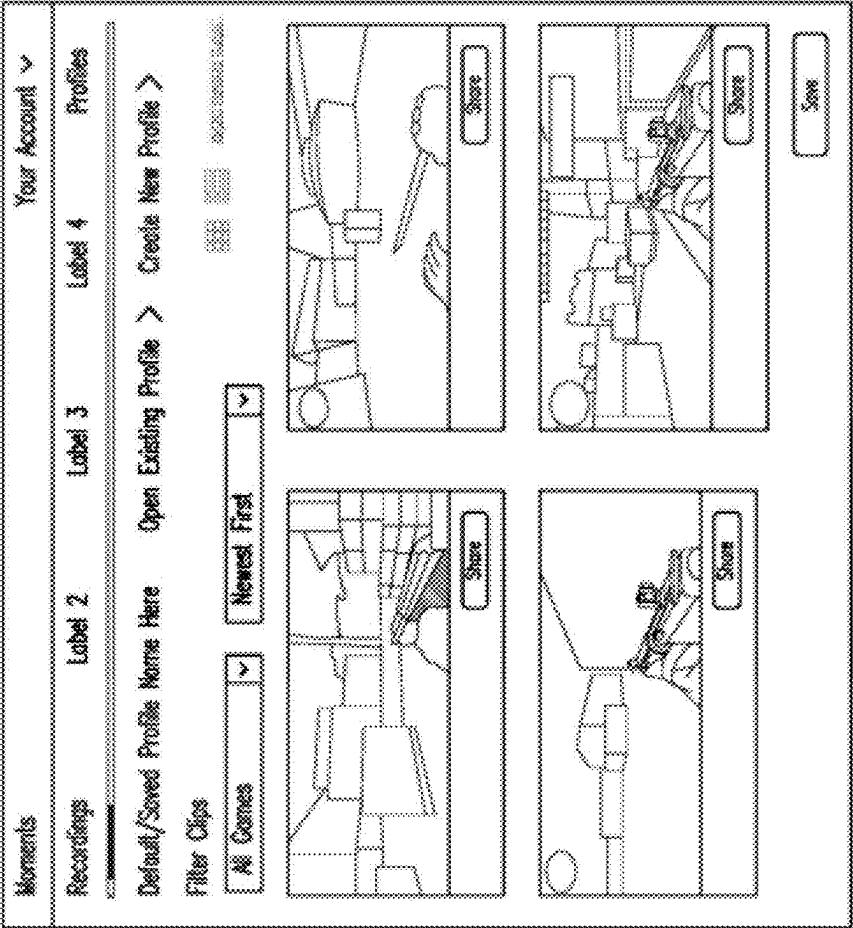


FIG. 7F

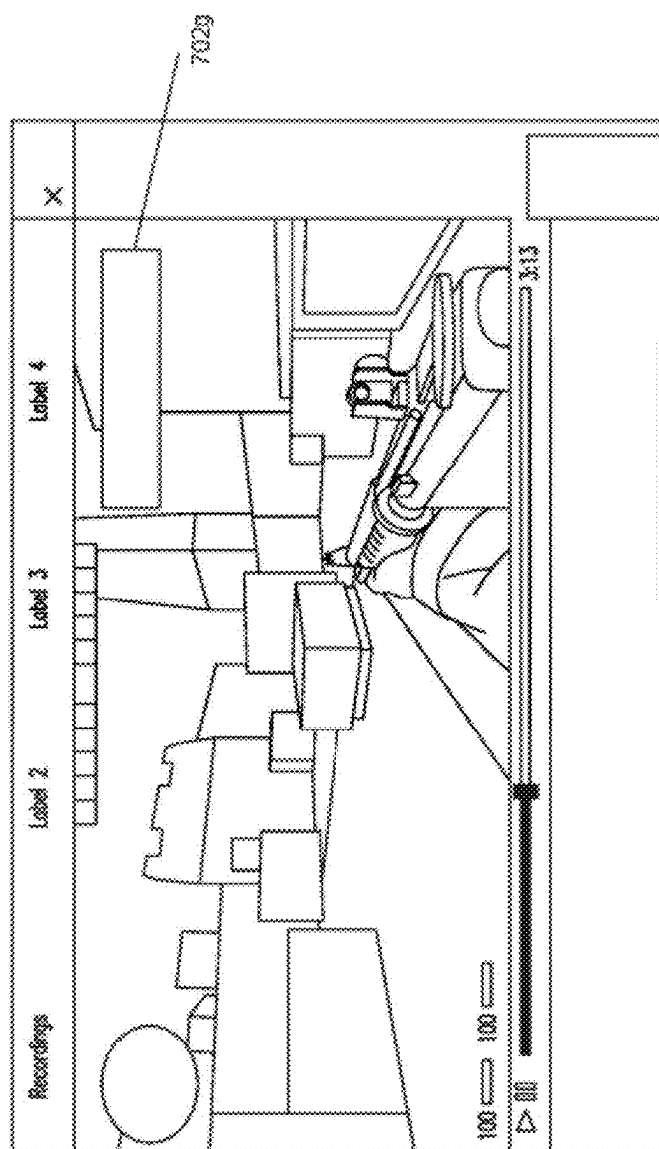


FIG. 7G

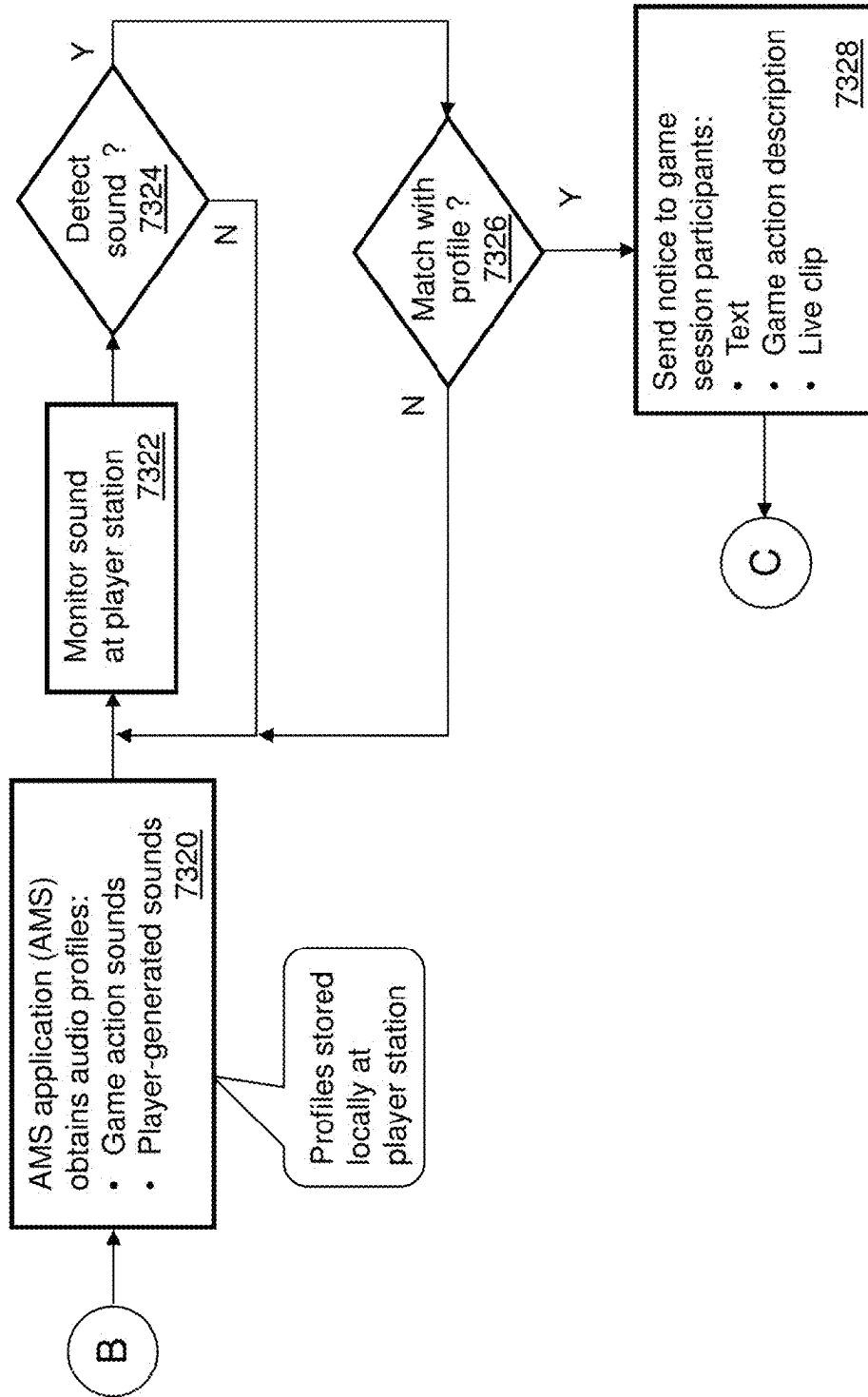


FIG. 7H

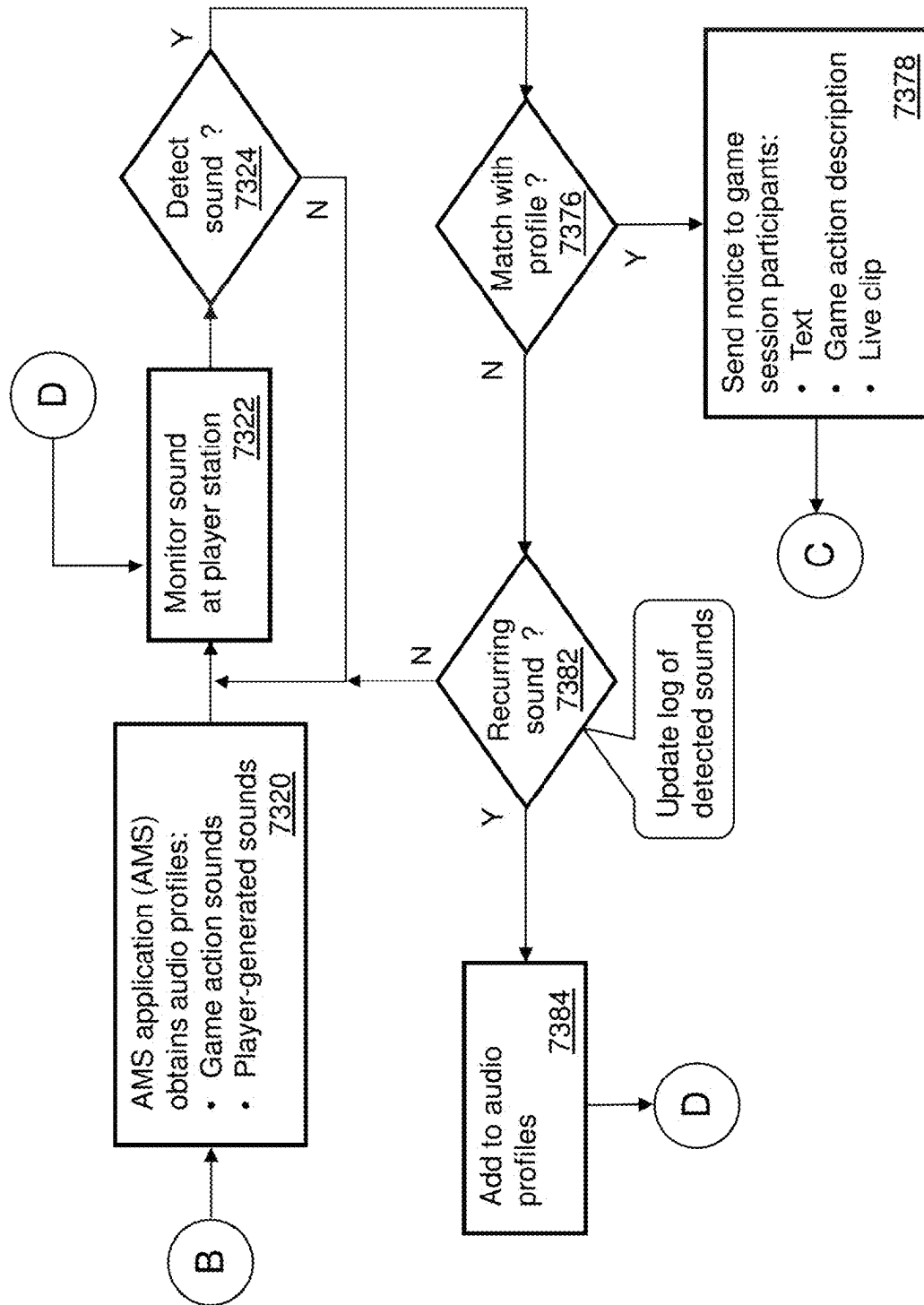


FIG. 71

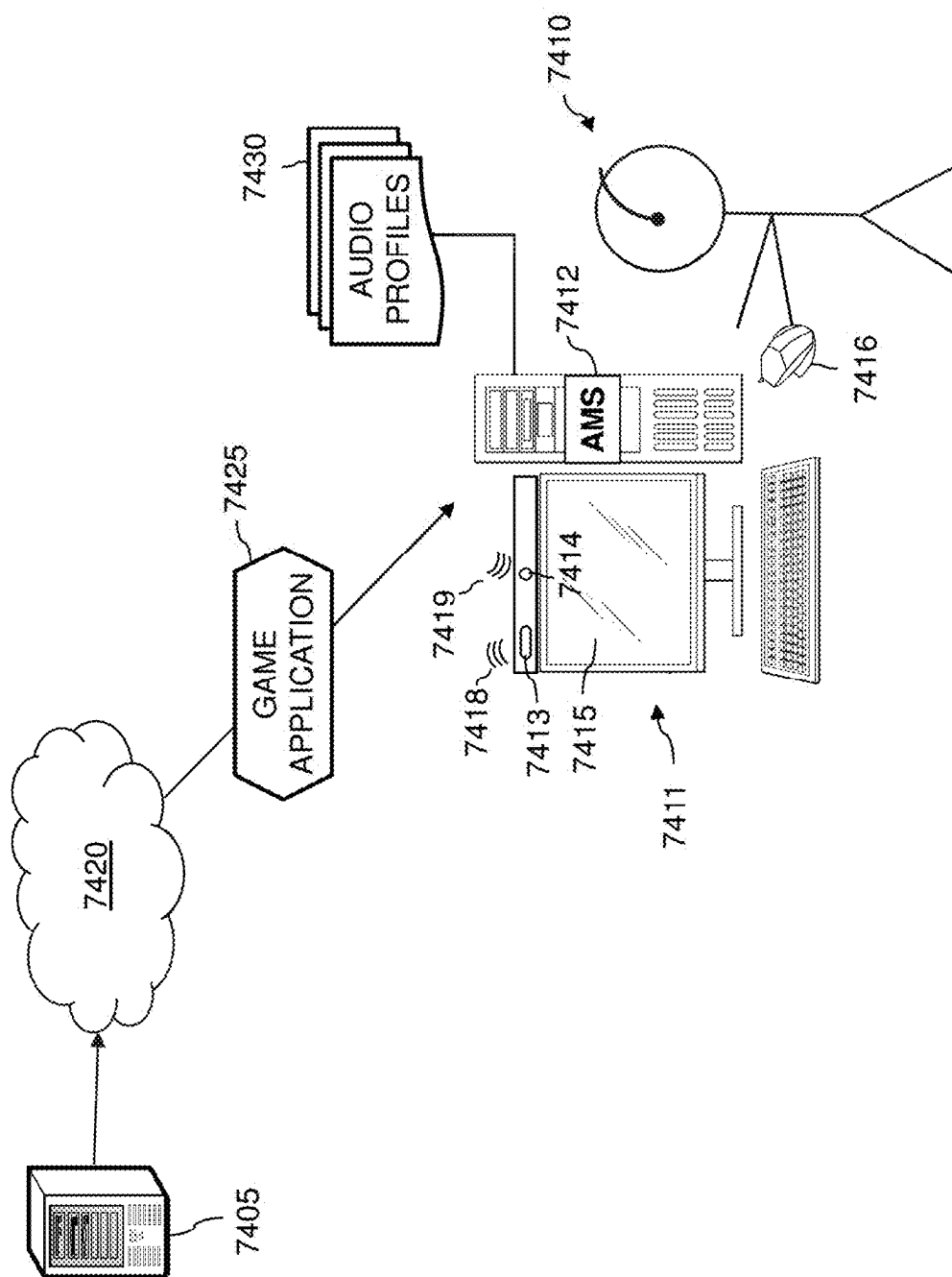


FIG. 7J

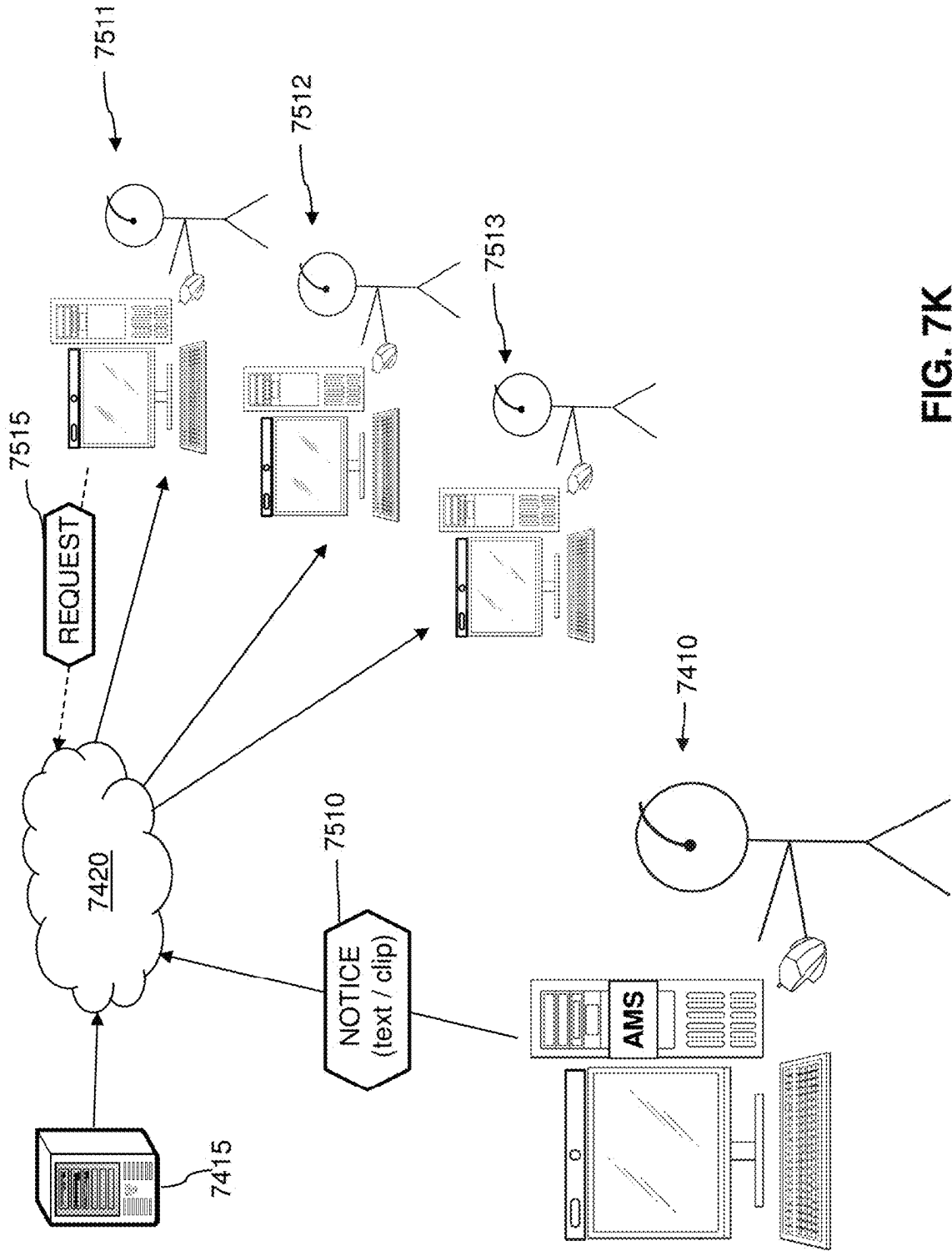
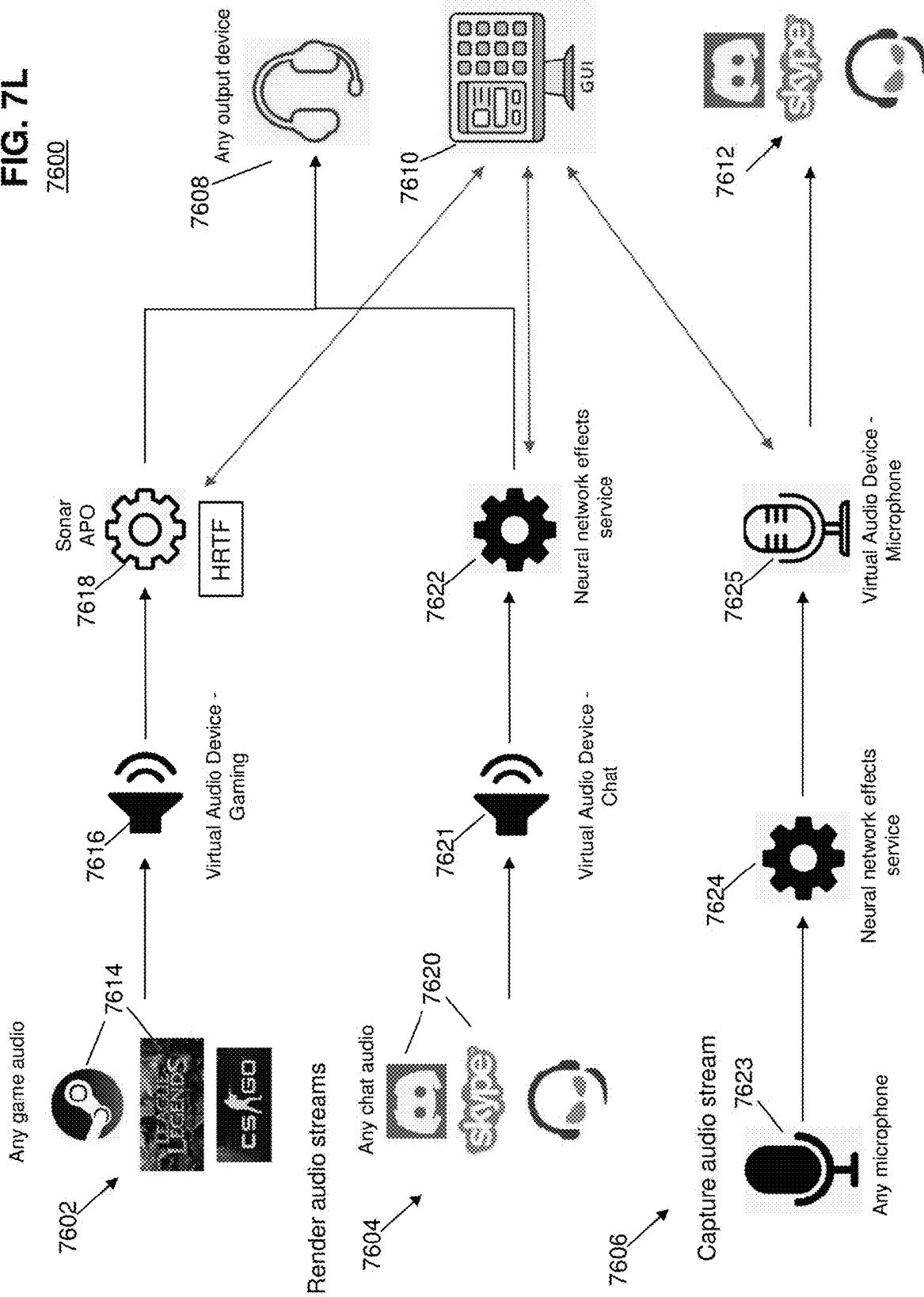


FIG. 7K

FIG. 7L



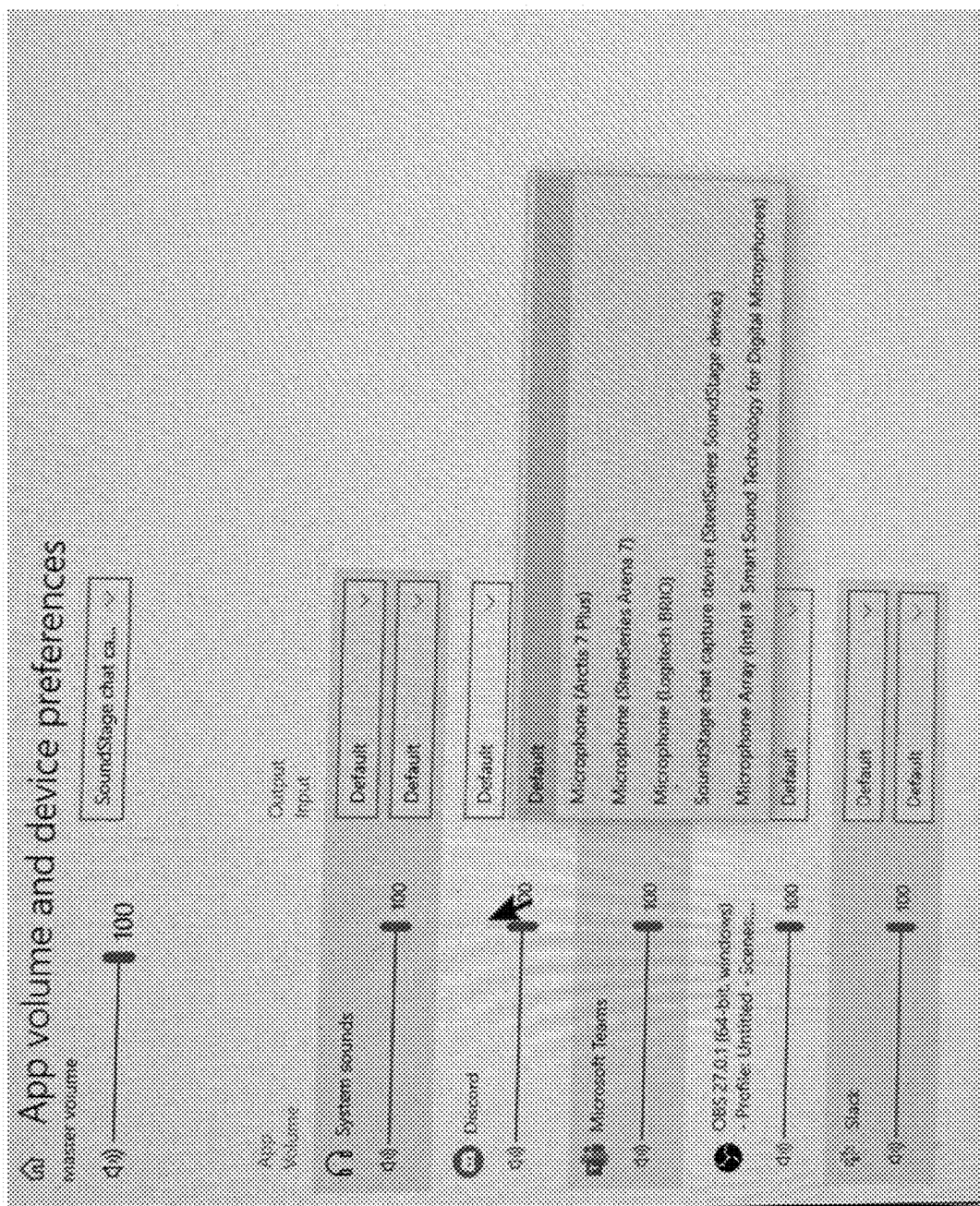


FIG. 7L-1

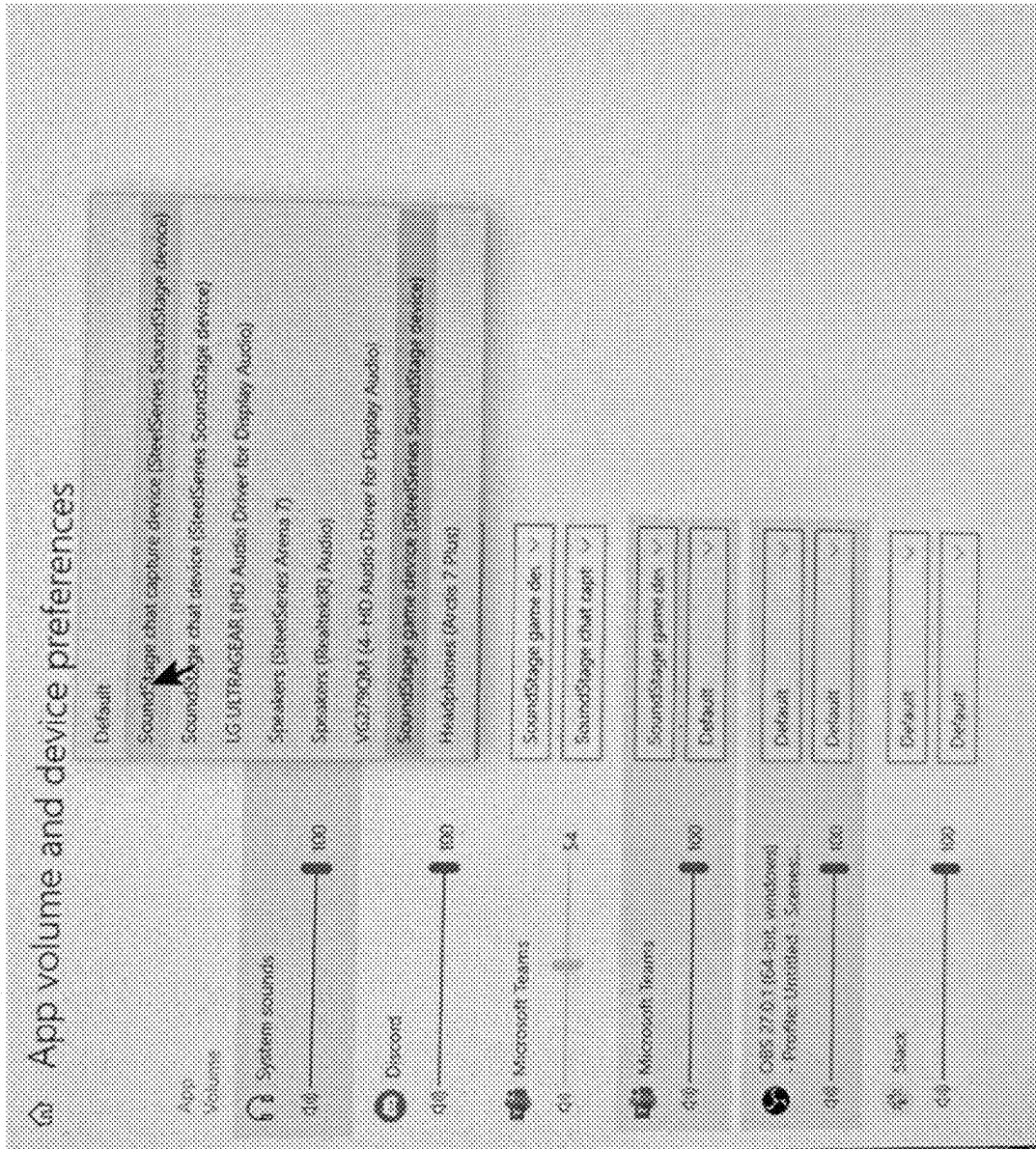


FIG. 7L-2

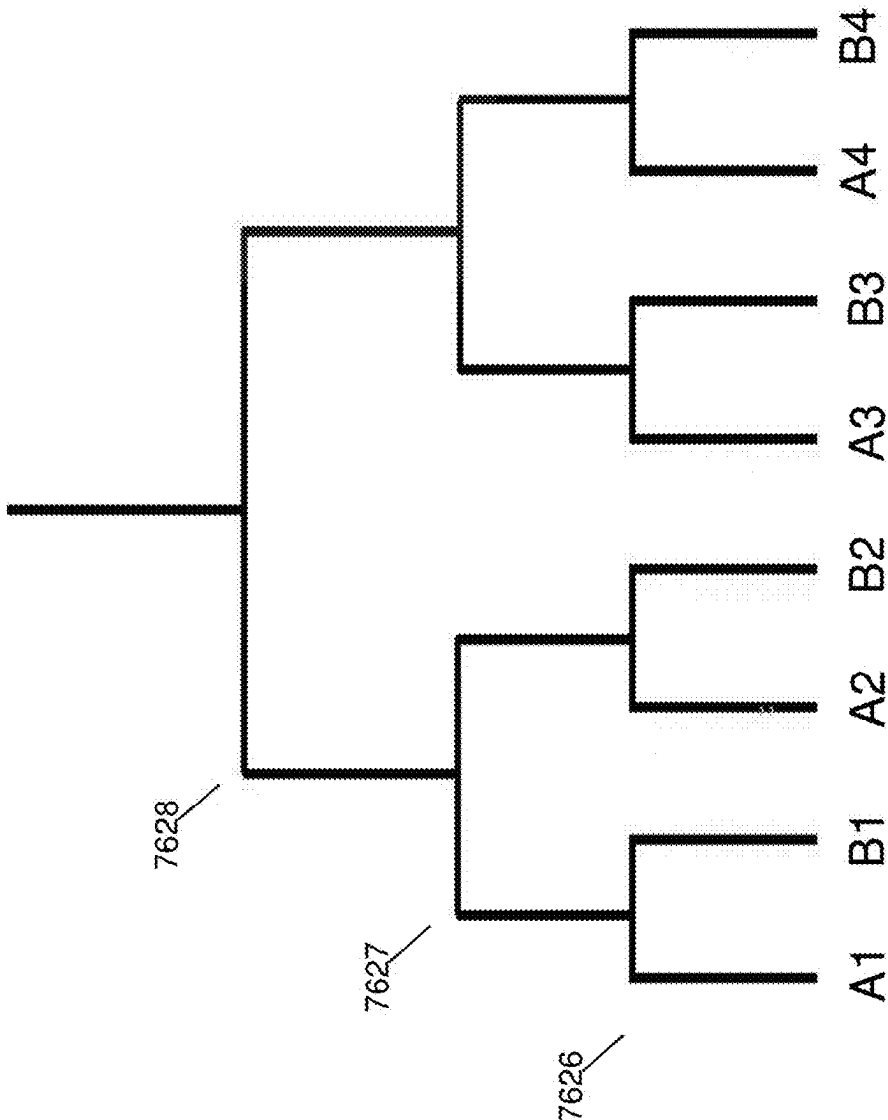


FIG. 7L-3

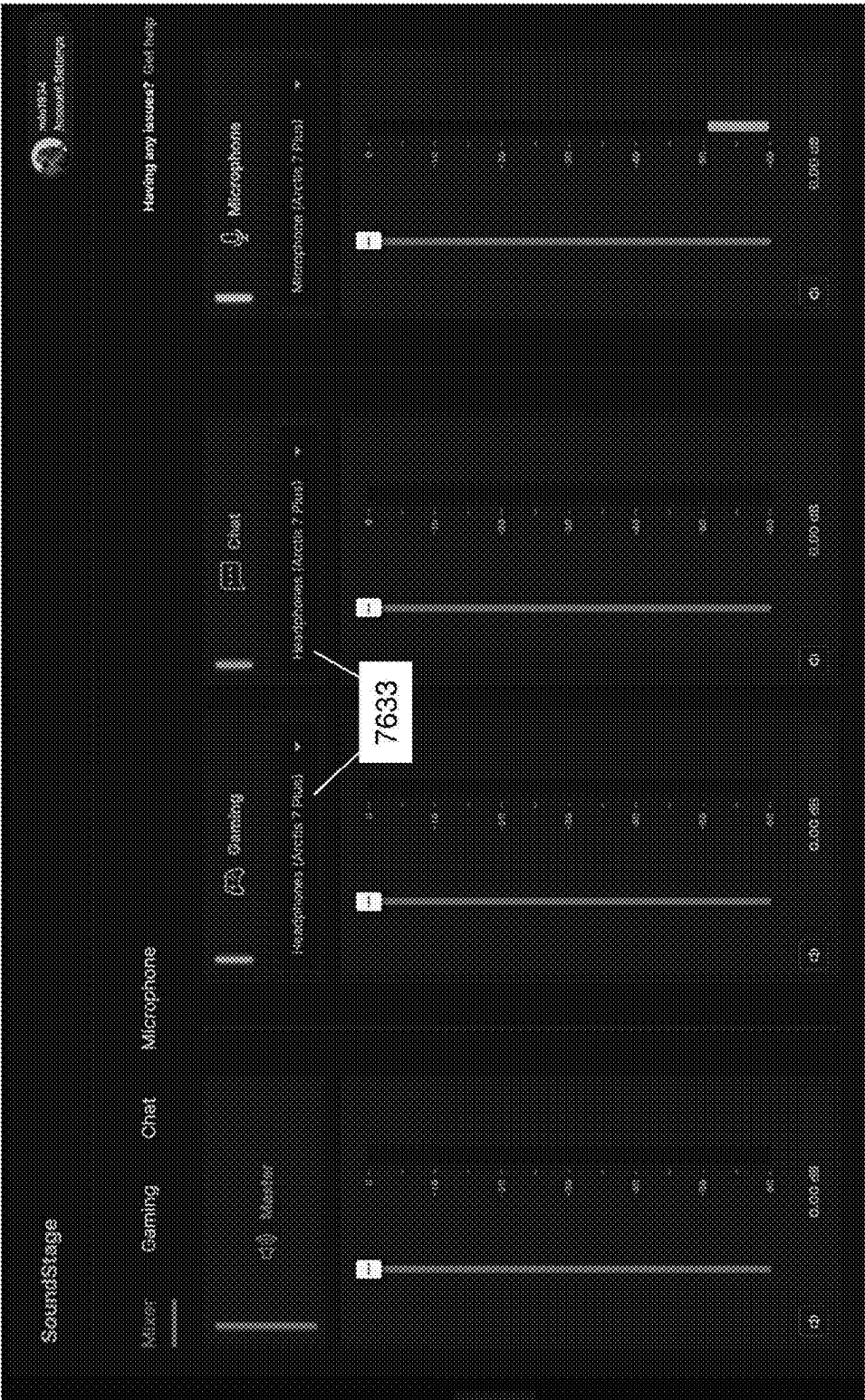


FIG. 7M

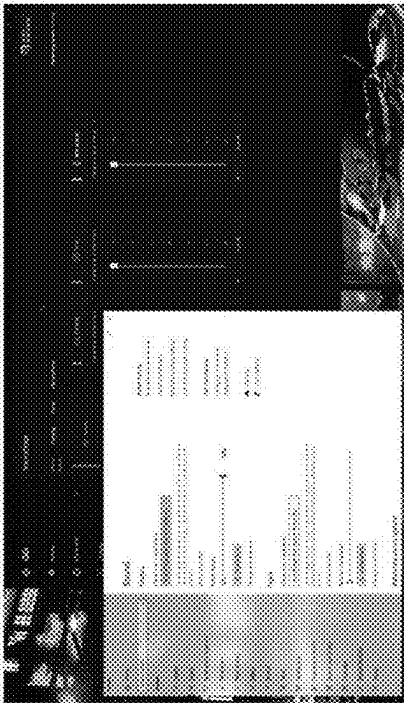


FIG. 7M-1

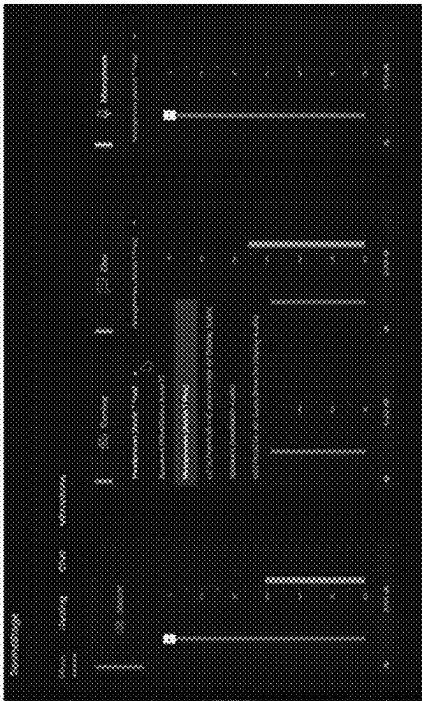


FIG. 7M-2

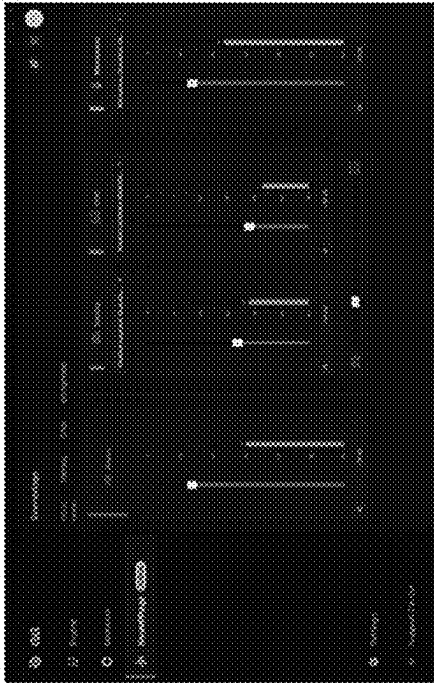


FIG. 7M-3

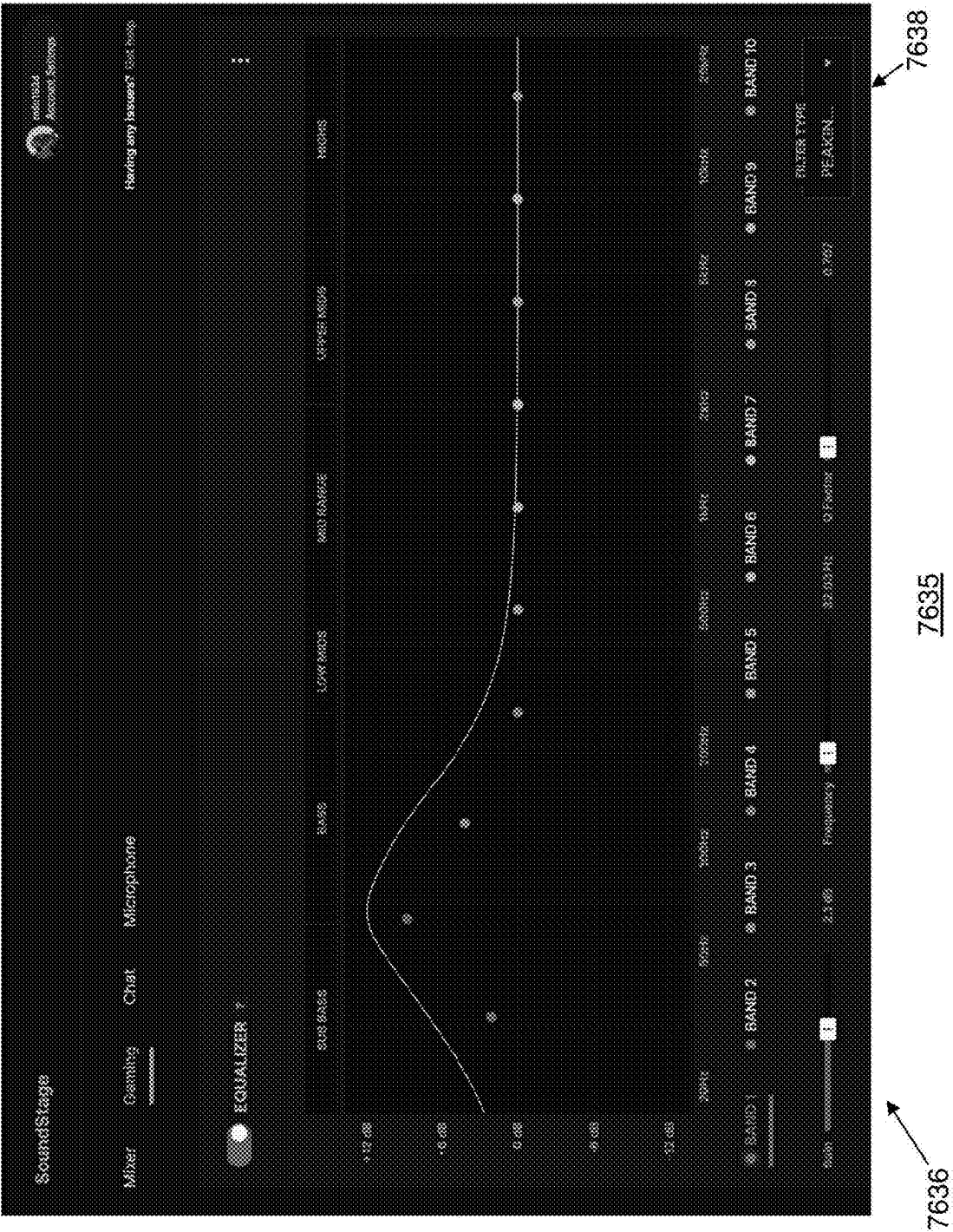


FIG. 7N

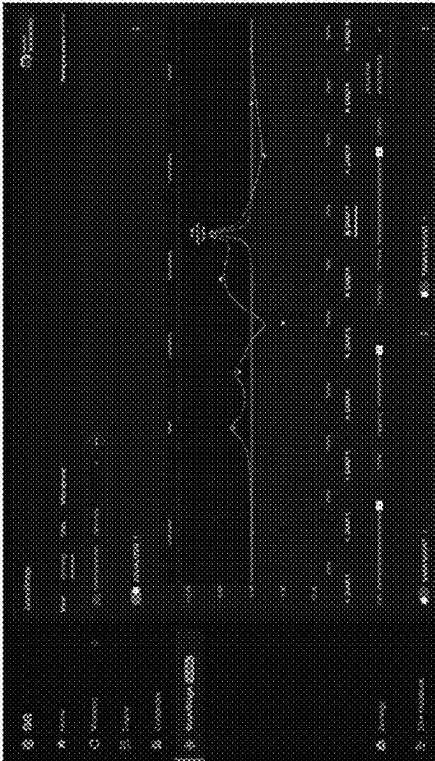


FIG. 7N-2

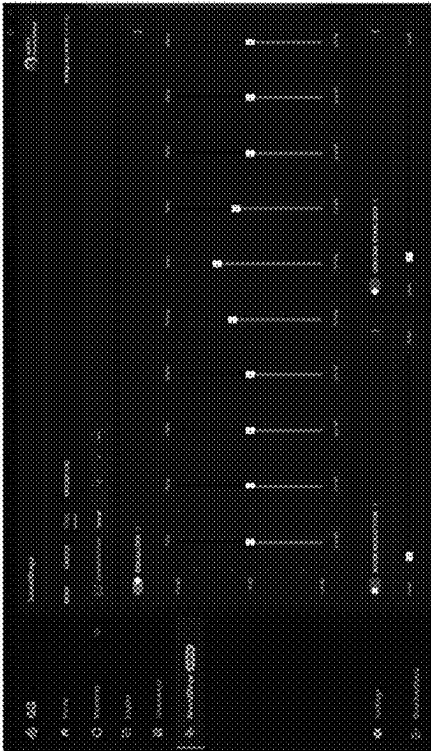


FIG. 7N-1

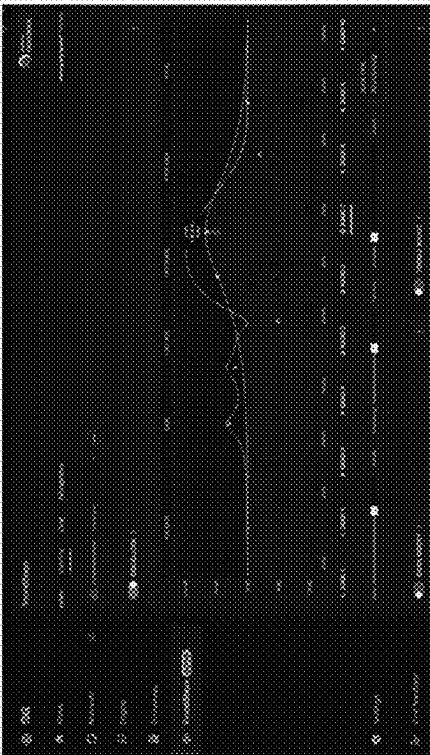
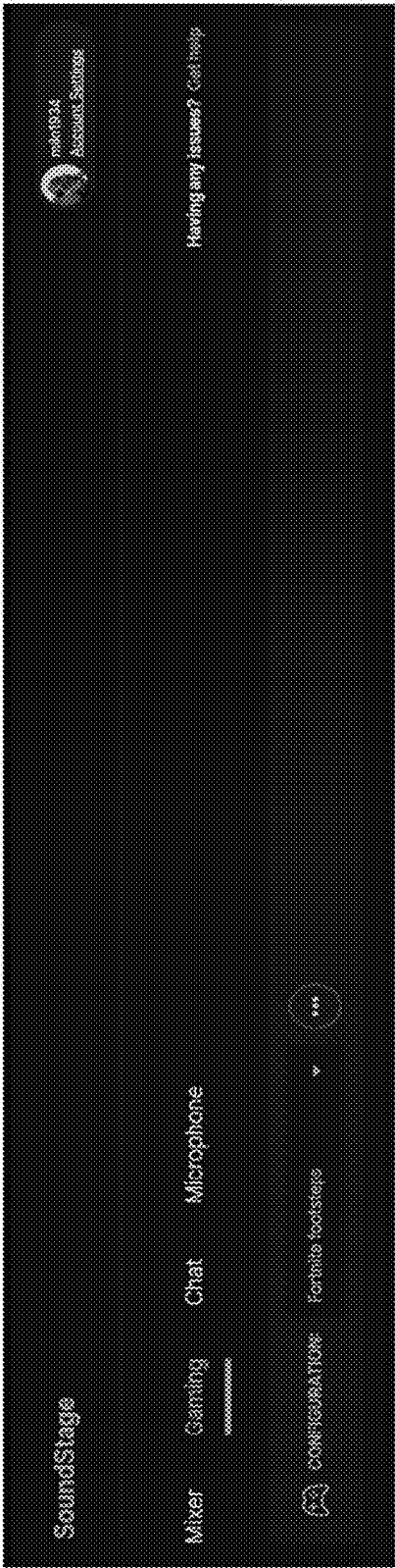
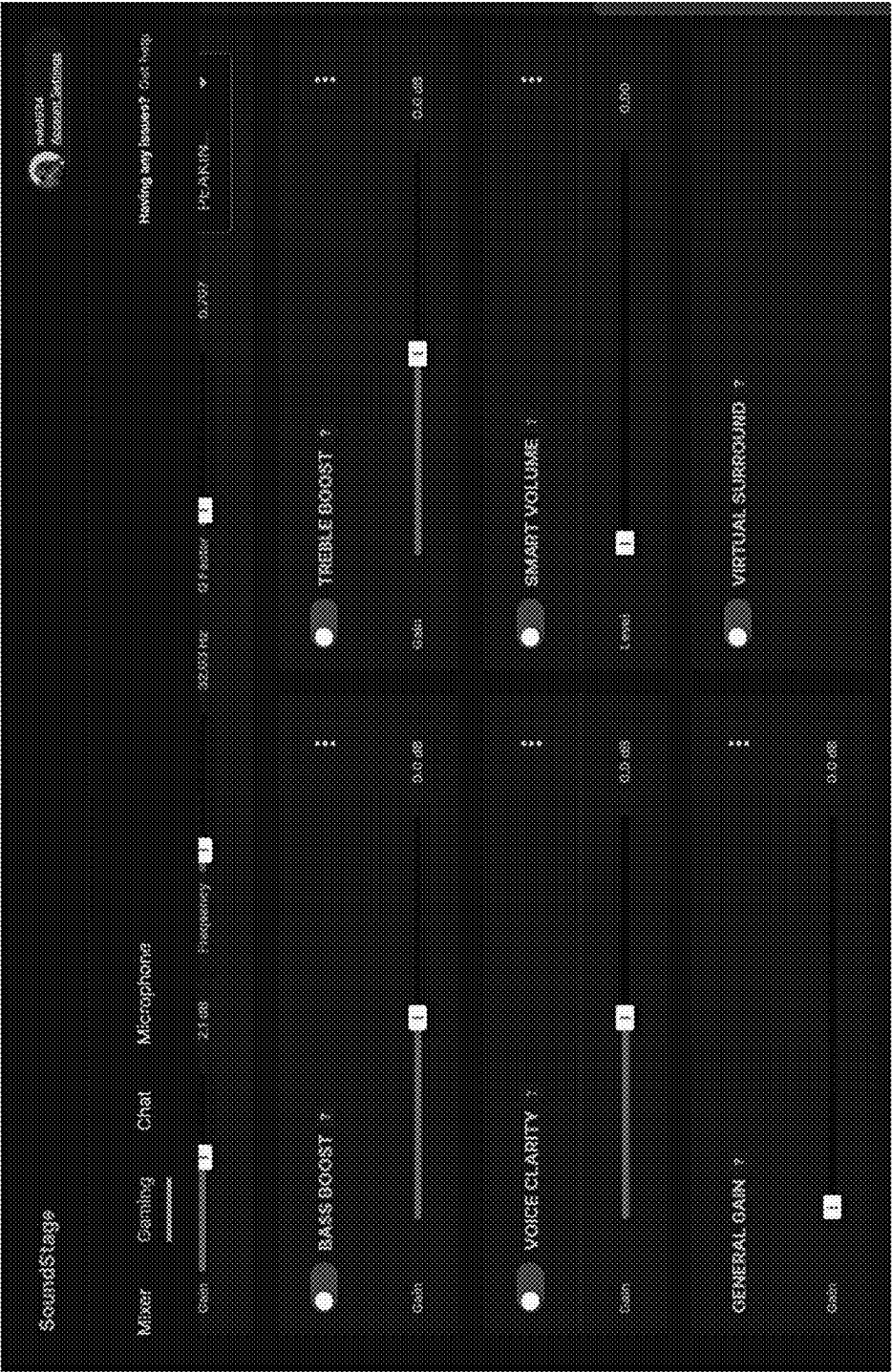


FIG. 7N-3



7640

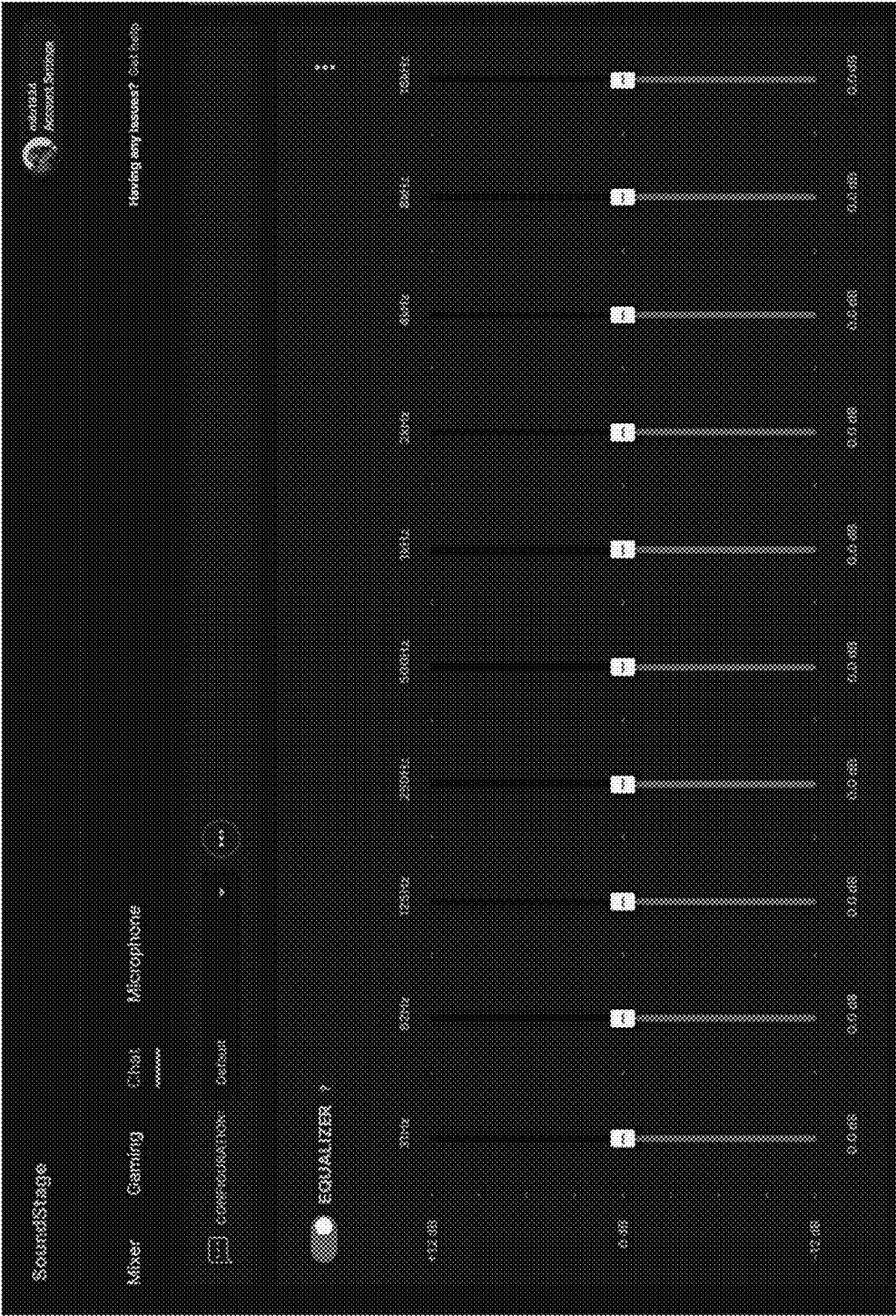
FIG. 70



7646 →

→ 7647

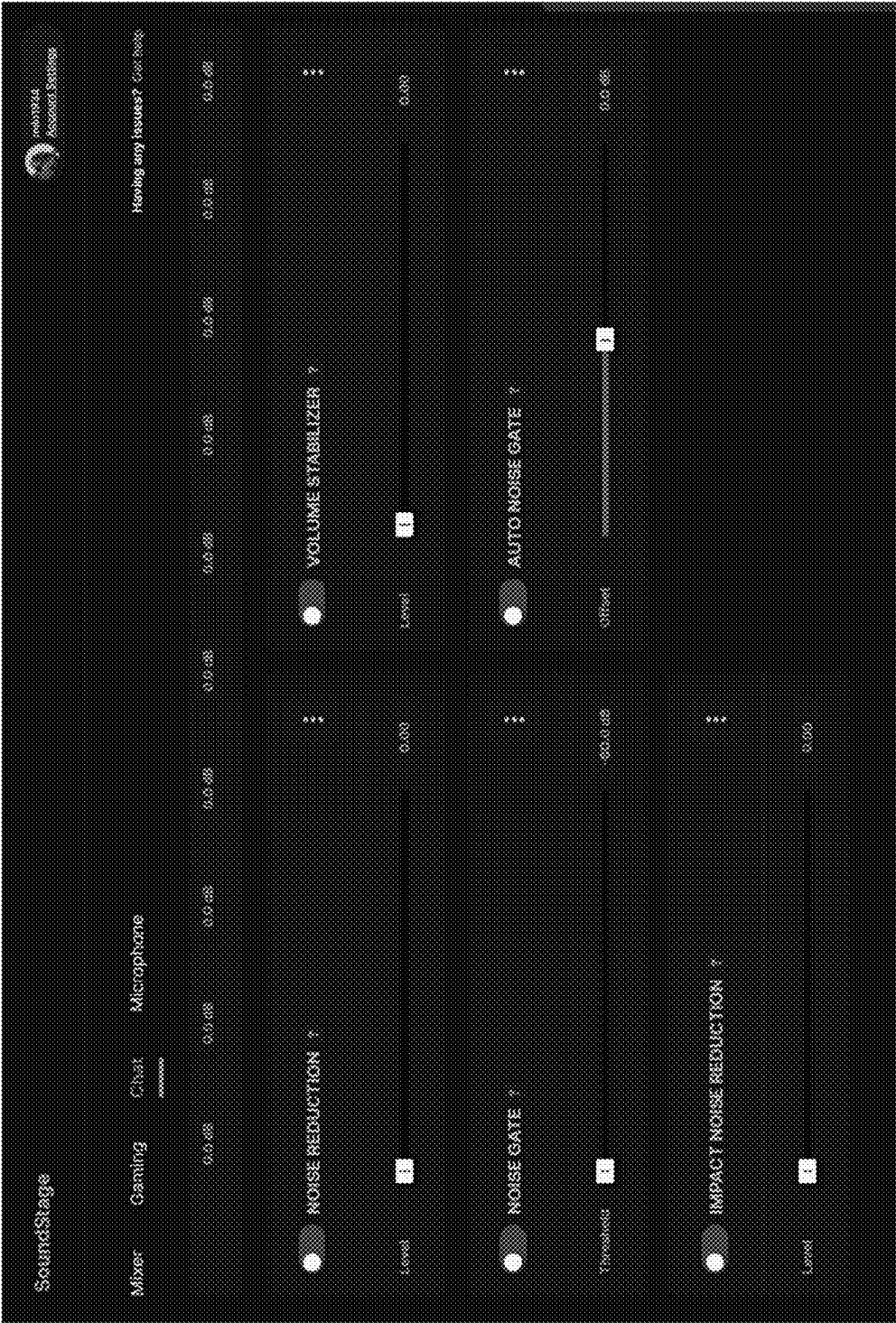
7645
FIG. 7P



7651
7653
7654

7652

7650
FIG. 7Q

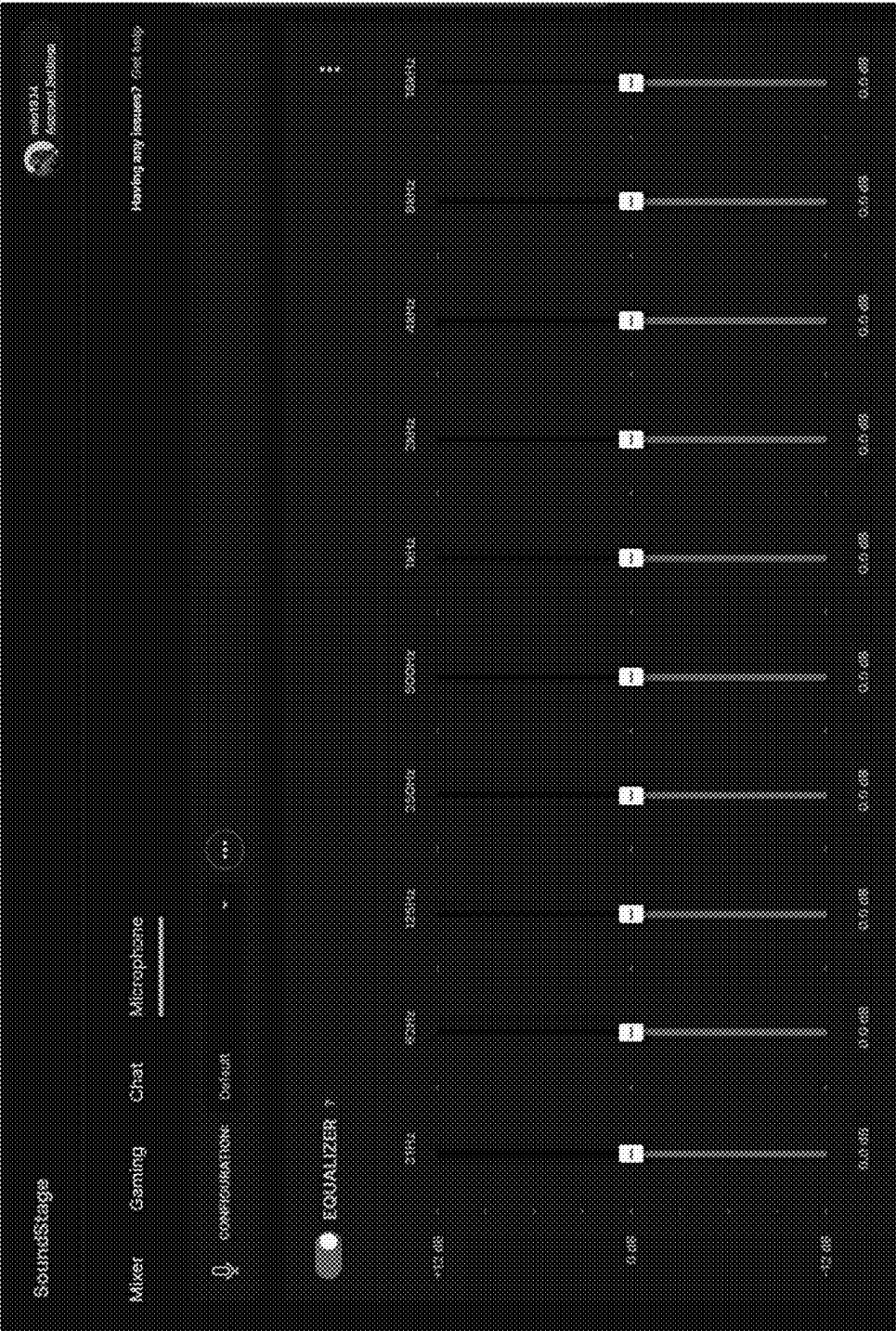


7656

7657

7655

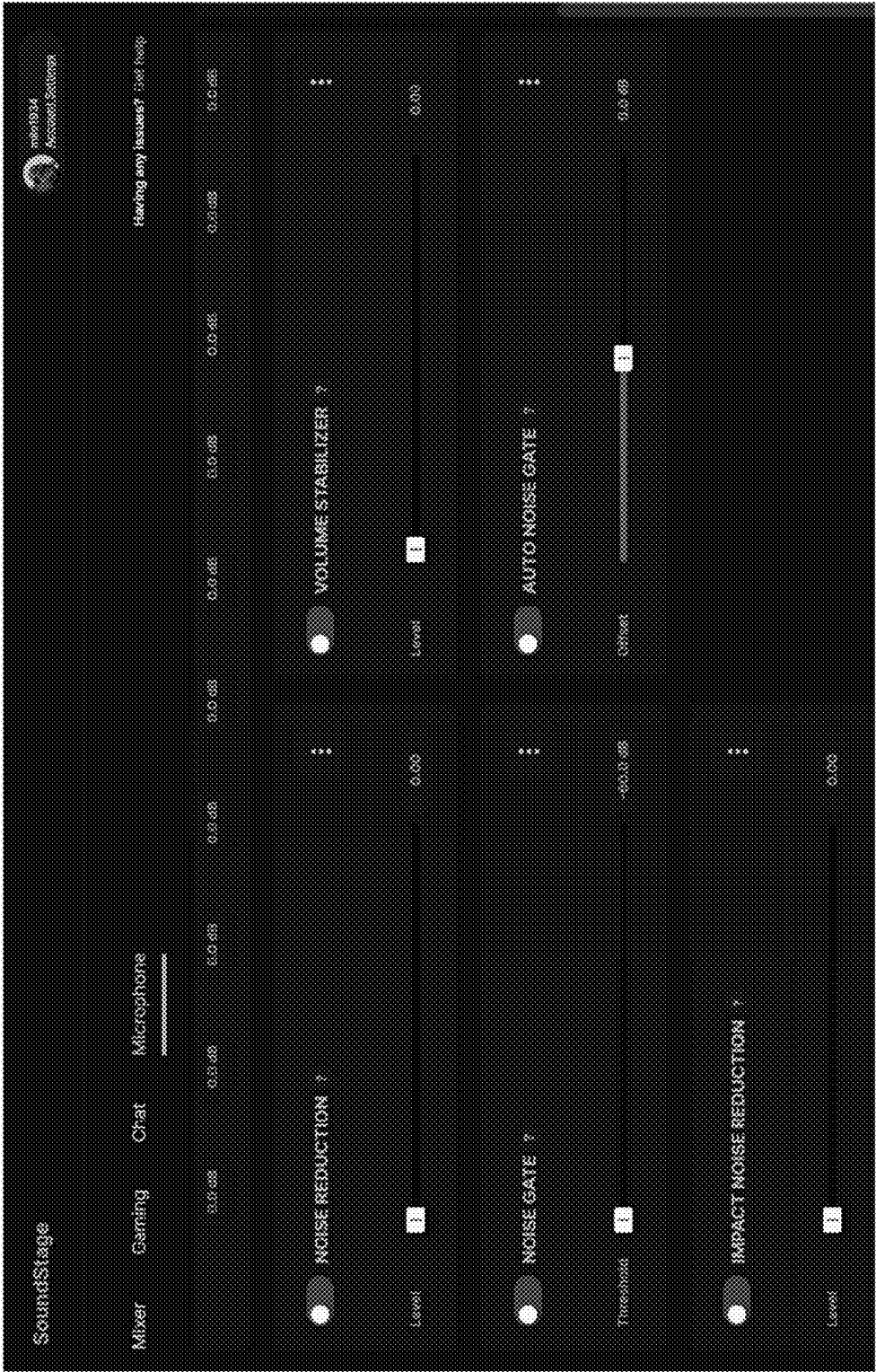
FIG. 7R



7661
7663
7664

7662

7660
FIG. 7S

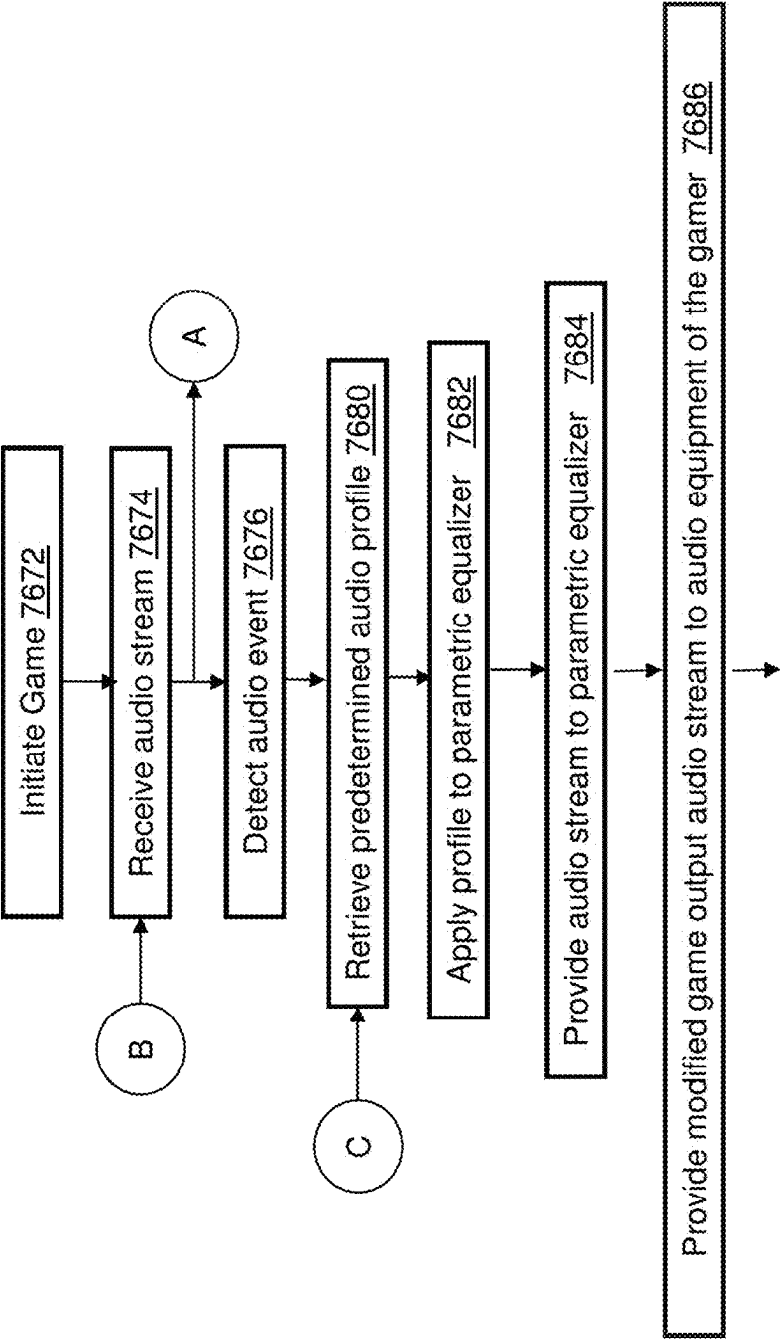


7666

7667

7665

FIG. 7T



7670

FIG. 7U

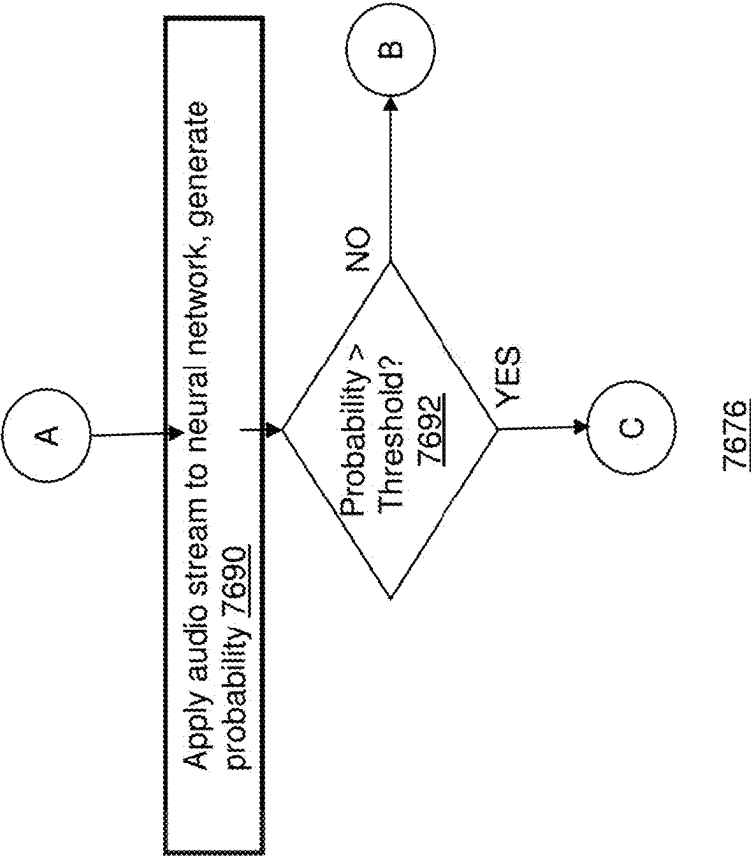


FIG. 7V

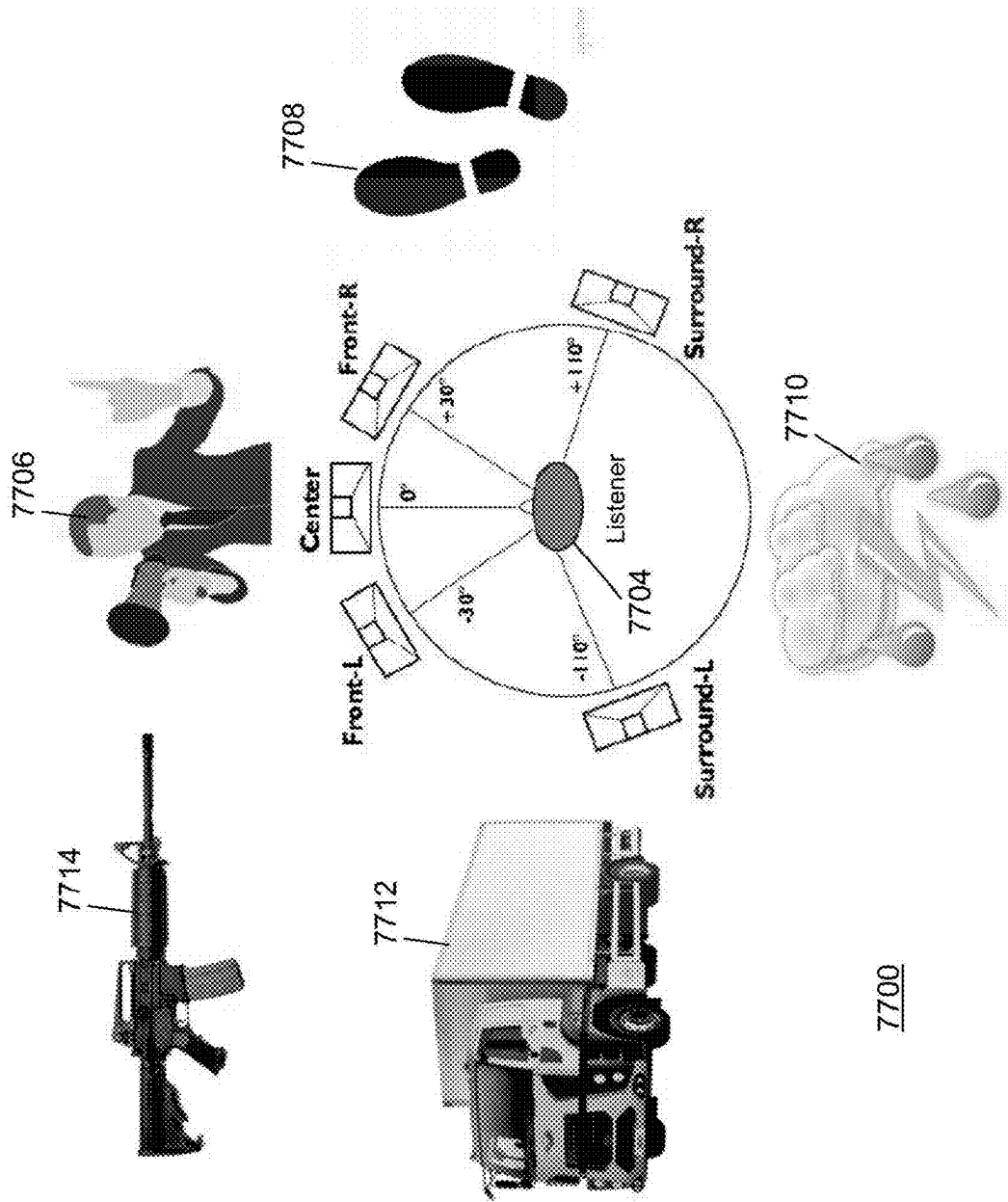
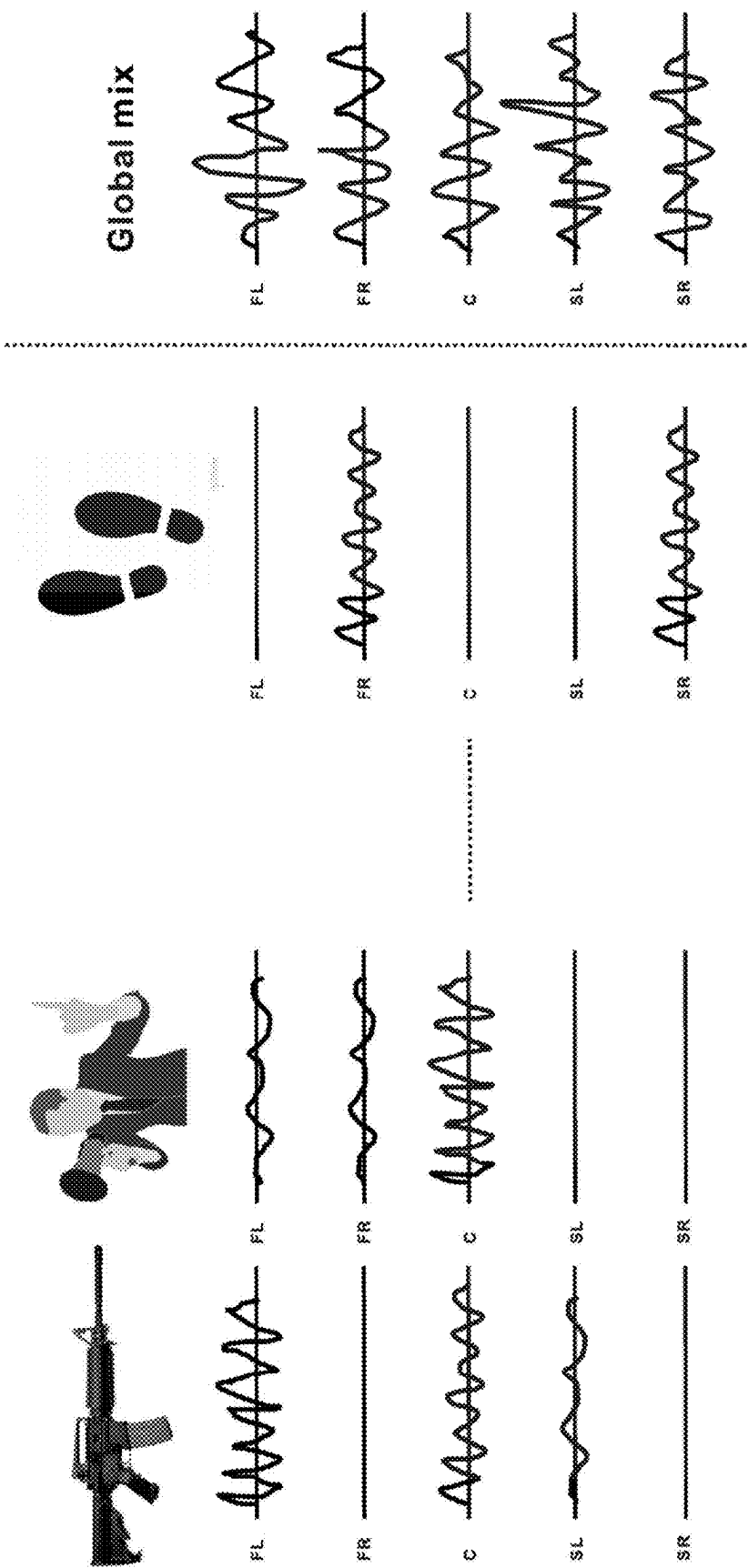


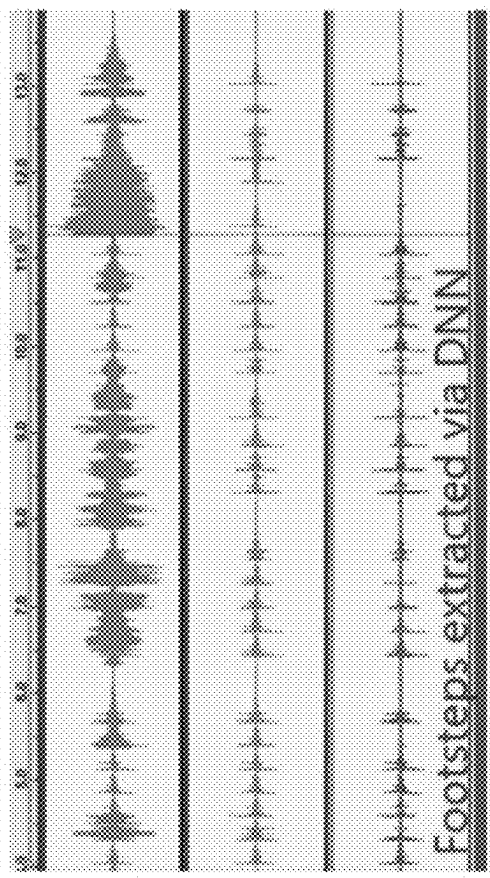
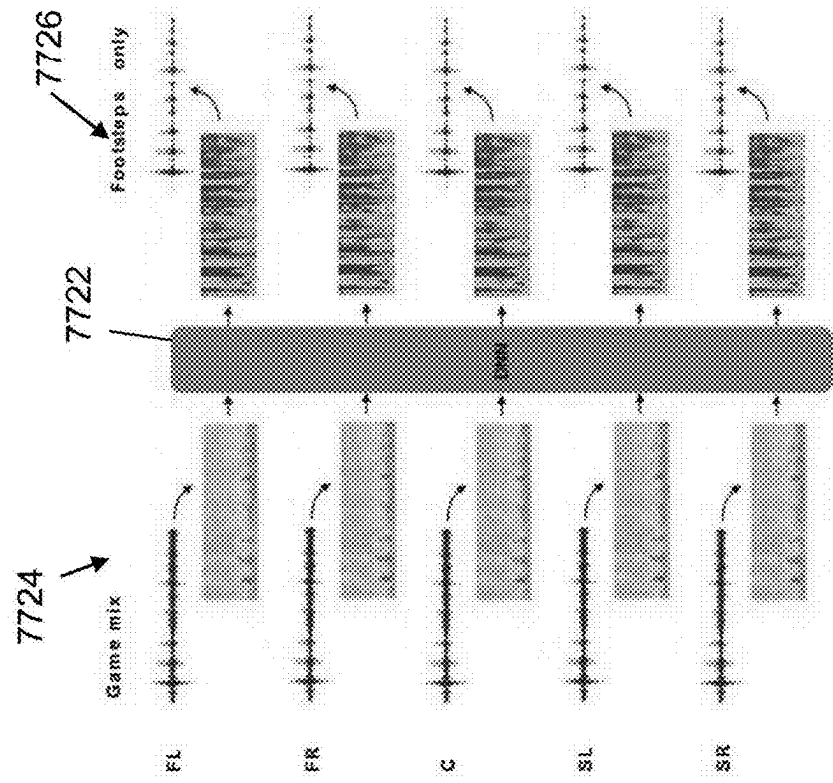
FIG. 7W

206



7702

FIG. 7X



7720

FIG. 7Y

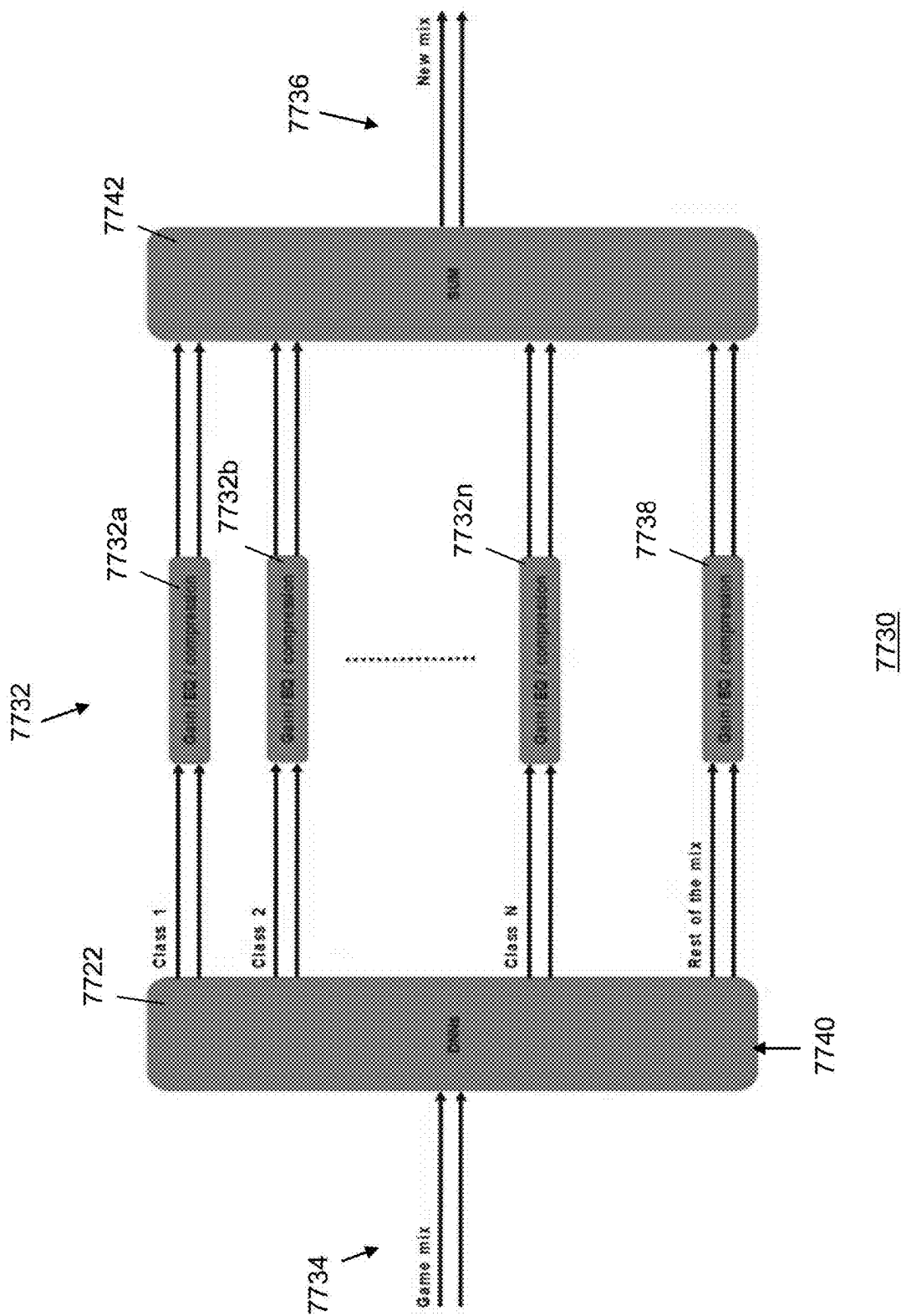
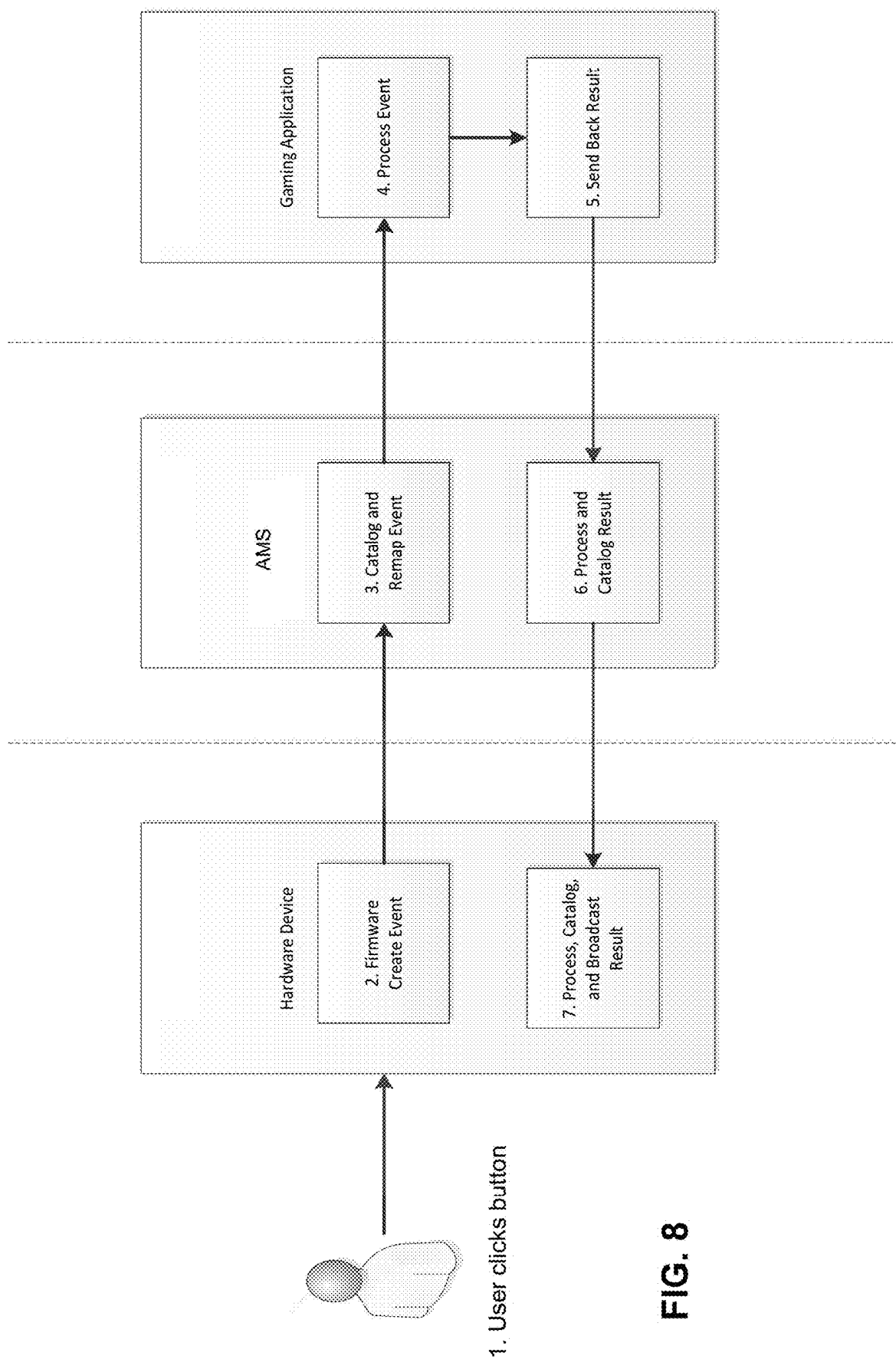
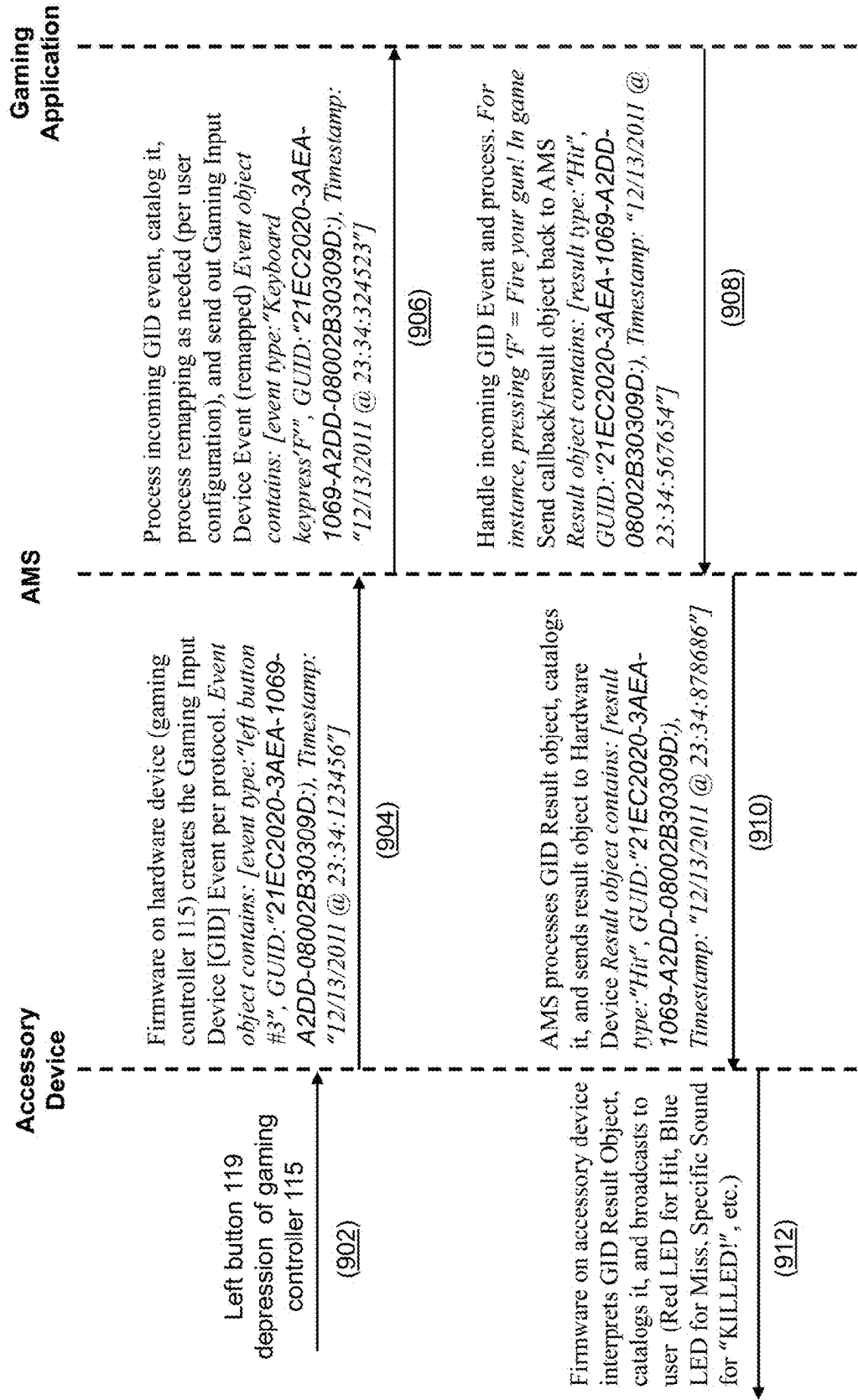
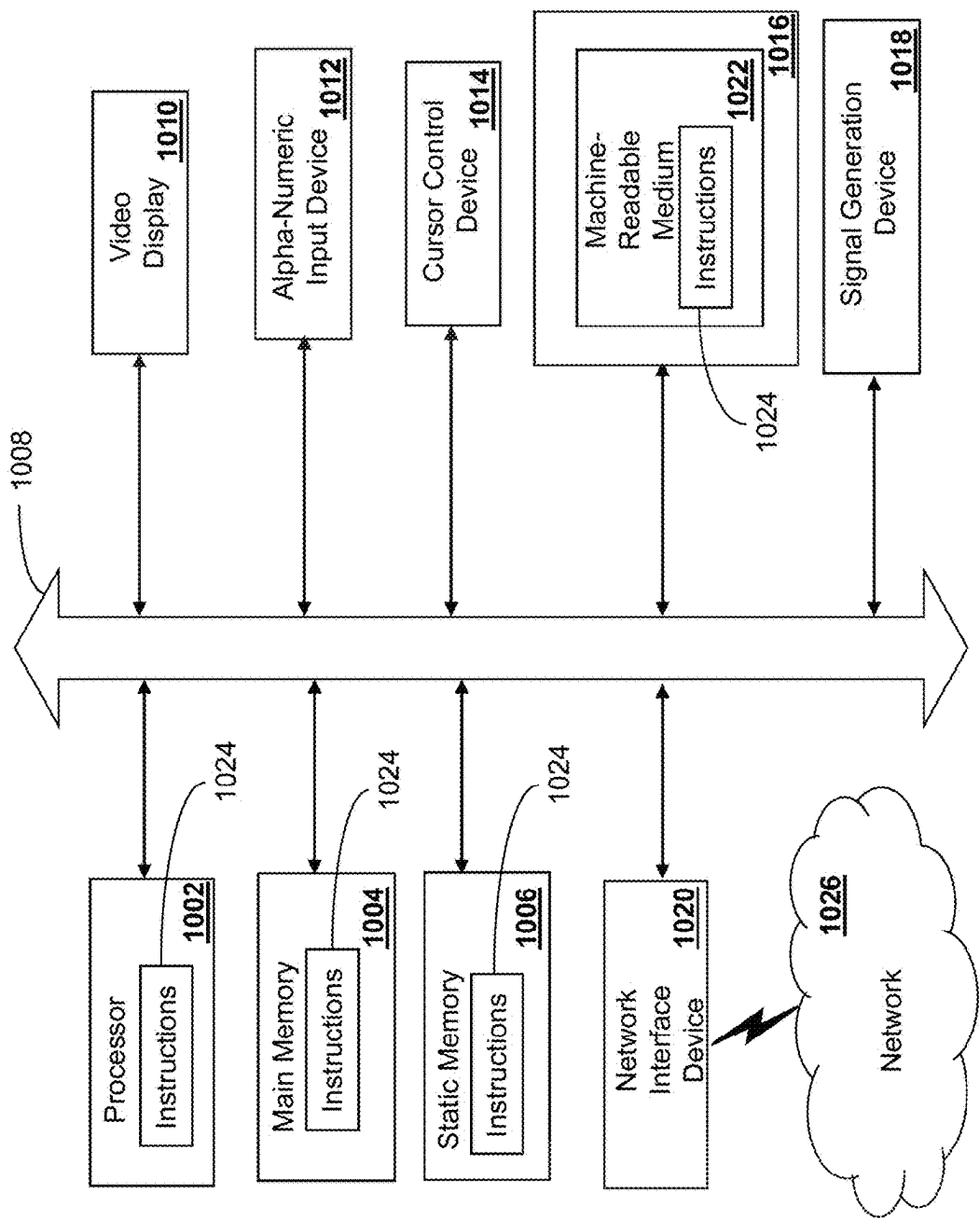


FIG. 7Z





900
FIG. 9



1000
FIG. 10

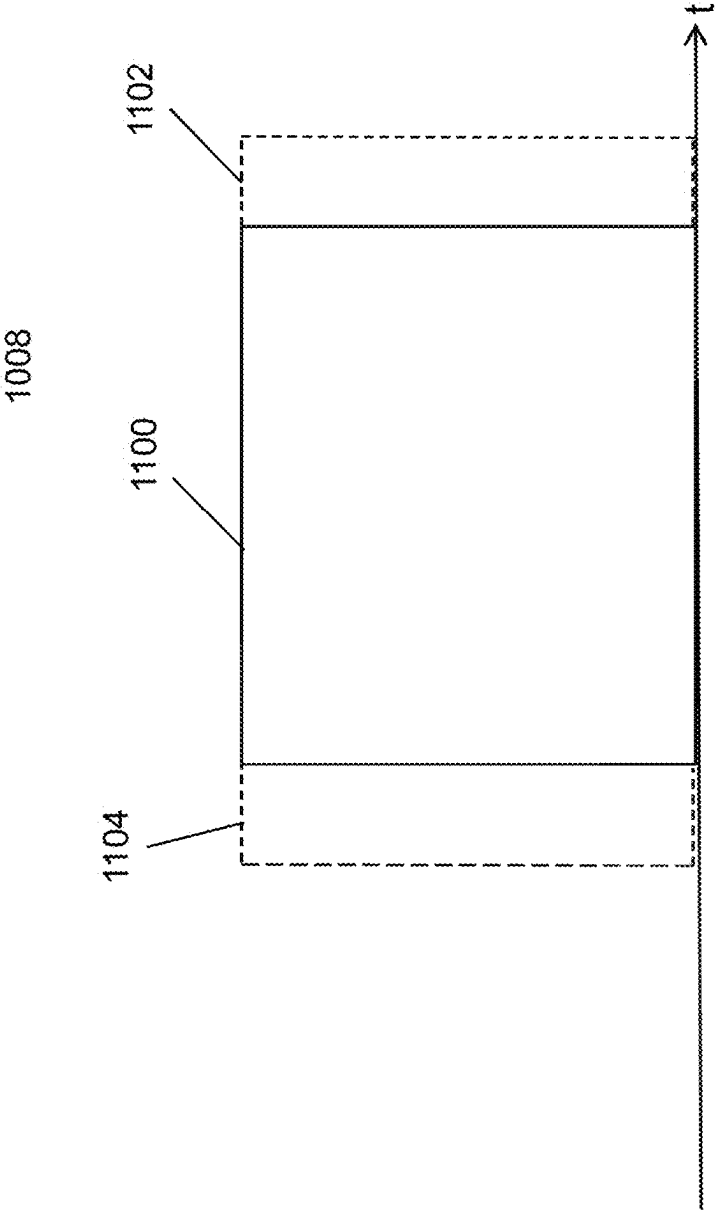


FIG. 11

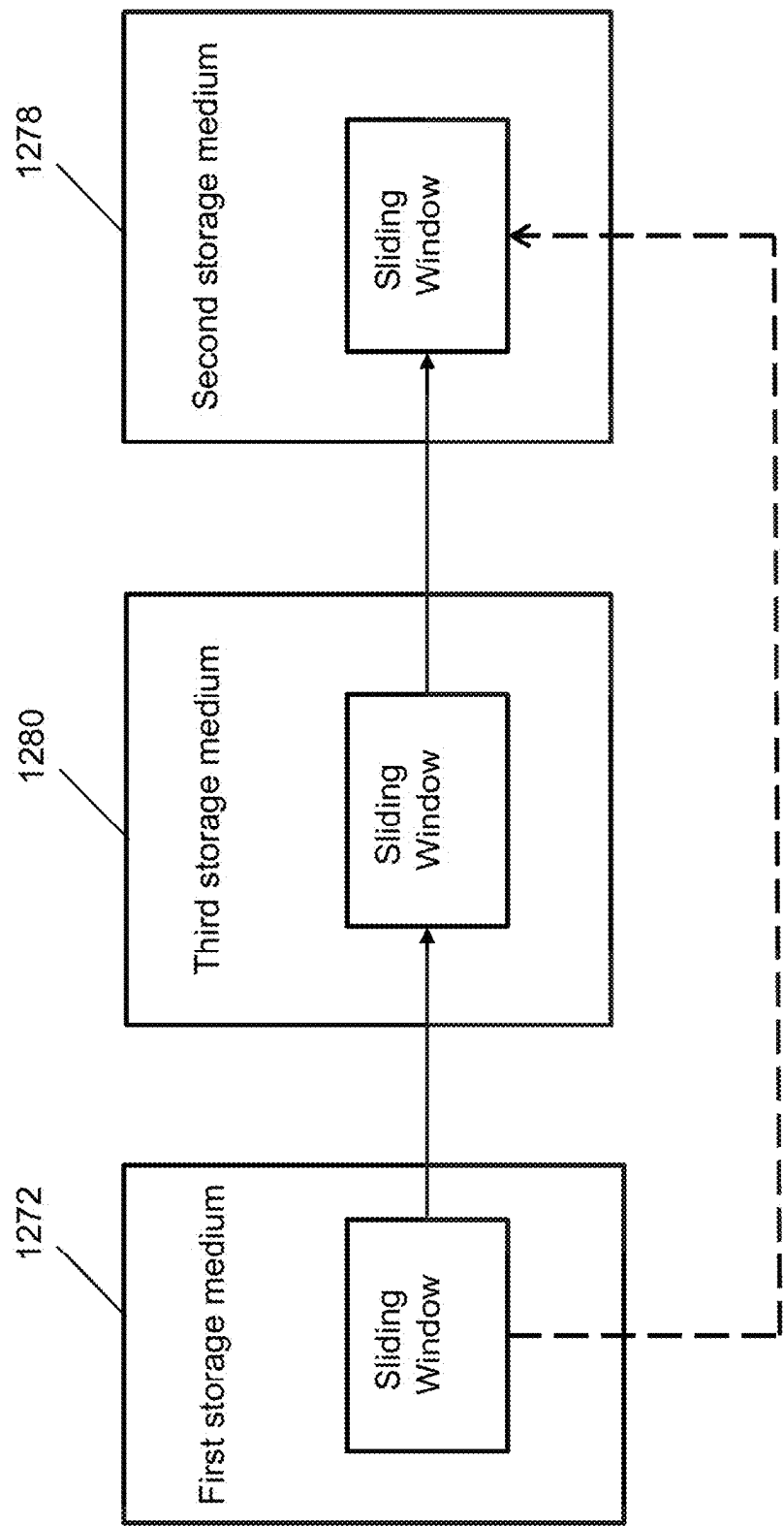
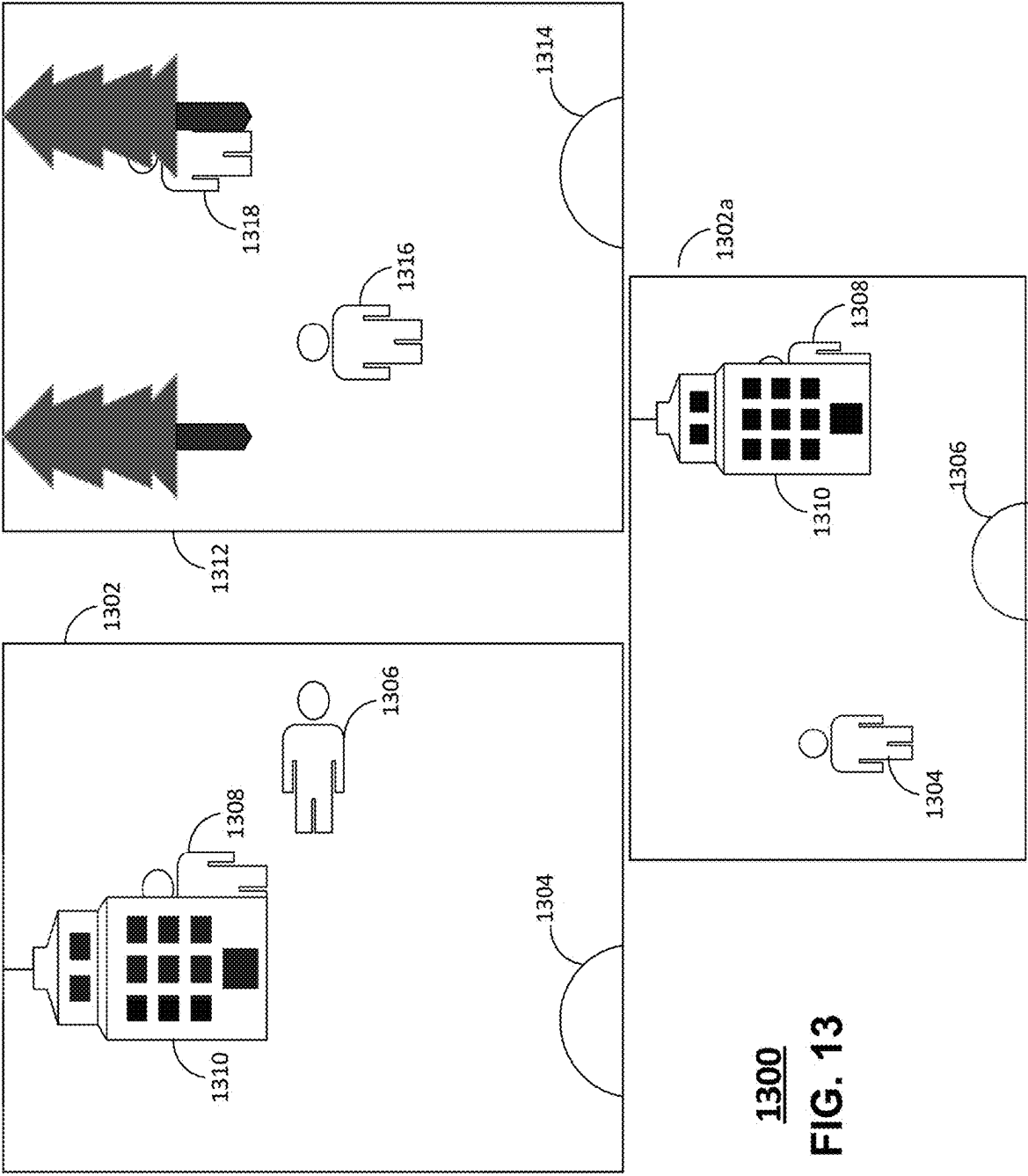
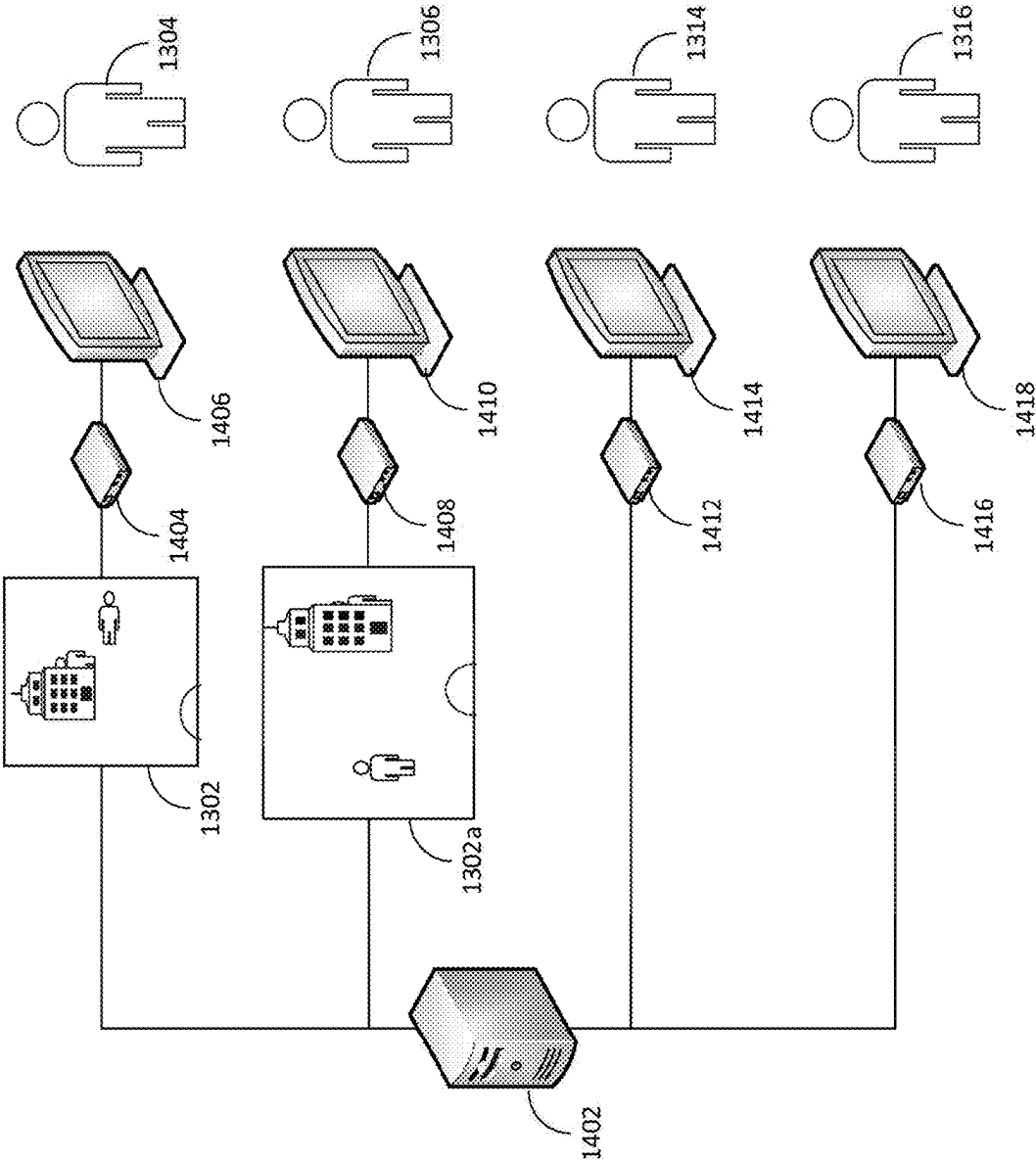


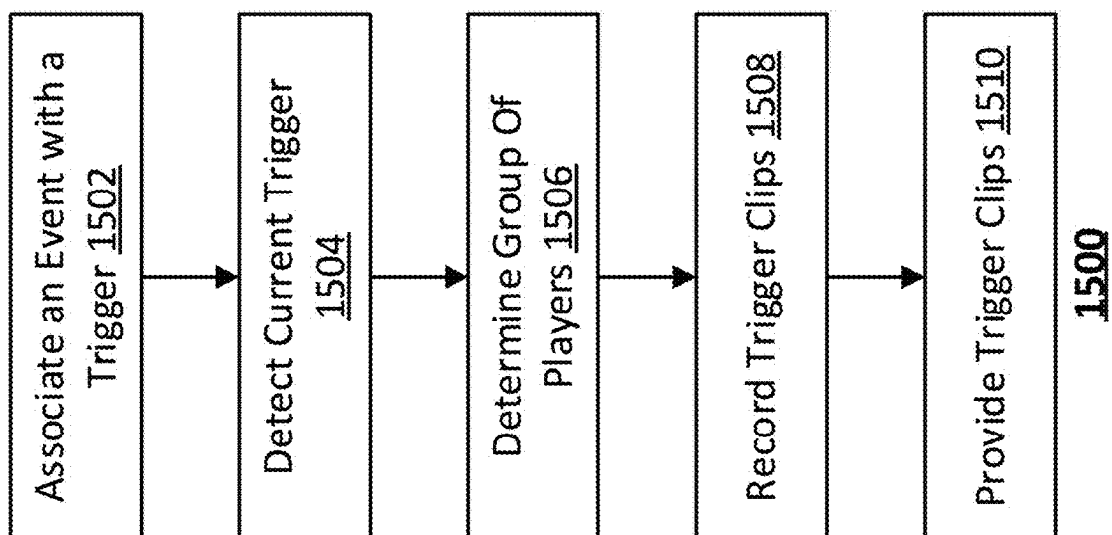
FIG. 12

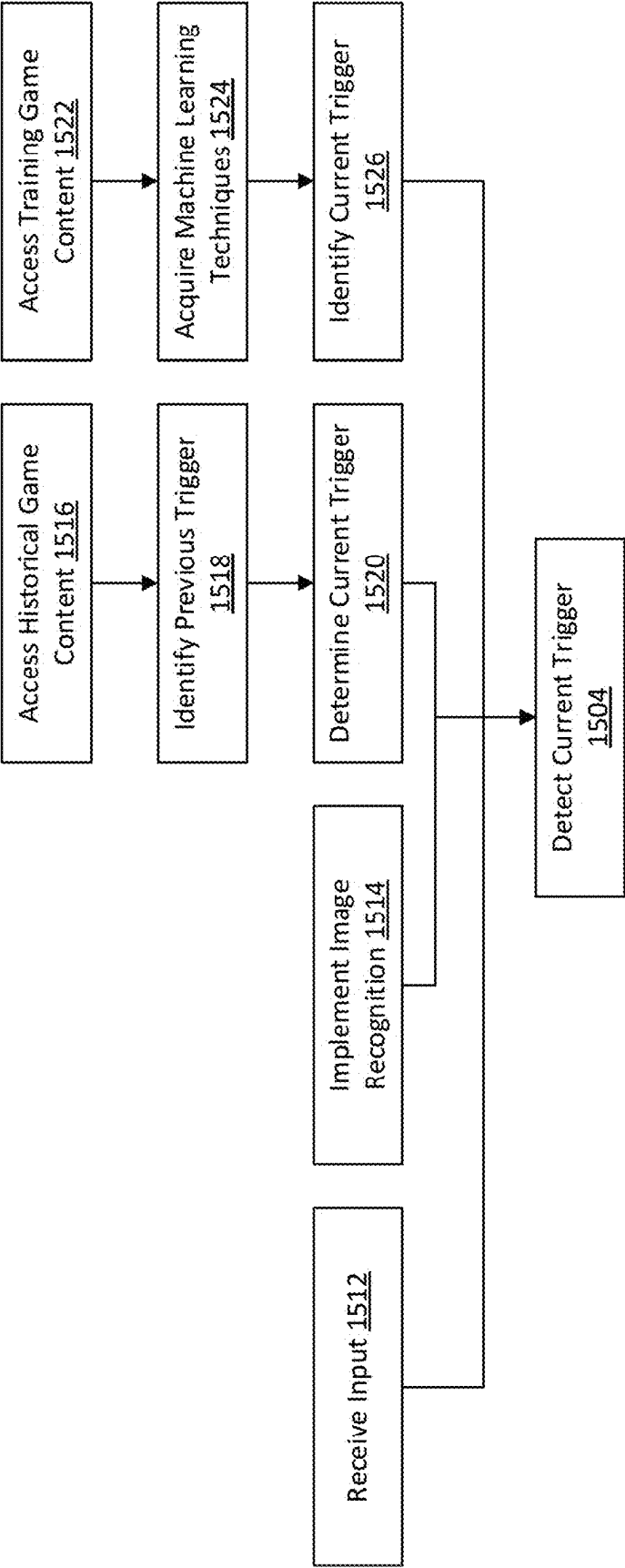


1300
FIG. 13

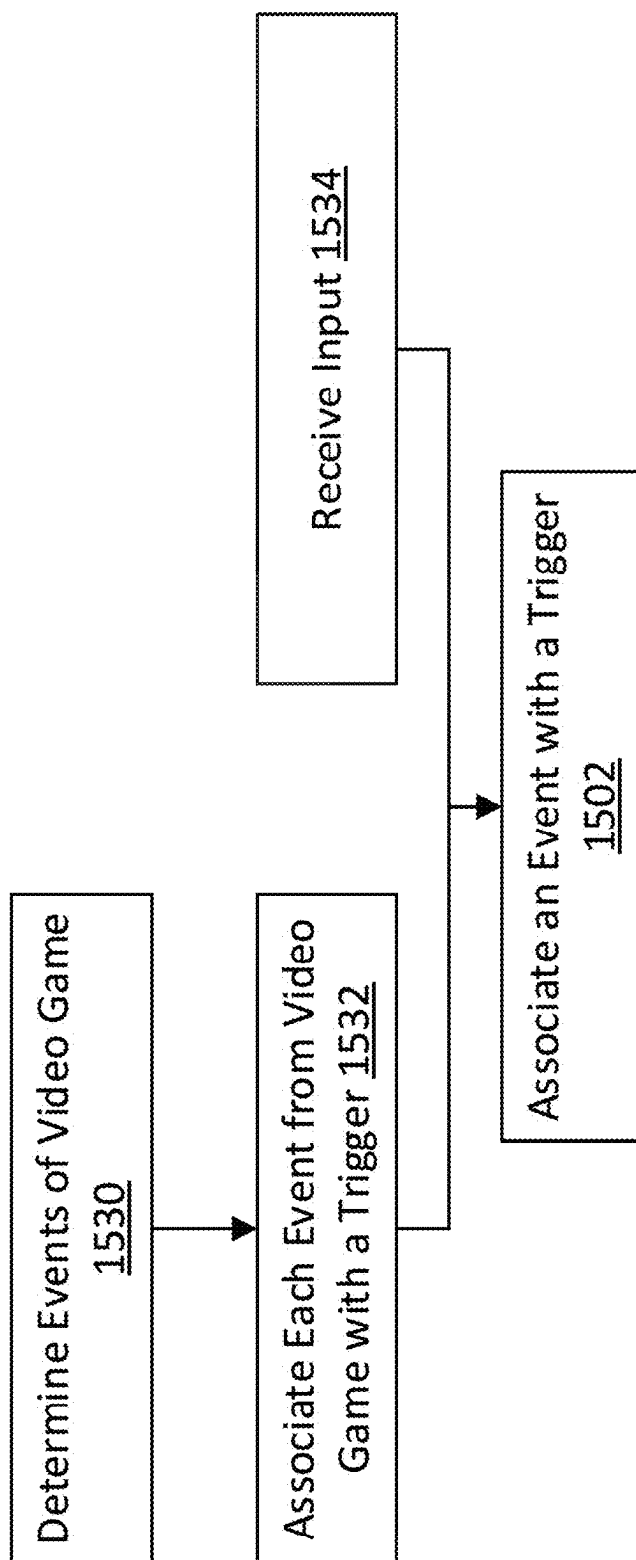


1400
FIG. 14

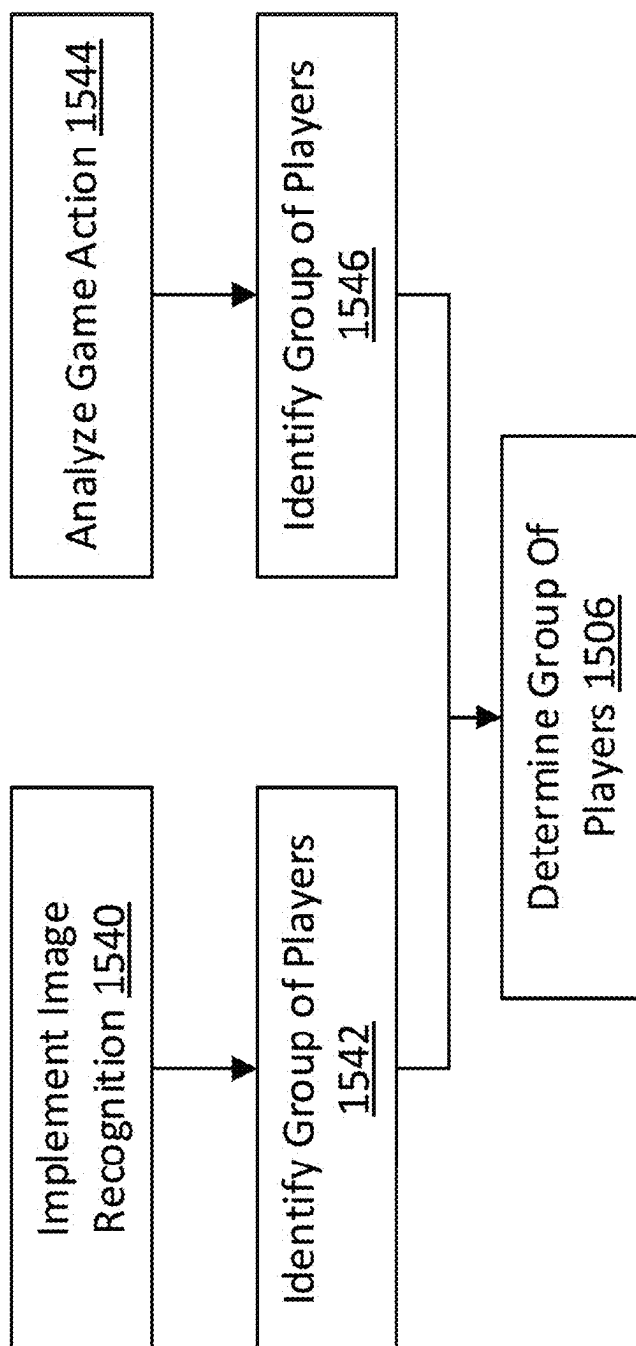
**FIG. 15A**



1511
FIG. 15B



1531
FIG. 15C



1541
FIG. 15D

EXTRACTION AND CLASSIFICATION OF AUDIO EVENTS IN GAMING SYSTEMS

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The present application is a continuation of U.S. patent application Ser. No. 17/899,766 filed Aug. 31, 2022 by Granger et al., entitled EXTRACTION AND CLASSIFICATION OF AUDIO EVENTS IN GAMING SYSTEMS, which claims the benefit of priority to U.S. Provisional Application No. 63/240,004 filed Sep. 2, 2021. All sections of the aforementioned applications are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates generally to a method and apparatus for extracting particular sounds or classes of sounds from gaming audio of a gaming system.

BACKGROUND

[0003] It is common today for gamers to utilize more than one gaming accessory. This is especially true of gamers who play on-line games or competitive games in a team or individual configuration. Gamers can have at their disposal accessories such as a keyboard, a general-purpose gaming pad, a mouse, a gaming console controller, a headset to communicate with other players, a joystick, a computer console, or other common gaming accessories.

[0004] A gamer can frequently use a combination of these accessories in a single game (e.g., headset, a keyboard, and mouse). Efficient management and utilization of these accessories can frequently impact a gamer's ability to compete.

[0005] Accessory management can have utility in other disciplines which may not relate to gaming applications. Efficient use of accessories in these other disciplines can be important to other users.

[0006] In addition, a player can play a video game with other players, as teammates or against opponents, all of which are remotely connected over a communication network. Players may communicate with teammates via spoken audio and players may hear game sounds generated by the video game. Further, the video game can monitor aspects of player interaction with the video game such that player performance can be reviewed to improve future performance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0008] FIG. 1 depicts an illustrative embodiment of a Graphical User Interface (GUI) generated by an Accessory Management Software (AMS) application according to the present disclosure;

[0009] FIGS. 2-3 depict illustrative embodiments for communicatively coupling a gaming controller to a computing device;

[0010] FIG. 4 depicts an illustrative embodiment of a communication device;

[0011] FIG. 5 depicts an illustrative embodiment of a first method utilized in the subject disclosure;

[0012] FIG. 6 depicts an illustrative embodiment of a second method utilized in the subject disclosure;

[0013] FIG. 7A depicts an illustrative embodiment of a third method utilized in the subject disclosure;

[0014] FIG. 7B depicts an illustrative embodiment of a fourth method utilized in the subject disclosure;

[0015] FIG. 7C depicts an illustrative embodiment of a fifth method utilized in the subject disclosure;

[0016] FIG. 7D depicts an illustrative embodiment of a sixth method utilized in the subject disclosure;

[0017] FIGS. 7E-7G depict illustrative embodiments of interfaces associated with a gameplay of a game;

[0018] FIG. 7H depicts an illustrative embodiment of a method for automatic triggering of a notification to one or more gameplayers based on a detected sound, in accordance with aspects of the subject disclosure;

[0019] FIG. 7I depicts an illustrative embodiment of an additional method for automatic triggering of a notification to one or more gameplayers based on a detected sound, in accordance with aspects of the subject disclosure;

[0020] FIGS. 7J and 7K schematically illustrate a system operating at least in part according to the methods of FIGS. 5, 6, 7A-7C and 7H-7I;

[0021] FIG. 7L depicts an illustrative embodiment of a block diagram for an audio processing system for a gaming system in accordance with aspects of the subject disclosure;

[0022] FIGS. 7L-1 and FIG. 7L-2 depict an illustrative embodiment of operating system controls of a user interface for selecting device preferences;

[0023] FIG. 7L-3 depicts an illustrative embodiment of a process for listener selection of a head-related transfer function (HRTF) profile for use by the user;

[0024] FIGS. 7M-7T depict an illustrative embodiment of a graphical user interface display of an audio processing system for a gaming system in accordance with aspects of the subject disclosure;

[0025] FIGS. 7U and 7V depict illustrative embodiments of methods utilized in the subject disclosure.

[0026] FIGS. 7W and 7X depict illustrative embodiments of an exemplary audio scene and audio mix;

[0027] FIG. 7Y is a block diagram of a system for extracting individual sounds from an audio stream in accordance with some aspects of the subject disclosure;

[0028] FIG. 7Z is a block diagram of a system 7730 for extracting individual sounds from an audio stream in accordance with some aspects of the subject disclosure;

[0029] FIG. 8 depicts an illustrative embodiment of a system operating at least in part according to the methods of FIGS. 5-7;

[0030] FIG. 9 depicts an illustrative embodiment of a communication flow diagram utilized by the system of FIG. 12; and

[0031] FIG. 10 depicts an illustrative diagrammatic representation of a machine in the form of a computer system within which a set of instructions, when executed, may cause the machine to perform any one or more of the methodologies disclosed herein.

[0032] FIG. 11 depicts an illustrative diagrammatic representation of a sliding window in accordance with aspects of this disclosure.

[0033] FIG. 12 depicts the storage media for storing a representation of a sliding window in accordance with aspects of this disclosure.

[0034] FIG. 13 depicts system to identify a group of possible trigger clips in accordance with aspects of this disclosure.

[0035] FIG. 14 depicts a system to distribute a group of trigger clips to a group of players in accordance with aspects of this disclosure.

[0036] FIGS. 15A-D depict illustrative embodiments of methods utilized in the subject disclosure.

DETAILED DESCRIPTION

[0037] The subject disclosure describes, among other things, illustrative embodiments of, for example, a gaming system that cooperates with a graphical user interface to enable user modification and enhancement of one or more audio streams associated with the gaming system. In embodiments, the audio streams may include a game audio stream, a chat audio stream of conversation among players of a video game, and a microphone audio stream of a player of the video game. A player may adjust audio of the audio streams independently to improve clarity, reduce noise, assist in identifying in-game sounds such as footsteps and gunshots. The graphical user interface may be displayed on a display device of the player and manually adjusted to preferred settings or player profile or configuration file may be retrieved for each audio stream. To further enhance sound and audio quality for a player, a head-related transfer function may be chosen from among a set of predefined transfer functions. Audio may be provided in a surround sound arrangement to enhance the audio experience of the user still further. Individual sounds within the game audio may be analyzed and identified, such as by a neural network, and enhanced to give the game player an improved audio experience and a competitive advantage in playing the video game. Further, individual sounds or classes of sounds may be extracted from the game audio. The extracted sounds may be separately provided to the play or may be enhanced to improve the player's enjoyment and competitiveness in the game. Other embodiments are described in the subject disclosure.

[0038] One or more aspects of the subject disclosure include receiving an input audio stream from a gaming system, the input audio stream including gaming audio of a video game played by a game player, the input audio stream including a plurality of classes of sounds, providing the input audio stream to a neural network, extracting, by the neural network, sounds of a selected class of sounds of the plurality of classes of sounds, and providing a plurality of output audio streams including providing a first audio stream including the sounds of the selected class of sounds of the input audio stream and a second audio stream including remaining sounds of the input audio stream.

[0039] One or more aspects of the subject disclosure include receiving an input audio stream from a gaming system, the input audio stream including gaming audio of a video game played by a game player, the input audio stream including a plurality of classes of sounds, extracting, from the input audio stream, sounds of a selected class of sounds of the plurality of classes of sounds, receiving audio processing control information from the game player, and providing a plurality of output audio streams to the game player according to the audio processing control information, including selectively providing one of a first audio stream including the sounds of the selected class of sounds

of the input audio stream and a second audio stream including remaining sounds of the input audio stream.

[0040] One or more aspects of the subject disclosure include receiving an input audio stream of a gaming system, the input audio stream including gaming audio of a video game played by a game player, the input audio stream including gaming sounds, the gaming sounds organizable as a plurality of classes of sounds, providing the input audio stream to a neural network, the neural network trained to recognize gaming sounds of at least one class of the plurality of classes of sounds, receiving from the neural network, extracted gaming sounds of the at least one class of the plurality of classes of sounds and providing to the game player, a first output audio stream including the extracted gaming sounds and a second output audio stream including remaining sounds of the gaming audio of the video game.

[0041] One embodiment of the subject disclosure includes receiving a plurality of audio streams including a game output audio stream of a gaming application, a gamer chat audio stream produced by a plurality of gamers participating in the gaming application, and a gamer microphone audio stream produced by one gamer of the plurality of gamers, processing the game output audio stream according to first audio processing settings of one gamer, forming a processed game output audio stream, and processing the gamer chat audio stream according to second audio processing settings of the one gamer, forming a processed gamer chat audio stream. Embodiments further include processing the gamer microphone audio stream according to third audio processing settings of the one gamer, forming a processed gamer microphone audio stream, providing the processed game output audio stream and the processed gamer chat audio stream to audio equipment of the one gamer, and providing, the processed gamer microphone audio stream to one or more gamers of the plurality of gamers.

[0042] One embodiment of the subject disclosure includes receiving a live game output audio stream of a gaming application currently being played by the plurality of gamers, providing on a display device of one gamer of the plurality of gamers, a graphical user interface, the graphical user interface including a frequency spectrum analyzer interface for selection of one or more live audio processing settings by the one gamer for the live game output audio stream, receiving from the graphical user interface, data defining the one or more live audio processing settings, and modifying the live game output audio stream of the gaming application according to the one or more live audio processing settings.

[0043] One embodiment of the subject disclosure includes receiving a live game output audio stream of a gaming application currently being played by the plurality of gamers, retrieving a stored configuration file, the stored configuration file including data defining stored live audio processing settings, and modifying the live game output audio stream of the gaming application according to the stored live audio processing settings.

[0044] One embodiment of the subject disclosure includes a processing system including a processor and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising a gaming application producing gaming output information including a game output audio stream and an audio processing module in data communication with the gaming application, the audio processing

module operative to receive the game output audio stream from the gaming application and to produce a plurality of game audio streams, the audio processing module further operative to apply respective audio customizations to each respective game audio stream of the plurality of game audio streams, according to gamer input defining the respective audio customizations for each respective game audio stream.

[0045] One embodiment of the subject disclosure includes a device including a gaming application producing gaming output information including a game output audio stream and a head-related transfer function (HRTF) module in data communication with the gaming application, the HRTF module operative to receive the game output audio stream from the gaming application and to apply a selected head-related transfer function to the game output audio stream to produce a surround sound audio stream, the selected head-related transfer function selected from a plurality of standardized head-related transfer functions.

[0046] One embodiment of the subject disclosure includes receiving a gaming system audio stream during a video game, detecting a current trigger in the gaming system audio stream during the video game, recording a trigger clip of game content, wherein the trigger clip comprises a portion of game content that includes the current trigger, and providing the clip to a group of gaming devices over a communication network, wherein each gaming device of the group of gaming devices is associated with one player of the video game.

[0047] One embodiment of the subject disclosure includes receiving a gaming system audio stream during a video game the gaming system audio stream comprising a plurality of audio streams including a chat audio stream, a game audio stream and a microphone audio stream, detecting, by the processing system, a predetermined event in the gaming system audio stream during the video game, modifying an audio spectrum of the gaming system audio stream to emphasize the predetermined event in the gaming system audio stream, producing an equalized audio stream, and providing the equalized audio stream to an audio device of a player of the video game.

[0048] One or more aspects of the subject disclosure include receiving a gaming system audio stream during a video game, the gaming system audio stream comprising a plurality of audio streams including a chat audio stream, a game audio stream and a microphone audio stream; detecting a predetermined event in the gaming system audio stream during the video game; modifying an audio spectrum of the gaming system audio stream to emphasize the predetermined event in the gaming system audio stream to produce an equalized audio stream; and providing the equalized audio stream to an audio device of a player of the video game.

[0049] One or more aspects of the subject disclosure include receiving a gaming system audio stream from a gaming system and detecting a predetermined sound in the gaming system audio stream. Aspects of the subject disclosure further include selecting a predetermined audio profile responsive to the detecting a predetermined event in the gaming system audio stream, modifying the gaming system audio stream according to the predetermined audio profile to produce a modified audio stream, and providing the modified audio stream to an audio device of a player of the gaming system.

[0050] One or more aspects of the subject disclosure include receiving a gaming system audio stream during a video game and separating the gaming system audio stream into a plurality of audio streams including a chat audio stream, a game audio stream, and a microphone audio stream. Aspects of the subject disclosure further include processing the game audio stream to detect a predetermined audio event in the game audio stream. Aspects of the subject disclosure further include providing, on a display device, a visual indication of detection of the predetermined audio event for a first player of the video game, modifying the game audio stream to enhance audibility of the predetermined audio event for the first player of the video game, producing a modified audio stream and providing the modified audio stream to audio equipment of the first player of the video game.

[0051] FIG. 1 depicts an illustrative embodiment of a Graphical User Interface (GUI) generated by an Accessory Management Software (AMS) application according to the present disclosure. The AMS application can be executed by a computing device such as a desktop computer, a laptop computer, a tablet, a server, a mainframe computer, a gaming console, a gaming accessory, or any combination or portions thereof. The AMS application can also be executed by portable computing devices such as a cellular phone, a personal digital assistant, or a media player. The AMS application can be executed by any device with suitable computing and communication resources.

[0052] FIG. 2 illustrates a number of embodiments for utilizing a gaming controller 115 with a computing device 206 in the form of a gaming console. In the illustration of FIG. 2, the gaming controller 115 can be communicatively coupled to the gaming console 206 with a tethered cable interface 202 such as a USB or proprietary cable, or a wireless interface 204 such as Wi-Fi, Bluetooth, ZigBee, or a proprietary wireless communications protocol. The cable interface 202 provides a means for communication that may be less susceptible to electromagnetic interference. It will be appreciated that the gaming controller 115 may further include a headset 114 (with or without a microphone not shown) utilized by a gamer to communicate with teammates and/or to listen to game sounds in high fidelity. In the illustration of FIG. 2, the AMS application can in whole or in part be executed by the gaming controller 115, the gaming console 206, or a combination thereof.

[0053] FIG. 3 illustrates a number of other embodiments for utilizing a gaming controller 115 with a computing device 206. In this embodiment, the gaming controller 115 comprises a mouse and the computing device 206 comprises a computer. The gaming controller 115 can be tethered to the computing device 206 by a cable interface 202 (e.g., USB cable or proprietary cable) or a wireless interface 204. The cable interface 202 provides a means for communication that may be less susceptible to electromagnetic interference. It will be appreciated that the gaming controller 115 may further include a headset (with or without a microphone not shown) utilized by a gamer to communicate with teammates and/or to listen to game sounds in high fidelity. In the illustration of FIG. 3, the AMS application can in whole or in part be executed by the gaming controller 115, the gaming console 206, or a combination thereof.

[0054] For illustration purposes, the terms gaming console 206 and computer 206 will be used hence forth interchangeably with the term computing device 206 with an under-

standing that a computing device **206** may represent a number of other devices such as a server, a tablet, a smart phone, and so on. Accordingly, a computing device **206** can represent any device with suitable computing resources to perform the methods described in the subject disclosure.

[0055] FIG. 4 depicts an illustrative embodiment of a communication device **400**. Communication device **400** can serve in whole or in part as an illustrative embodiment of devices described in the subject disclosure. The communication device **400** can comprise a wireline and/or wireless transceiver **402** (herein transceiver **402**), a user interface (UI) **404**, a power supply **414**, a proximity sensor **416**, a motion sensor **418**, an orientation sensor **420**, and a controller **406** for managing operations thereof. The transceiver **402** can support short-range or long-range wireless access technologies such as Bluetooth, Wi-Fi, Digital Enhanced Cordless Telecommunications (DECT), or cellular communication technologies, just to mention a few. Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, software defined radio (SDR), Long Term Evolution (LTE), as well as other next generation wireless communication technologies as they arise. The transceiver **402** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VOIP, etc.), and combinations thereof.

[0056] The UI **404** can include a depressible or touch-sensitive keypad **408** coupled to a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **400**. The keypad **408** can be an integral part of a housing assembly of the communication device **400** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth. The keypad **408** can represent a numeric keypad, and/or a QWERTY keypad with alphanumeric keys. The UI **404** can further include a display **410** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **400**.

[0057] In an embodiment where the display **410** utilizes touch-sensitive technology, a portion or all of the keypad **408** can be presented by way of the display **410** with navigation features. As a touch screen display, the communication device **400** can be adapted to present a user interface with graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The touch screen display **410** can be equipped with capacitive, resistive, or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements.

[0058] The UI **404** can also include an audio system **412** that utilizes common audio technology for conveying low volume audio (such as audio heard only in the proximity of a human ear) and high-volume audio (such as speakerphone for hands free operation, stereo or surround sound system). The audio system **412** can further include a microphone for receiving audible signals of an end user. The audio system **412** can also be used for voice recognition applications. The UI **404** can further include an image sensor **413** such as a

charged coupled device (CCD) camera for capturing still or moving images and performing image recognition therefrom.

[0059] The power supply **414** can utilize common power management technologies such as replaceable or rechargeable batteries, supply regulation technologies, and charging system technologies for supplying energy to the components of the communication device **400** to facilitate long-range or short-range portable applications. Alternatively, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or by way of a power cord attached to a transformer that converts AC to DC power.

[0060] The proximity sensor **416** can utilize proximity sensing technology such as an electromagnetic sensor, a capacitive sensor, an inductive sensor, an image sensor or combinations thereof. The motion sensor **418** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect movement of the communication device **400** in three-dimensional space. The orientation sensor **420** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **400** (North, South, West, East, combined orientations thereof in degrees, minutes, or other suitable orientation metrics).

[0061] The communication device **400** can use the transceiver **402** to also determine a proximity to a cellular, Wi-Fi, Bluetooth, or other wireless access points by common sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or a signal time of arrival (TOA) or time of flight (TOF). The controller **406** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies.

[0062] The communication device **400** as described herein can operate with more or less components described in FIG. 4 to accommodate the implementation of devices described by the subject disclosure. These variant embodiments are contemplated by the subject disclosure.

[0063] FIGS. 5-7A depict methods **500-700** describing illustrative embodiments of the AMS application. Method **500** can begin with step **502** in which the AMS application is invoked in a computing device. The computing device can be a remote server (not shown), the gaming console **206** or computer **206** of FIGS. 2-3, or any other computing device with suitable computing resources. The invocation step can result from a user selection of the AMS application from a menu or iconic symbol presented by the computing device **206**, or when a user communicatively couples a gaming controller **115** or other form of accessory device with the computing device **206**. In step **504**, the AMS application can detect by way of software drivers in an operating system (OS) of the computing device **206** a plurality of operationally distinct accessories communicatively coupled to the computing device **206**. The accessories can be coupled to the computing device **206** by a tethered interface (e.g., USB cable), a wireless interface (e.g., Bluetooth or Wireless Fidelity-Wi-Fi), or combinations thereof.

[0064] In the present context, an accessory can represent any type of device which can be communicatively coupled to the computing device **206** (or which can be an integral part of the computing device) and which can control aspects of the OS and/or a software application operating from the

computing device **206**. An accessory can represent for example a keyboard, a touch screen display, a gaming pad, a gaming controller, a mouse, a joystick, a microphone, or a headset with a microphone—just to mention a few.

[0065] In step **506**, the AMS application presents a GUI **101** such as depicted in FIG. **1** depicting operationally distinct accessories such as a keyboard **108**, and a gaming controller **115**. The GUI **101** presents the accessories **108-116** in a scrollable section **117**. One or more accessories can be selected by a user with a mouse pointer. In this illustration, the keyboard **108** and the gaming controller **115** were selected for customization. Upon selecting the keyboard **108** and the gaming controller **115** from the scrollable window of section **117**, the AMS application presents the keyboard **108** and the gaming controller **115** in split windows **118, 120**, respectively, to assist the user during the customization process.

[0066] In step **508**, the AMS application can be programmed to detect a user-selection of a particular software application such as a video game. This step can be the result of the user entering in a Quick Search field **160** the name of a gaming application (e.g., World of Warcraft™ or WoW). Upon identifying a gaming application, the AMS application can retrieve in step **510** from a remote or local database gaming application actions which can be presented in a scrollable section **139** of the GUI represented as “Actions” **130**. The actions can be tactical actions **132**, communication actions **134**, menu actions **136**, and movement actions **138** which can be used to invoke and manage features of the gaming application.

[0067] The actions presented descriptively in section **130** of the GUI can represent a sequence of accessory input functions which a user can stimulate by button depressions, navigation or speech. For example, depressing the left button on the mouse **110** can represent the tactical action “Reload”, while the simultaneous keyboard depressions “Ctrl A” can represent the tactical action “Melee Attack”. For ease of use, the “Actions” **130** section of the GUI is presented descriptively rather than by a description of the input function(s) of a particular accessory.

[0068] Any one of the Actions **130** can be associated with one or more input functions of the accessories being customized in windows **118** and **120** by way of a drag and drop action or other customization options. For instance, a user can select a “Melee Attack” by placing a mouse pointer **133** over an iconic symbol associated with this action. Upon doing so, the symbol can be highlighted to indicate to the user that the icon is selectable. At this point, the user can select the icon by holding the left mouse button and drag the symbol to any of the input functions (e.g., buttons) of the keyboard **108** or selectable options of the gaming controller **115** to make an association with an input function of one of these accessories. Actions of one accessory can also be associated with another accessory that is of a different category. For example, key depressions “Ctrl A” of the keyboard **108** can be associated with one of the buttons of the gaming controller **115** (e.g., the left button **119**).

[0069] In one embodiment, a Melee Attack action can be associated by dragging this action to either the left button **119** or right button **121** of the gaming controller **115**. Thus, when the selected button is depressed, the stimulus signal that is generated by the selected button of the gaming controller **115** can be substituted by the AMS application with the Melee Attack action. In another embodiment, the

AMS application can be configured so that the Melee Action can be associated with a combination of key button presses (e.g., simultaneous depression of the left and right buttons **119, 121**, or a sequence of button depressions: two rapid left button depressions followed by a right button depression).

[0070] In yet another embodiment, the AMS application can be configured so that the Melee Action can be associated with movement of the gaming controller **115** such as, for example, rapid movement or shaking of the gaming controller **115**. In a further embodiment, the AMS application can be adapted to make associations with two dimensional or three-dimensional movements of the gaming controller **115** according to a gaming venue state. For example, suppose the player’s avatar enters a fighter jet. In this gaming venue state, moving the left navigation knob forward can be associated by the AMS application with controlling the throttle of the jet engines. Rapidly moving the gaming controller **115** downward can represent release of munitions such as a bomb.

[0071] In a gaming venue state where the gamer’s avatar has entered a building, lifting of the gaming controller **115** above a first displacement threshold can be associated with a rapid movement of the avatar up one floor. A second displacement threshold can be associated with a rapid movement of the avatar down one floor—the opposite of the first displacement threshold. Alternatively, the second displacement threshold could be associated with a different action such as jumping between buildings when the avatar is on the roof of a building.

[0072] The AMS application can monitor gaming venue states by analyzing captured images produced by the gaming application (e.g., one or more still images of a tank, or a video of an avatar entering a tank), and/or by receiving messages from the gaming application by way of an application programming interface (API) thereby enabling the AMS application to identify the occurrence of a particular gaming venue state.

[0073] At step **512** the AMS application can also respond to a user selection of a profile. A profile can be a device profile or master profile invoked by selecting GUI button **156** or **158**, each of which can identify the association of gaming actions with input functions of one or more accessories. If a profile selection is detected in step **512**, the AMS application can retrieve in step **514** macro(s) and/or prior associations defined by the profile. The actions and/or macros defined in the profile can also be presented in step **516** by the AMS application in the actions section **130** of the GUI **101** to modify existing profile associations or create new associations.

[0074] In step **518**, the AMS application can also respond to a user selection to create a macro. A macro in the present context can mean any actionable command which can be recorded by the AMS application. An actionable command can represent a sequence of stimuli generated by manipulating input functions of an accessory, a combination of actions in the Action section **130**, an identification of a software application to be initiated by the OS of the computing device **206**, or any other recordable stimulus to initiate, control or manipulate software applications. For instance, a macro can represent a user entering the identity of a software application (e.g., instant messaging tool) to be initiated by the OS upon the AMS application detecting a speech command using speech recognition technology.

[0075] A macro can also represent recordable speech delivered by a microphone singly or in combination with a headset for detection by another software application through speech recognition or for delivery of the recorded speech to other parties. In yet another embodiment a macro can represent recordable navigation of an accessory such as a joystick of the gaming controller 115, recordable selections of buttons of the gaming controller 115, and so on. Macros can also be combinations of the above illustrations with selected actions from the Actions 130 menu. Macros can be created from the GUI 101 by selecting a “Record Macro” button 148. The macro can be given a name and category in user-defined fields 140 and 142.

[0076] Upon selecting the Record Macro button 148, a macro can be generated by selection of input functions on an accessory (e.g., Ctrl A, speech, navigation knob movements of the gaming controller 115, etc.) and/or by manual entry in field 144 (e.g., typing the name and location of a software application to be initiated by an OS, such as an instant messaging application, keyboard entries such as Ctrl A, etc.). Once the macro is created, it can be tested by selecting button 150 which can repeat the sequence specified in field 144. The clone button 152 can be selected to replicate the macro sequence if desired. Fields 152 can also present timing characteristics of the stimulation sequence in the macro with the ability to modify and thereby customize the timing of one or more stimulations in the stimulation sequence. Once the macro has been fully defined, selection of button 154 records the macro in step 520. The recording step can be combined with a step for adding the macro to the associable items Actions column 130, thereby providing the user the means to associate the macro with input functions of the accessories (e.g., one or more keys of the keyboard 108, buttons of the gaming controller 115, etc.).

[0077] In step 522, the AMS application can respond to drag and drop associations of actions with input functions of the keyboard 108 or the gaming controller 115. Associations can also be made based on the two- or three-dimensional movements of the gaming controller 115. If user input indicates that a user is performing an association, the AMS application can proceed to step 524 where it can determine if a profile has been identified in step 512 to record the association(s) detected. If a profile has been identified, the associations are recorded/stored in the profile in step 526. If a profile has not been identified in step 512, the AMS application can create a profile in step 528 for recording the detected associations. In the same step, the user can name the newly created profile as desired. The newly created profile can also be associated with one or more gaming software applications in step 530 for future reference. The AMS application can also record in a profile in step 526 associations based on gaming venue states. In this embodiment the same stimuli generated by the gaming controller 115 can result in different substitutions based on the gaming venue state detected by the AMS application.

[0078] Referring back to step 526, once the associations have been recorded in a profile, the AMS application can determine in step 532 which of the accessories shown illustratively in FIGS. 1-3 are programmable and available for programming. If the AMS application detects that an accessory (e.g., keyboard 108, gaming controller 115) is communicatively coupled to the computing device 206 and determines that the accessory is capable of performing stimulus substitutions locally, the AMS application can

proceed to step 534 of FIG. 5 where it submits the profile and its contents for storage in the accessory (e.g., the gaming controller 115 in FIGS. 2-3). Once the accessory (e.g., the gaming controller 115) is programmed with the profile, the accessory can perform stimuli substitutions according to the associations recorded by the AMS application in the profile. Alternatively, the AMS application can store the profile in the computing device 206 of FIGS. 2-3 and perform substitutions of stimuli supplied by the gaming controller 115 according to associations recorded in the profile by the AMS application.

[0079] The GUI 101 of FIG. 1 presented by the AMS application can have other functions. For example, the GUI 101 can present a layout of the accessory (button 122), how the accessory is illuminated when associations between input functions and actions are made (button 124), and configuration options for the accessory (button 126). The AMS application can adapt the GUI 101 to present more than one functional GUI page. For instance, by selecting button 102, the AMS application can adapt the GUI 101 to present a means to create macros and associate actions to accessory input functions as depicted in FIG. 1. Selecting button 104 can cause the AMS application to adapt the GUI 101 to present statistics from stimulation information and/or gaming action results captured by the AMS application as described in the subject disclosure. Selecting button 106 can also cause the AMS application to adapt the GUI 101 to present promotional offers and software updates.

[0080] The steps of method 500 in whole or in part can be repeated until a desirable pattern is achieved of associations between stimulus signals generated by accessories and substitute stimuli. It would be apparent to an artisan with ordinary skill in the art that there can be numerous other approaches to accomplish the embodiments described by method 500 or variants thereof. These undisclosed approaches are contemplated by the subject disclosure.

[0081] FIG. 6 depicts a method 600 for illustrating additional operations of the AMS application. In the configurations of FIGS. 2-3, the AMS application can be operating in whole or in part from the gaming controller 115, a gaming console 206, a computer 206, or a remote server (not shown). For illustration purposes, it is assumed the AMS application operates from the gaming console 206. Method 600 can begin with the AMS application establishing communications in steps 602 and 604 between the gaming console 206 and a gaming accessory such as the gaming controller 115, and a headset 114 such as shown in FIG. 1. These steps can represent for example a user starting the AMS application from the gaming console 206 and/or the user inserting at a USB port of the gaming console 206 a connector of a USB cable tethered to the gaming controller 115, which invokes the AMS application. In step 606, the gaming controller 115 and/or headset 114 can in turn provide the AMS application one or more accessory ID's, or the user can provide by way of a keyboard or the gaming controller 115 user identification. With the accessory ID's, or user input the AMS application can identify in step 608 a user account associated with the gaming controller 115 and/or headset 114. In step 610, the AMS application can retrieve one or more profiles associated with the user account.

[0082] In step 612, the user can be presented by way of a display coupled to the gaming console 206 profiles available to the user to choose from. If the user makes a selection, the AMS application proceeds to step 614 where it retrieves

from the selected profiles the association(s) stored therein. If a selection is not made, the AMS application can proceed to step 616 where it can determine whether a software gaming application (e.g., video game) is operating from the gaming console 206 or whether the gaming console 206 is communicating with the software gaming application by way of a remote system communicatively coupled to the gaming console 206 (e.g., on-line gaming server(s) presenting, for example, World of Warcraft™). If a gaming software application is detected, the AMS application proceeds to step 617 where it retrieves a profile that matches the gaming application detected and the association(s) contained in the profile. As noted earlier, association(s) can represent accessory stimulations, navigation, speech, the invocation of other software applications, macros or other suitable associations that result in substitute stimulations. The accessory stimulations can be stimulations that are generated by the gaming controller 115, as well as stimulations from other accessories (e.g., headset 114), or combinations thereof.

[0083] Once a profile and its contents have been retrieved in either of steps 614 or step 617, the AMS application can proceed to step 719 of FIG. 7 where it monitors for a change in a gaming venue state based on the presentations made by the gaming application, or API messages supplied by the gaming application. At the start of a game, for example, the gaming venue state can be determined immediately depending on the gaming options chosen by the gamer. The AMS application can determine the gaming venue state by tracking the gaming options chosen by a gamer, receiving an API instruction from the gaming application, or by performing image processing on the video presentation generated by the gaming application. For example, the AMS application can detect that the gamer has directed an avatar to enter a tank. The AMS application can retrieve in step 719 associations for the gaming controller 115 for controlling the tank.

[0084] The AMS application can process movements of the gaming controller 115 forwards, backwards, or sideways in two or three dimensions to control the tanks movement. Similarly, rotating the gaming controller 115 or tilting the gaming controller 115 forward can cause an accelerometer, gyro or magnetometer of the gaming controller 115 to provide navigational data to the AMS application which can be substituted with an action to cause the tank to turn and/or move forward. The profile retrieved by the AMS application can indicate that the greater the forward tilt of the gaming controller 115, the greater the speed of the tank should be moving forward. Similarly, a rear tilt can generate navigation data that is substituted with a reverse motion and/or deceleration of the forward motion to stop or slow down the tank. A three-dimensional lift of the mouse can cause the tank to steer according to the three-dimensional navigation data provided by the gaming controller 115. For example, navigation data associated with a combination of a forward tilt and right bank of the gaming controller 115 can be substituted by the AMS application to cause an increase in forward speed of the tank with a turn to the right determined by the AMS application according to a degree of banking of the gaming controller 115 to the right. In the above embodiment, the three-dimensional navigation data allows a gamer to control any directional vector of the tank including speed, direction, acceleration and deceleration.

[0085] In another illustration, the AMS application can detect a new gaming venue state as a result of the gamer directing the avatar to leave the tank and travel on foot. Once

again, the AMS application retrieves in step 719 associations related to the gaming venue state. In this embodiment, selection of buttons of the gaming controller 115 can be associated by the AMS application with weaponry selection, firing, reloading and so on. The movement of the gaming controller 115 in two or three dimensions can control the direction of the avatar and/or selection or use of weaponry. Once the gaming venue state is detected in step 719, the AMS application retrieves the associations related to the venue state and can perform substitutions of stimuli generated by the gaming controller 115, and/or speech commands received by microphone of the headset 114.

[0086] In one embodiment, the AMS application can be configured in step 719 to retrieve a profile that provides substitute stimuli for replacing certain stimuli generated by accessories. The associations recorded in the profile can be venue independent. In another embodiment, the AMS application can retrieve a combination of profiles, where one or more profiles provide substitute stimuli that are venue dependent and one or more other profiles provide substitute stimuli that are venue independent.

[0087] The AMS application can monitor in step 720 stimulations generated by the accessories coupled to the gaming console 206. The stimulations can be generated by the gamer manipulating the gaming controller 115, and/or by generating speech commands detected by a microphone of the headset 114. If a stimulation is detected at step 720, the AMS application can determine in step 722 whether to forward the detected stimulation(s) to an Operating System (OS) of the gaming console 206 or the gaming application directly without substitutions. This determination can be made by comparing the detected stimulation(s) to corresponding associations in one or more profiles retrieved by the AMS application. If the detected stimulation(s) match the associations, then the AMS application proceeds to step 740 where it retrieves substitute stimulation(s) in the profile(s). In step 742, the AMS application can substitute the detected stimulation(s) with the substitute stimulations in the profile(s).

[0088] In one embodiment, the AMS application can track in step 744 the substitute stimulations by updating the stimulations with a unique identifier such as a globally unique identifier (GUID). In this embodiment, the AMS application can also add a time stamp to each substitute stimulation to track when the substitution was performed. In another embodiment, the AMS application can track each substitute stimulation according to its order of submission to the gaming application. For instance, sequence numbers can be generated for the substitute stimulations to track the order in which they were submitted to the gaming application. In this embodiment, the substitute stimulations do not need to be updated with sequence numbers or identifiers so long as the order of gaming action results submitted by the gaming application to the AMS application remain in the same order as the substitute stimulations were originally submitted.

[0089] For example, if a first stimulation sent to the gaming application by the AMS application is a command to shoot, and a second stimulation sent to the gaming application is a command to shoot again, then so long as the gaming application provides a first a game action result for the first shot, followed by a game action result for the second shot, then the substitute stimulations will not require updating with sequence numbers since the game action results are reported in the order that the stimulations were sent. If on the

other hand, the game action results can be submitted out of order, then updating the stimulations with sequence numbers or another suitable identifier would be required to enable the AMS application to properly track and correlate stimulations and corresponding gaming action results.

[0090] Referring back to step **722**, if the detected stimulation(s) do not match an association in the profile(s), then the AMS application proceeds to one of steps **744** or **746** in order to track the stimulations of the accessory as described above. In another embodiment, tracking of original stimulations or substitute stimulations can be bypassed by skipping steps **744** or **746** and proceeding to step **770** of FIG. 7B.

[0091] Once the stimulations received in step **720** have been substituted with other stimulations at step **742** responsive to a detected association or maintained unchanged responsive to detecting no association with substitute stimuli, and (optionally) the AMS application has chosen a proper tracking methodology for correlating gaming action results with stimulations, the AMS application can proceed to step **770** of FIG. 7B. Referring to FIG. 7B, at step **770**, the AMS application can obtain an identification of an action to monitor during a gameplay associated with a game. The identification of the action may include a specification of a sound volume level associated with a user (e.g., a gamer). The identification of the action may include a specification of a number of user inputs exceeding a threshold. The number of user inputs may include a number of messages that are submitted, an identification of a content of the messages, an identification of an emoji, or a combination thereof. The identification of an action may include a gaming action provided by the game-see FIGS. **8-9** and accompanying descriptions.

[0092] At step **772**, the AMS application can store a representation of a sliding window of the gameplay in a first storage medium (e.g., first storage medium **1272** of FIG. **12**). The storage of step **772** may occur in real-time during the gameplay. The representation of the sliding window of the gameplay may include a video, an image, an audio track, or a combination thereof. The first storage medium may include a buffer of a graphics card, a random-access memory, or a combination thereof.

[0093] The sliding window may be of a substantially fixed duration, such that the sliding window progresses as the user/gamer continues to play a game. For example, and briefly referring to FIG. **11**, a sliding window **1100** (as a function of time *t*) is shown. As gameplay progresses, a new/supplemental representation of the gameplay may be added as shown via reference character/dashed portion **1102**. In order to accommodate storage of the portion **1102**, another portion **1104** may be deleted/overwritten. In the embodiment shown in FIG. **11**, the portion **1104** to be deleted/overwritten corresponds to the oldest/earliest-in-time portion of the window **1100**. In some embodiments, a portion other than, or in addition to, the oldest portion may be identified for being deleted/overwritten. Still further, in some embodiments the sliding window **1100** may be of a variable duration. For example, the duration/length of the sliding window may be a function of network traffic, a capability of a device (e.g., storage capacity), user/gamer inputs, etc.

[0094] Referring back to FIG. 7B, at step **774**, the AMS application can monitor for the identification of the action during the gameplay.

[0095] At step **776**, the AMS application can detect the identification during the gameplay responsive to the monitoring. In some embodiments, whether an event has occurred or not, as reflected by the detection of step **776**, may be based on a comparison of game action with one or more thresholds. Such thresholds may be specified by users/gamers (e.g., in accordance with user inputs/preferences), may be predetermined based on one or more rules/configurations associated with a game, etc.

[0096] At step **778**, the AMS application can store at least a portion of the representation of the sliding window of the gameplay in a second storage medium. The second storage medium may be the same as, or different from, the first storage medium. The second storage medium may include a server associated with a social media platform, a server associated with a virtual machine, a memory contained within a common housing as the first storage medium, a network element (e.g., a router, a gateway, a switch, etc.), or a combination thereof.

[0097] The storing of step **778** may include storing a video of a gamer, an image of the gamer (e.g., a thumbnail or icon representation of the gamer), an audio track of the gamer, or a combination thereof.

[0098] The storing of step **778** may include presenting a prompt (potentially responsive to the monitoring of step **774**), placing a copy of the representation of the sliding window of the gameplay in a third storage medium, which may be different from the first storage medium and/or the second storage medium, receiving a user input in response to the prompt, and storing the copy in the second storage medium responsive to the user input.

[0099] The placement of the representation/copy of the sliding window of the gameplay in the third storage medium may free/alleviate the first storage medium, such that the first storage medium can continue capturing gameplay/action as the gameplay continues subsequent to the detection of step **776**. Also, the placement of the representation/copy of the sliding window of the gameplay in the third storage medium may free the user/gamer of not having to commit to placing the representation/copy of the sliding window of the gameplay into more permanent storage (e.g., the second storage medium). For example, placement in the third storage medium may facilitate editing or review operations of the representation/copy of the sliding window prior to uploading the same to the second storage medium.

[0100] In some embodiments, the placing of the copy of the representation of the sliding window of the gameplay in the third storage medium may include initiating a timer to store a second sliding window of the representation after detecting the action, thereby resulting in an updated representation of the sliding window of the gameplay. Responsive to detecting an expiration of the timer, the updated representation may be stored in the third storage medium. A length of the timer may be based on a user input.

[0101] In some embodiments, the storing of step **778** may include storing a new representation of the sliding window of the gameplay in the first storage medium during the gameplay after placing the copy in the third storage medium; in some embodiments, the storage of the new representation may coincide with a step that is separate from step **778**.

[0102] At step **780**, the AMS application may present (e.g., simultaneously present) the representation of the sliding window and/or the video, image, and/or audio track of the gamer, or a combination thereof. In some embodiments,

a user/gamer may generate media that may be shared on one or more platforms (e.g., social media platforms) as a game is occurring, where the media may include the representation of the sliding window and/or the video, image, and/or audio track of the gamer, or a combination thereof. Alternatively, the user/gamer may generate the media following the conclusion of the game in order to avoid distractions during the game.

[0103] One or more of the steps shown in conjunction with FIG. 7B may be executed more than once. For example, subsequent to storing the representation of the sliding window of the gameplay in the second storage medium as part of step 778, a second representation of the sliding window of the gameplay may be stored in the first storage medium (as part of a second execution of step 772). The storing of the second representation of the sliding window of the gameplay may overwrite at least a portion of the representation of the sliding window of the gameplay in the first storage medium as described above.

[0104] FIG. 7C illustrates another embodiment of a method that may be executed in conjunction with the flow shown in FIG. 7A. As shown in FIG. 7C, in step 7170 the AMS application can obtain an identification of an action to monitor during a gameplay associated with a game. The identification of the action may include a specification of a number of actions per unit time.

[0105] In step 7172, the AMS application can store a representation of a portion of the gameplay in a storage medium.

[0106] In step 7174, the AMS application can monitor the gameplay for the identification of the action.

[0107] In step 7176, the AMS application can apply a tag to the representation of the portion of the gameplay in the storage medium responsive to the monitoring.

[0108] The representation of the portion of the gameplay may include a first video clip that occurs prior to an occurrence of the action and a second video clip that occurs subsequent to the action. A first-time duration of the first video clip, a first resolution of the first video clip, a second time duration of the second video clip, and a second resolution of the second video clip may be based on one or more user preferences, network traffic, a device capability, etc.

[0109] The representation of the portion of the gameplay may include a video clip. The tag may include a watermark that is applied to the video clip. The watermark may include the identification of the action. The tag may include metadata that is associated with the video clip. The metadata may be searchable via a search engine. The metadata may include a selectable link that, when selected, causes a client device to obtain the video clip.

[0110] FIG. 7D illustrates another embodiment of a method that may be executed in conjunction with the flow shown in FIG. 7A. As shown in FIG. 7D, in step 7270 the AMS application can monitor for an identification of an action during a gameplay.

[0111] In step 7272, the AMS application can detect the identification of the action during the gameplay responsive to the monitoring.

[0112] In step 7274, the AMS application can present a prompt responsive to the detecting.

[0113] In step 7276, the AMS application can store a representation of a portion of the gameplay, a representation of a gamer controlling the gameplay, or a combination

thereof, in a storage medium according to a user-generated input associated with the prompt.

[0114] In some embodiments, machine-learning/artificial intelligence may be applied to identify portions of a gameplay that are memorable or are of interest to a user (e.g., a gamer). For example, responsive to the user-generated input associated with the prompt in step 7276, the AMS application can monitor for a second identification of the action (or an alternative action) in step 7278.

[0115] In step 7280, the AMS application can detect the second identification of the action (or the alternative action) during the gameplay responsive to the monitoring for the second identification.

[0116] In step 7282, the AMS application can store a second representation of a second portion of the gameplay, a second representation of the gamer, or a combination thereof, in the storage medium without presenting a second prompt.

[0117] Once the AMS application at step 748 supplies to the OS of the computing device 206 a gaming action (i.e., one or more stimulations) from the method of FIG. 7B, the method of FIG. 7C, the method of FIG. 7D, or a combination thereof, the AMS application can proceed to step 734. The gaming action supplied to the OS at step 748 can be the unadulterated “original” gaming action of step 720, or an alternative gaming action generated by steps 744 or 746. At step 734, the OS determines whether to invoke in step 736 a software application identified in the stimulation(s) (e.g., gamer says “turn on team chat”, which invokes a chat application), whether to forward the received stimulation(s) to the gaming software application in step 738, or combinations thereof.

[0118] Contemporaneous to the embodiments described above, the AMS application can monitor in step 750 for game action results supplied by the gaming application via API messages previously described. For instance, suppose the stimulation sent to the gaming application in step 738 is a command to shoot a pistol. The gaming application can determine that the shot fired resulted in a miss of a target or a hit. The gaming application can respond with a message which is submitted by way of the API to the AMS application that indicates the shot fired resulted in a miss or a hit. If IDs such as GUIDs were sent with each stimulation, the gaming application can submit game action results with their corresponding GUID to enable the AMS application to correlate the gaming action results with stimulations having the same GUID.

[0119] For example, if the command to shoot included the ID “1234”, then the game action result indicating a miss will include the ID “1234”, enabling the AMS application in step 752 to correlate the game action result to the stimulation having the same ID. If on other hand, the order of game action results can be maintained consistent with the order of the stimulations, then the AMS application can correlate in step 754 stimulations with game action results by the order in which stimulation were submitted and the order in which game action results are received. In step 756, the AMS application can catalogue stimulations and game action results. In another embodiment, the AMS application can be adapted to catalogue the stimulations in step 760. In this embodiment, step 760 can be performed as an alternative to steps 750 through 756. In another embodiment, step 760 can be performed in combination with steps 750 through 756 in

order to generate a catalogue of stimulations, and a catalogue for gaming action results correlated to the stimulations.

[0120] FIG. 7E illustrates an interface that may be used to present at least a portion of a gameplay associated with a game. Various controls/commands, such as for example VCR types/styles of controls/commands, may be presented as a part of the interface to facilitate a recording or capture of one or more portions of the gameplay.

[0121] FIG. 7F illustrates an interface that may provide control over a recording or sharing of one or more representations (e.g., clips) of a gameplay associated with a game. Various controls, such as for example a “share” button or the like, may be provided to enable a user (e.g., a gamer) to post or otherwise share the representation(s). In some embodiments, editing controls may be provided to allow the user to customize the representation prior to, or subsequent to, sharing the representation.

[0122] In some embodiments, a user/gamer may have an ability to supplement the representation of the gameplay with commentary that describes, for example, what the user’s thought process was during the captured/represented portion of the gameplay. In this respect, and assuming that the user/gamer is viewed or otherwise characterized as an expert in the game, a sharing of the representation of the gameplay may serve as a tutorial for novice users.

[0123] FIG. 7G illustrates an interface that may present a tag **702g** (e.g., a watermark and/or metadata) associated with a representation of a gameplay. The tag **702g** may include data acquired/obtained during the gameplay, such as for example a statement or other indication of results obtained by the gamer during the gameplay.

[0124] Such a statement or other indication may be received via, e.g., a microphone, a keyboard, a mobile device, a computing/gaming console, etc.

[0125] The methods described herein (e.g., the methods described above in conjunction with FIGS. 7A-7D) may incorporate additional aspects. For example, in some embodiments a clip may be generated based on a user defined keybind (on a keyboard, mouse, or controller). Keybinds to trigger the clipping of a buffer to save to a local file system may be customized (e.g., may be based on user preferences). The gamer will be able to choose: the actual key to bind to the action, and the time slice to save (N seconds before and N’ seconds after).

[0126] In some embodiments, clips may be auto-generated based on some event, such as for example a detected event, an audible input (e.g., screaming), messages associated with a chat client, etc. In some embodiments, default settings may be provided, and those settings may be at least partially overridden/replaced based on affirmative user inputs and/or based on artificial intelligence/machine-learned user preferences.

[0127] In some embodiments, one or more filtering techniques may be applied to remove content from a representation of a gameplay that is not of interest. Such filtering may be based on one or more user inputs/preferences, may be learned over time via machine learning/artificial intelligence, etc. If multiple events/actions that are being monitored for happen within a threshold amount of time (which may coincide with a buffer time), an event/action endpoint may be extended to create one long time slice/representation of the gameplay. Alternatively, separate representations may be generated in some embodiments.

[0128] In some embodiments, tagging (e.g., watermarking) may be overlaid on a representation (e.g., a video) of a gameplay. A watermark may have a given level of transparency associated with it to avoid obscuring/blocking the representation of the gameplay. One or more logos may be applied as part of the tagging. In some embodiments, a watermark may pulsate or otherwise fade in-and-out. In this respect, dynamic watermarks may be used. The use of a dynamic watermark may serve to draw additional additional/incremental attention to the watermark, which may be useful for promotional/marketing/branding purposes

[0129] Aspects of sharing the representation of the gameplay may be controlled via one or more control parameters. Such control parameters may condition the sharing on a size of the representation (e.g., a video length), the content of the representation (e.g., controls may be present to limit a dissemination of the representation in view of intellectual property rights or other rights), etc. In some embodiments, a sharing of the representation of the gameplay may be limited to users that the gamer (or other entity) authorizes. For example, the sharing may be based on identifying a contact (e.g., a friend) of the gamer in one or more applications (e.g., a phone application, an email application, a text message application, a social media application, etc.).

[0130] In some embodiments, audio from the game can be monitored and analyzed in real time to generate game events. In addition, sounds that occur during a game can be analyzed to generate notices to one or more game participants; these notices can include live clips of the gameplay (sometimes referred to as moment clips), and/or statistics related to one or more players’ performance, equipment, health status, etc.

[0131] Referring to FIG. 7H, at step **7320**, the AMS application can obtain one or more audio profiles (e.g. spectral profiles) of sounds that may be received from a gaming software application at a player’s station during a gameplay associated with a game. The sounds may correspond to game action (for example, machine-gun fire, explosives detonating, etc.) or be generated by the player (for example, the player saying specific words during a game session). Game action sounds can be based on the player’s actions (for example, the player shooting) or on another player’s actions (for example, the other player detonating an explosive). In an embodiment, the audio profiles are stored locally at the player’s station. In other embodiments, the audio profiles can be stored on a server or gaming console executing the gaming software application associated with the game.

[0132] The AMS application monitors sounds generated at the player’s station (step **7322**). In an embodiment, the AMS application can distinguish between sounds emitted by the gaming software application and presented audibly by the player’s station and sounds produced in the environment of the player. (For example, in player station or equipment **7411** of FIG. 7J, if a sound **7419** detected via microphone **7414** has a similar profile and is in large part contemporaneous with a sound **7418** emitted from speaker **7413**, the sound is understood as a game sound; otherwise, a player environment sound, which can be a sound produced by the player.) In a particular embodiment, the player station includes a spectral filter that can be tuned to detect player environment sounds with known characteristics.

[0133] When a sound is detected (step **7324**), its spectral profile is analyzed and compared with the audio profiles. In

an embodiment, the monitoring can include a comparison to determine a correlation; the correlation can be based on a threshold of similarity of characteristics of the detected sound to characteristics of a reference sound in the audio profiles. If the profile of the detected sound matches a stored profile (step 7326), a gaming event can be identified (e.g., explosion detected). In an embodiment, the gaming event can be distributed as a notice corresponding to the detected sound that is distributed to other participants in the game session (step 7328). In additional embodiments, the gaming event can trigger detection of an image on the display 7415 of the player station to further analyze the significance of the gaming event and generate further information that may be useful to other players (e.g., explosion detected, but health of player near explosion has not been affected). Such enhanced information can be conveyed in the notice to other players. In further embodiments, the notice can be triggered on detection of a displayed image and/or a game-produced sound, in combination with a player's speech, which can be conveyed with the notice. In other embodiments, the AMS the gaming event can cause the AMS to generate a gaming action that can be sent to the game and/or the players (e.g., initiate shield protection for affected player, send message to teammates that player's shield has been initiated due to an explosion). The gaming action can be pre-stored and associated with the detected sound and/or image recognition at the time of the sound via the AMS application interface 100 of FIG. 1.

[0134] The notice referred to early can take several forms, in accordance with embodiments of the disclosure. The AMS application can send a text message such as "Player A taking fire" or "Bomb detonated at Player A." In an embodiment, this message can be shown on the display of the player station, superimposed on the gaming video being presented; in another embodiment, the message can be sent to a different player device. The AMS application can also analyze the current state of the game action as seen by the player, and provide additional information (e.g., statistics) regarding the game action associated with the detected sound; for example, "Bomb detonated near player A, player A's health has diminished by x, bomb type is b, player A is engaged with n opponents."

[0135] In additional embodiments, the gaming event detected by sound (or combination of sound and imaging) can trigger a recording of a video clip before and after the gaming event as described in the embodiments of FIGS. 7A-7G. The notice can be included in the video clip representing the gameplay occurring at the player's station at the time the sound is detected (that is, a live clip with respect to the detected sound). In some embodiments, the live clip is automatically generated when the sound is detected. The AMS application can distribute the live clip as part of the notice (for example, text "bomb near player A" superimposed on the live clip). In an embodiment, the AMS application generates the text notice by performing image processing of the live clip.

[0136] The notice can also include a sequence of two or more video clips (for example, a live clip combined with a clip of game action from 30 seconds earlier). In an embodiment, the triggering sound and/or image is analyzed to determine which video clips are to be combined and distributed to other participants.

[0137] In a further embodiment, the AMS application shares the player's live clip (and/or additional clips in a

sequence) with another player upon request by that player. Alternatively, the AMS application can select one or more other players for sharing the video clip(s), based on analysis of the triggering sound (or combination of sound/image/speech) and statistics regarding the player and other players. For example, a sound can trigger analysis of the player's health status, equipment status (e.g. remaining ammunition) and location, correlated with corresponding data regarding other players; the notice can then be sent to the closest player having resources that can be shared.

[0138] In an additional embodiment, the AMS application can analyze the game action shown in the live clip together with recent movements by the player's avatar, and provide recommendations associated with the detected gaming action (for example, "Run east 100 yards").

[0139] Referring to FIG. 7I, the AMS application can maintain a log of sounds detected during a game session. If a detected sound does not have a profile matching the stored audio profiles (step 7376/N) but has recurred during the game session (step 7382/Y), the AMS application can add the profile for that sound to the stored audio profiles (step 7384). In an embodiment, the recurring sound has its profile added after it has been detected a prescribed number of times. The AMS application can analyze the clips corresponding to the recurring sound to identify the sound, so that the profile of the sound can be stored with a text label identifying the sound.

[0140] FIGS. 7J and 7K schematically illustrate a system operating at least in part according to the methods of FIGS. 5, 6, and 7A-7C. As shown in FIG. 7J, a player 7410 uses equipment 7411 (generally referred to as a player station), which includes a local processing system, for playing a game; in this embodiment, AMS application 7412 and game application 7425 execute on the player station. Game application 7425 can be downloaded from a game server 7405 via a network 7420. Player 7410 can engage with the game action using any of several accessories including, for example, a mouse 7416.

[0141] In this embodiment, the player station stores audio profiles 7430 for game-action sounds (e.g. sound 7418 from speaker 7413) and player-generated sounds 7419 (e.g. words spoken by the player) detected by microphone 7414. As shown in FIG. 7K, a notice 7510 regarding a detected sound can be distributed via network 7420 to equipment of other players 7511, 7512, 7513 participating in the game session. In an embodiment, the notice includes a live clip associated with occurrence of the sound. In a further embodiment, the live clip is sent to equipment of a player in response to a request (e.g. request 7515 from equipment of player 7511).

[0142] FIG. 7L depicts an illustrative embodiment of a block diagram for an audio processing system 7600 for a gaming system in accordance with aspects of the subject disclosure. The audio processing system 7600 may cooperate with a gaming system to provide audio mixing and equalization for multiple audio streams associated with the gaming system. In embodiments, the gaming system may be embodied as an accessory such as gaming controller 115 or computing device 206 (FIG. 2, FIG. 3). The computing device 206 operates according to an operating system (OS) for controlling various functions of the computing device 206 including input of data and other information and output of data or other information. The audio processing system 7600 further cooperates with the OS of the computing

device to control audio processing, including mixing and equalization of audio streams associated with the computing device 206.

[0143] The audio processing system 7600 includes a plurality of audio input channels including, in this embodiment, a game audio input channel 7602, a chat audio input channel 7604 and a capture audio channel 7606. Further, the audio processing system 7600 includes one or more audio output devices 7608, a graphical user interface (GUI) 7610 and remote audio devices 7612. Other embodiments may include additional or alternative features and functions.

[0144] The game audio input channel 7602 operates under control of the GUI 7610 to select one or more audio streams from game audio source 7614 and to control audio processing of the selected audio streams in the game audio processing module 7618. Further, the game audio input channel 7602 operates under control of the GUI 7610 to route processed audio of the game audio input channel 7602 to an output device 7608. The game audio input channel 7602 includes a game audio input source 7614, a gaming virtual audio device 7616, and an audio processing module 7618. Other embodiments may include other or alternative features or functions. The game audio source 7614 may be, for example, a gaming system, one or more application programs running on a device such as computing device 206 and other sources accessed by a user of the gaming system, such as a network source. In the illustrated embodiment, game audio source 7614 may be an application or online source such as Steam® for gaming software and applications (available online at steampowered.com) or League of Legends® or Counter-Strike: Global Offensive (CS GO)®. The audio from the gaming system may include game sounds associated with video displayed on a video display device of the computing device 206. In the case of a war simulation game, for example, the game audio input stream 7602 may include sounds of gun shots, footsteps of teammates or opponents, explosions, vehicle noises, and others.

[0145] The game audio input stream 7602 may also include audio from other sources operating as game audio source 7614. Some applications generate audio and may be active on a computing device 206 along with the gaming system. For example, a gamer may be listening to music while playing a game of the gaming system. The game audio input stream 7602 may include the music audio stream from a music application. Further, an audio stream may be received from a remote location, over a network such as the public internet. For example, the game may be watching a movie provided by an online provider such as Netflix® or other source. The game audio input stream 7602 may include the audio received over the network.

[0146] In some applications, the game audio input stream 7602 includes multichannel audio or surround sound. Multichannel audio may include stereo audio with sound separation into right (R) and left (L) channels. Stereo audio may readily be provided to speakers built-in to a gaming device such as computing device 206. Stereo audio may also be readily provided to headphones worn by a gamer or other user.

[0147] Some applications such as games or videos produce surround sound audio. For example, surround sound audio may include techniques for enriching the fidelity and depth of sound reproduction by using multiple audio channels from multiple speakers that surround the listener, referred to as surround channels. Surround sound audio may

use four to seven or more independent audio channels and speakers placed in front of and behind the listener in order to surround the listener with sound. Surround sound audio may be produced according to a standard which specifies encoding and decoding of audio data. One example of a surround sound standard is referred to as 5.1 surround sound. The 5.1 standard calls for six speakers including center (C) in front of the listener; front left (FL) and front right (FR) in front of the listener at angles of thirty degrees off center to the listener; left surround (Ls) and right surround (Rs) at angles of 100 to 120 degrees off center to the listener; and a subwoofer, the position of which is not specified. The 5.1 surround sound standard is published as Recommendation ITU-R BS.775-3 (August 2012 by the International Telecommunications Union (ITU)). Another example of surround sound is referred to as 7.1 surround sound. 7.1 surround sound is similar to 5.1 surround sound but with four surround channels instead of two, the rear left (RL) and rear right (RR) at angles of 135 to 150 degrees off center to the listener; the side left (SL) and side right (SR) at angles of 90 to 110 degrees off center to the listener. The 7.1 surround sound standard is published as Report ITU-R BS.2159-7, February 2015, by the ITU.

[0148] Surround sound audio is typically directed to a listener position where the audio effects work best. The surround sound encoding presents a fixed or forward perspective of the sound field to the listener at this location. The technique enhances the perception of sound spatialization by exploiting sound localization. Sound localization refers to a listener's ability to identify the location or origin of a detected sound in direction and distance. In a surround sound system, sound localization is achieved by using multiple discrete audio channels routed to an array of loudspeakers. Each respective audio channel is routed to a respective loudspeaker.

[0149] Multi-channel or surround sound audio can enhance the enjoyment for a listener. This is particularly true for a listener enjoying a video, film, or audio such as a concert in a home theater experience. This is also particularly true for a gamer participating in a game using a gaming system on computing device 206.

[0150] The gaming virtual audio device 7616 provides device routing for the game audio source 7614. The gaming virtual audio device 7616 may form an audio logical device. The gaming virtual audio device 7616 may be a device used for example for recording the received audio data or for sending the received audio data on a network. In contrast to a virtual audio device, a real audio device is a physical device such as a laptop computer audio system or a Bluetooth speaker. On a computer such as a laptop computer forming the computing device 206, an operating system will expose a digital device. The operating system will see the digital device as an addressable endpoint. There is no real, physical system behind the addressable endpoint, including no hardware and no firmware. The gaming virtual audio device 7616 is merely a device in the sense of the operating system. The gaming virtual audio device 7616 may present itself to the operating system of the computing device 206 as a sound card or other physical device. It may merely be an address to which audio data is written by the processing system under control of the operating system. The audio data may be subsequently read and processed by other devices. However, the gaming virtual audio device 7616 is a virtual device because there is no physical device associated with it.

[0151] In an embodiment, the gaming virtual audio device 7616 may be configured as a portion of code stored in memory for performing suitable functions such as controlling a processing system. The gaming virtual audio device 7616 may receive multiple audio streams of the game audio source 7614 from the game audio input stream 7602 and combine them to form a target audio stream. The gaming virtual audio device 7616 is configured to route the target audio stream to the audio processing module 7618.

[0152] In an embodiment, the gaming virtual audio device 7616 includes a source code that embeds a functionality capable of routing the target audio stream to the audio processing module 7618. In practice, the gaming virtual audio device 7616 re-emits the audio data received on its audio receiver endpoint in the form of the plurality of first audio streams on an audio emitter endpoint that will be read by the audio processing module 7618. In some embodiments in conjunction with the Windows operating system the audio endpoint receiver of the gaming virtual audio device 7616 typically corresponds to the so-called Windows operating system audio endpoint sink, and the audio endpoint emitter of the gaming virtual audio device 7616 typically corresponds to the so-called Windows operating system audio endpoint source.

[0153] The audio processing module 7618 enables custom digital signal processing effects. In some embodiments, the audio processing module 7618 may be embodied as a Windows audio processing object (APO). In the Windows® operating system, for example, audio processing objects (APOs) provide software based digital signal processing (DSP) for Windows audio streams. An APO is a COM host object that contains an algorithm to provide a specific DSP effect referred to as an audio effect. Examples of APOs include graphic equalizers, reverb, tremolo, acoustic echo cancellation (AEC) and automatic gain control (AGC). APOs are COM-based, real-time, in-process objects. The APO provides programmable audio processing.

[0154] The chat audio input stream 7604 includes one or more chat audio sources 7620, a chat virtual audio device 7621 and a chat neural network effects service 7622. The chat audio input stream 7604 operates under control of the GUI 7610 to select one or more audio streams from the chat audio sources 7620 and to control audio processing of the selected audio streams in the chat neural network effects service 7622. Further, the chat audio input stream 7604 operates under control of the GUI 7610 to route processed audio of the chat audio input stream 7604 to an output device 7608.

[0155] In accordance with various aspects described herein, a gamer using computing device 206 to participate in a game or gaming application may communicate verbally with one or more teammates or others using a chat channel. For example, the teammates may use a voice over internet protocol (VOIP) application to communicate with each other. Examples of suitable applications include the Skype® application and the TeamSpeak® application. In other examples, the gaming application may include a built-in voice communication channel. The gamer or other user may access the GUI 7610 or the operating system of the computing device 206 to select one or more chat audio sources 7620 for communication. Audio from the chat audio sources 7620 is provided to the audio processing system 7600 at the chat audio input stream 7604 and may include voice con-

versation by teammates of the gamer or other user associated with the audio processing system 7600.

[0156] The chat virtual audio device 7621 operates on principles similar to those described herein for the gaming audio device 7616. The chat virtual audio device 7621 may be embodied as a module including software code. The chat virtual audio device 7621 may operate in response to control inputs from the GUI 7610 to select one or more of the more chat audio sources 7620. The chat virtual audio device 7621 routes an audio stream from the more chat audio sources 7620 to a selected destination such as the neural network effects service 7622 in response to control data from the GUI 7610. In some embodiments, multiple instances of the chat virtual audio device 7621 may be used to access and route multiple respective audio streams from multiple respective chat audio sources 7620.

[0157] The neural network effects service 7622 provides audio processing of the user audio stream from the chat audio input stream 7604. In some embodiments, the neural network effects service 7624 implements an artificial intelligence routine for customizing one or more audio streams such as the audio stream from the chat audio sources 7620. For example, the neural network effects service 7624 may implement an artificial intelligence-based noise cancellation for the user audio stream from the microphone 7623 that operates to remove background noise originating in the environment of the user or gamer. The neural network effects service 7622 responds to control information from the GUI 7610 to tailor audio processing in the audio stream received from the chat virtual audio device 7621. FIGS. 7M-7T depict an illustrative embodiment of a GUI that may embody the GUI 7610. Further, the GUI 7610 provides control information to route audio stream from the neural network effects service 7622 to one or more output devices 7608.

[0158] The capture audio stream 7606 includes a microphone 7623, a neural network effects service 7624 and a microphone virtual audio device 7625. The capture audio stream 7606 corresponds to audio originating with a gamer or user interacting locally with the gaming system on the computing device 206. For example, the user may interact with one or more gaming accessories such as a gaming console to participate in the game. As noted, the user may access other audio sources such as music and video including audio from a local application or over a network such as the internet. As part of the interaction, the user provides spoken audio input which may be chat shared with gaming teammates, for example. Chat or other conversation from the teammates is received at the chat audio input stream 7604. Similarly, the reverse channel with chat or other conversation from the gamer to teammates is provided on the capture audio stream 7606.

[0159] The microphone 7623 may include any microphone or other device for capturing audio locally with the user. In an example, the user wears headphones or a headset equipped with a microphone. In another example, a laptop computer or other computing device 206 includes a microphone for capturing user audio. The user audio stream from the microphone 7623 is routed and processed according to the operating system of the computing device 206.

[0160] The neural network effects service 7624 provides audio processing of the user audio stream from the microphone 7623. In some embodiments, the neural network effects service 7624 implements an artificial intelligence

routine for customizing one or more audio streams such as the user audio stream. For example, the neural network effects service **7624** may implement an artificial intelligence-based noise cancellation for the user audio stream from the microphone **7623** that operates to remove background noise originating in the environment of the user or gamer. Further, the neural network effects service **7624** responds to control information from the GUI **7610** to tailor audio processing in the user audio stream received from the microphone **7623**. FIGS. 7M-7T depict an illustrative embodiment of a GUI that may embody the GUI **7610**.

[0161] The user may wish to adjust how the user sounds on team chat to teammates or others who may hear the user. The GUI **7610** may be used to adjust processing by the neural network effects service **7624** or other audio processing system to adjust how the user sounds to others. For example, some users want to have their voice sound a little deeper to others. Some users may want their voice to sound a little clearer to others, such as by adding noise reduction to the microphone audio stream. In another example, a user may select a pre-set audio processing configuration such as a broadcaster pre-set, which modifies the user's voice to sound like a professional broadcast audio stream. In yet other examples, the user's voice may be completely modified to sound different, such as a male user who wishes to sound female or a very young user who wishes to sound very old. In another example, a female user may modify her voice to sound male to avoid harassment that can occur online. The user may choose to add effects such as compression to remove relatively high frequencies or relatively low frequencies. Moreover, the user may choose to apply parametric equalization to the user's voice audio stream to further tailor the user's output audio. This may improve intelligibility of the user's voice. In another example, the user may add an accent or remove an accent, such as the user who adds an Australian accent to sound more charming or intelligent to American teammates. As indicated, the user may be provided with a number of selectable pre-set audio modifications that may be applied by the neural network effects service **7624**. The selectable pre-set audio modifications may be accessed and selected by actuation of the GUI **7610**.

[0162] The microphone virtual audio device **7625** operates similarly to those described herein for the gaming virtual audio device **7616**. The microphone virtual audio device **7625** may be embodied as a module including software code. The microphone virtual audio device **7625** may operate in response to control inputs from the GUI **7610** to select one or more of the more destinations for an audio stream from the neural network effects service **7624**.

[0163] Thus, the GUI **7610** provides control information to route an audio stream from the neural network effects service **7622** to remote audio devices **7612**. In the example, the remote audio devices **7612** may include any of the chat audio sources **7620**, such as a VoIP application. Examples of suitable VOIP applications include the Skype® application and the TeamSpeak® application. Others may be specified, as well.

[0164] The audio processing system **7600** may create virtual audio devices such as the gaming virtual audio device **7616**, the chat virtual audio device **7621** and the microphone virtual audio device **7625**. Such virtual audio devices may be accessed in the Windows operating system. Applications can be assigned to the virtual audio devices. The virtual audio

device may be assigned names or other designations and may be actively selected by the user or gamer. In the example of FIG. 7L, a single virtual audio device **7616** is illustrated for processing audio from games such as Steam as well as audio streams from video sources such as the Twitch® live video streaming service or a sports video program as well as audio streams from audio sources such as the Spotify® audio streaming service. In other embodiments, a respective virtual audio device may be established for each respective audio stream handled by the audio processing system **7600**.

[0165] In embodiments, the audio processing system can be running on a device such as computing device **206** and processing audio from some or all open applications on the device. For example, if five applications are current running on the device. If three applications of the five open applications are routed to the same virtual audio device **7616** and audio processing module **7618**, the three applications will have the same effects applied, such as volume and equalization and filtering. Similarly, the other two applications, if routed to a different virtual audio device analogous to virtual audio device **7616** and a different audio processing module analogous to audio processing module **7618**, the other two applications will have the same effects applied. Thus, the virtual audio device **7616** and audio processing module **7618**, and analogous devices for other audio streams, may form a switching module for mapping or routing audio streams from applications or other sources to any audio destination such as audio output devices **7608**, the graphical user interface (GUI) **7610** and the remote audio devices **7612**.

[0166] FIG. 7L-1 and FIG. 7L-2 illustrate Windows® operating system controls of a user interface for selecting device preferences. In particular, FIG. 7L-1 and FIG. 7L-2 illustrate control of audio routing within a Windows device such as computing device **206**. A user may access the Windows Control Panel to control audio stream routing. In FIG. 7L-1 and FIG. 7L-2, various audio sources are illustrated including in this example, System sounds, Discord, Microsoft Teams, OBS 27.0.1 and Slack. System sounds refers to audio generated internally by the Windows operating system. Discord, Microsoft Teams, Open Broadcaster Software (OBS) and Slack are application programs for sharing audio and video within the computing device **206** and between the computing device **206** and other devices over a network. For example, in a gaming environment, teammates may chat or communicate verbally using a feature of the game or they may use a third-party app such as Discord or Slack.

[0167] The Windows control panel of FIG. 7L-1 and FIG. 7L-2 allows the user to control association of audio sources and audio destinations. The user may select an input device and an output device for each application. In the example of FIG. 7L-1, an audio stream from the Discord application is selected to come from a Default input. Further, in the example of FIG. 7L-2, an audio stream from the Discord application is selected for routing to a destination designated as SoundStage chat capture device (SteelSeries SoundStage device). In other examples, the applications listed on the left of FIG. 7L-1 and FIG. 7L-2 may include other applications for music, streaming video, broadcast television and other sources.

[0168] The operating system of the computing device **206**, in this example, the Windows® operating system, may thus

be used to map audio streams from audio sources to audio destinations. The audio sources include games, applications, network locations and any other source of audio to the computing device. Mapping the audio streams may further include mapping or routing particular audio streams to a particular audio processing module **7618** such as APO or neural network effects service **7622**.

[0169] In some embodiments, the audio processing system **7600** may automatically characterize audio events occurring in the game audio stream, the chat audio stream, the microphone audio stream, or some combination of these. In some embodiments, machine learning techniques may be used to characterize or profile audio events in the audio stream to identify a predetermined audio event such as a footstep or a gunshot associated with another player. In an example, the live game audio stream may be provided to a neural network. The neural network may be trained using any suitable data such as audio from other games or conversations. The training data may be tagged, for example, to identify predetermined audio events such as a gunshot or a footstep. Different characteristics may be further trained and identified, such as a footstep on wet pavement or a gunshot with ricochet. Once trained and provided with the live audio, the neural network or other processing module may produce an indication when the predetermined audio event has occurred. In some embodiments, the indication is a value corresponding to the probability that the predetermined audio event has occurred. If the probability exceeds a threshold, such as 75 percent or 95 percent, the audio processing system may conclude that the predetermined event has been detected.

[0170] In response to detecting the predetermined audio event, the audio processing system may take further steps to enhance or adjust the audio corresponding to the predetermined audio event. In an example, the audio processing system may automatically adjust a parametric equalizer to emphasize spectral components of the predetermined audio event. For example, if a footstep is detected, the spectrum may be adjusted in the parametric equalizer to emphasize to the listener, the player, the sound of the footstep. Further, if the player is using a headset or other audio equipment that provides directionality or other surround sound effect, the audio may be automatically adjusted to emphasize the direction of origin of the predetermined event. For example, if the footstep in the game comes from behind the player, the audio of the footstep in the surround sound arrangement will emphasize the direction so that the audio from behind is louder or clearer and the footstep is more distinct. Other sounds from that area may be suppressed to emphasize the footprint. In a video game in which the player has an associated avatar, the directionality of the sound may be processed according to the position and orientation of the avatar in the game.

[0171] In other embodiments, a graphical user interface or other visual interface may be activated to provide the player with a visual indication of the detected predetermined audio event. For example, the graphical user interface may flash text advising of “footstep behind you,” “possible gunshot detected to your right.” The nature of the visual indication may be based on the nature of the detected audio event and the probability or likelihood of detection, plus any other information that can be derived from the audio stream.

[0172] Further, in some embodiments, the GUI **7610** may be used to control the association of audio sources and audio

destinations. FIG. 7M and FIG. 7M-1 illustrate an example in which the GUI **7610** of the audio processing system **7600** may be used to access the Windows Control panel to manage audio stream configurations. In the illustrated embodiment, the Windows Control panel image is produced on a display screen which displays the GUI **7610**. In other embodiments, a custom graphical user interface display (not shown) may be produced with suitable controls enabling a user to access and modify mapping or routing of particular audio streams. Any suitable interface controls may be provided including mouse-based controls, touch-based controls on a touch-sensitive screen and voice-based controls.

[0173] In some embodiment, the audio processing system **7600** may automatically detect one or more audio sources and one or more audio destinations and automatically route audio streams from source to destination. In some embodiments, the audio processing system **7600** may pre-populate a routing matrix which can then be modified manually by the user, such as by accessing one or more pop-up menus similar to those illustrated in FIG. 7L-1 and FIG. 7L-2 or similar to those illustrated in FIG. 7M-1 or 7M-2. The audio processing system **7600** may select a group of audio applications known to be the most popular applications and pre-populate the menus with those applications. For example, the most popular voice over internet protocol (VOIP) applications may be selected as defaults. In another example, if the user is known to have a preference for an application, the audio processing system **7600** may select the preferred application as a default. The user can access the GUI **7610** and modify the selection as desired. The default selections for audio routing and audio processing are automatically selected for the user upon start-up of the gaming system or other applications to simplify and streamline the process for the user. The user starts the game and his preferred audio settings are established automatically. Moreover, if the user has different preferences for different games or different types of games, or for different types of applications such as music, video and streaming video, the audio processing system **7600** automatically detects the type of game or type of application and selects the user's preferred settings for each audio stream. Information defining the user's preferred settings may be stored at any convenient location such as locally at the computing device **206** or remotely at a server or other storage location accessible over a network such as the public internet. The user's preferred setting may be part of a profile or configuration file of the user.

[0174] In some examples, the audio processing system **7600** may apply a head related transfer function (HRTF). HRTF is a methodology used to generate virtual surround sound. It may be used to simulate virtual objects in a virtual surround environment using two speakers. Human sound localization may depend on the HRTF in the frequency domain and a head-related Impulse Response (HRIR) in the time domain. The HRTF relates to an acoustical footprint from a location in space to the ear of a person. A soundwave traveling from a location in space to the ears of a person will be modified due to acoustical reflection and diffraction phenomena by the listener's body, shoulders, head, and ear pinnae. The HRTFs contain the so-called localization cues, i.e. the information used by the human brain to decode the direction of a sound event. Hearing a sound is affect by interaural time differences (ITD) and interaural level differences (ILD). These are mainly due to an acoustical head-shadowing effect. They enable lateral localization (left/right)

by the listener, but they do not provide enough information to precisely locate a sound source.

[0175] The HRTF includes various spectral cues used by the user to localize sound. These include notches and peaks in the audio spectrum. These are primarily due to reflection on the listener's shoulders (low frequency) and reflection or diffraction by the ear pinnae of the listener (high frequency). Such spectral cues provide information used by the brain of the listener to decode the front/back and up/down location of a sound source.

[0176] Spatial sound over headphones uses HRTFs to simulate a sound source at a precise location around the listener. Most conventional spatial sound or virtual surround sound solutions use one unique generic HRTF profile, developed from one unique person or a dummy head. In an example, if $S(f)$ is the signal of the sound source, the stereo signal playback by the headphones is obtained by filtering it with the appropriate left HRTF and right HRTF. This may be described as follows:

$$S_L(f) = S(f) * \text{HRTF}_L(f, \text{direction person})$$

$$S_R(f) = S(f) * \text{HRTF}_R(f, \text{direction person})$$

[0177] for the left (L) ear or channel and right (R) ear or channel, respectively.

[0178] A generic HRTF profile cannot fit every listener, the localization cues will be the right ones for some listeners, but not for others. The localization cues are dependent on the anatomy of the listener. The typical drawback of using a generic HRTF profile is that the front/back and up/down localization is distorted. For example, sounds intended to be heard in the front of the listener are heard by the listener in the rear of the listener. Acquiring a personal HRTF for a user is cumbersome, expensive and requires substantial time and equipment, and is not realistic in most instances. Some conventional approaches to HRTF customization include use of pictures of the listener's ears to derive an adapted HRTF profile or allowing a user to select among tweaked versions of a single HRTF profile.

[0179] FIG. 7L-3 illustrates a process for listener selection of a HRTF profile for use by the user. The technique of FIG. 7L-3 employs a version of A/B testing to allow the user to select an HRTF that sounds best to the listener. In the process illustrated in FIG. 7L-3, several HRTF profiles are presented to the user. The user is guided through a series of A/B comparison tests to select the best or most appropriate HRTF for that user. The HRTF is then applied to an audio processing system such as the audio processing system 7600 of FIG. 7L.

[0180] In the example embodiment, the listener is presented with a series of eight HRTF profiles. Eight profiles are arbitrarily chosen. Any suitable number can be used. In FIG. 7L-3, the HRTF profiles are labelled A1, B1, A2, B2, A3, B3, A4 and B4. Each HRTF proposal in the example is based on different a HRTF developed for another person and is related to different morphologies, or persons with varying anatomy that will affect the HRTF. In the example embodiment, the eight HRTF profiles are representative of a large variety of morphologies or persons.

[0181] The process of FIG. 7L-3 is a guided selection by the user of a best or optimal or closest-fit HRTF profile. The process may be described as a tournament of several A/B listening tests. Thus, in the first round 7626 of the tournament, the user is presented with a listening test and comparison between HRTF profile A1 and HRTF profile B1.

Further, in the first round 7626, the user is presented with a listening test and comparison between HRTF profile A2 and HRTF profile B2. Still further in the first round 7626, the user is presented with a listening test and comparison between HRTF profile A3 and HRTF profile B3 and a listening test and comparison between HRTF profile A4 and HRTF profile B4.

[0182] Any suitable listening test may be provided. In some embodiments, audio test files are presented to the listener that highlight common perceptual drawbacks such as front/back discrimination difficulties. In the example, the listener first hears one or more audio test files as modified by HRTF profile A1 and the same one or more audio test files as modified by HRTF profile B1. The user selects or indicates whether profile A1 or profile B1 sounded better or more realistic or more natural to the listener, focusing preferably on the ability of the listener to spatially locate sounds played in the one or more audio test files. The selected profile, A1 or B1, is judged a winner of the first round 7626. The user continues with the other A/B tests of the first round, including profile A2 versus profile B2; profile A3 versus profile B3; and profile A4 versus profile B4. For each A/B test of the first round 7626, a winner advances to the second round 7627.

[0183] The A/B testing process is repeated for the second round 7627. The four winners from the first round 7626 are subjected to two A/B tests and two winners of the second round 7627 are advanced to the final round. Again, the winner of each A/B test is the HRTF profile that is judged best by the listener for that listener, focusing on spatial location of sounds of the audio test files. A final round 7628 is conducted with a similar A/B test and an ultimate winner HRTF profile is selected as the best or optimal HRTF profile for the listener among the eight HRTF profiles that were initially selected.

[0184] Throughout the process illustrated in FIG. 7L-3, the user may interact with a graphical user interface such as the GUI 7610 of FIG. 7L. The graphical user interface may provide written or audible instructions to the listener to lead the listener through each A/B test and through each round including the first round 7626, the second round 7627 and the final round 7628. Directions and guidance may be as detailed as necessary for the listener. Generally, the process of FIG. 7L-3 needs to be performed a single time to select an optimal HRTF profile for the listener.

[0185] Using a tournament format for selecting an optimal HRTF profile for the listener simplifies a process of selecting a best-fit or optimal HRTF profile for the listener. The tournament format reduces the number of A/B listening tests required of the user to identify the best fit for this listener. The listening tests allow the listener to make an HRTF profile selection that is the best fit for that listener, by directly comparing with another HRTF profile. As the tournament advances through the first round 7626, the second round 7627 and the final round 7628, the HRTF profiles should generally be closer to the actual ideal HRTF profile for this particular listener. The winner of the final round 7628 is unlikely to be precisely the ideal HRTF profile for this listener but is likely to be a good fit and the best fit from a wide selection of eight pre-selected HRTF profiles.

[0186] Once an HRTF profile for a user has been established, such as by the process illustrated in FIG. 7L-3, the HRTF profile may be used in the audio processing system 7600 of FIG. 7L. In some embodiments, the game audio

input stream **7602** includes audio processing to establish HRTF processing. HRTF processing may be applied, for example to audio streams encoded with 5.1 surround sound or 7.1 surround sound to provide the virtual surround sound effect when the user listens through two stereo speakers. For example, the audio processing module **7618** may include an HRTF module operative to enable HRTF processing for the audio stream received from game audio, movie or video audio, and other streams to create a virtual surround sound effect for the user. The HRTF processing may be applied after audio processing including parametric equalization and other functions. The user may activate and control the HRTF module and HRTF processing using the GUI **7610** or in any other suitable manner.

[0187] FIGS. 7M-7T depict an illustrative embodiment of a graphical user interface display of an audio processing system for a gaming system in accordance with aspects of the subject disclosure. A user of the audio processing system **7600** of FIG. 7L can interact with the graphical user interface to control aspects of the audio processing system **7600**. In the exemplary embodiment, the user may participate in gameplay with the gaming system with other players including teammates. For example, the user may interact with a personal computer, laptop computer or other device that implements the gaming system and is in data communication with a network such the public internet. The gaming system generates an audio stream of gaming audio that is presented to the user through an audio output system such as headphones. The player and the teammates of the player converse together in an audio stream referred to as gaming chat. The gaming chat audio stream is received by the user's computer and is presented to the user through an audio output system such as headphones. The user generates an audio stream by speaking into a microphone, for example as part of the gaming chat. The speech of the user is converted to data by the microphone and communicated over the network to teammates of the user.

[0188] FIG. 7M depicts an illustrative embodiment of a mixer panel **7630** of the graphical user interface. The mixer panel **7630** includes an audio stream selector **7631**, a series of audio stream controllers **7632** and a series of output selectors **7633**. Other embodiments may include other features in addition to those illustrated in FIG. 7M.

[0189] The audio processing system of FIG. 7L is configured to process three separate audio streams, Game, Chat, Microphone. Each audio stream is processed independently from the others. In the mixer panel **7630**, the user can assign a different audio device, such as a headset, speakers, microphone, or others, to each respective audio stream and tune finely the volume of the respective audio stream. Furthermore, the user also has a master volume. Separately processing the respective audio streams allows the user to finely tune the game sound without impacting teammates' voice audio on the chat audio stream.

[0190] The audio stream selector **7631** allows the user to select and control an audio stream. The audio stream selector **7631** may be used to select for display the mixer panel **7630**, a gaming audio control panel, a chat audio control panel and a microphone audio control panel. The audio stream selector **7631** may be used, for example to select using a mouse or other accessory, or by touching a touch-sensitive screen, to select a particular audio stream. In the illustrated embodiment, the mixer panel **7630** enables volume control over three separate audio streams of the gaming system, includ-

ing a gaming audio stream, a team chat audio stream and a user's microphone audio stream, as well as a master audio volume. The user may select a particular audio stream and, using a mouse for example, may drag the volume slider control for the selected audio stream to a desired level. In this manner, the relative volume of each separate audio stream may be tailored to the user's preferences or needs at a given time. The master volume slider allows the user to control overall game audio volume from all sources.

[0191] The output selectors **7633** are embodied as a pop-up menu that may be selected by the user with a mouse or other accessory. The pop-up menu is populated by the system with the audio devices that the operating system of the computer has identified. FIG. 7M-1 shows an example of selecting an audio output device on a pop-up menu. Generally, recognized audio devices include speaker systems and microphone systems that are detected by the operating system.

[0192] One of the recognized audio devices is designated as the default audio devices. By default, the Windows® operating system has a default multimedia device. The default multimedia device is the stream used for gaming. The dropdown menu of the output selectors **7633** may be used to control where audio output is directed. To select chat, the user can use the menu to select where to users wants to redirect the audio, such as headphones, speakers, audio driver. On the output selectors **7633**, the list of a dropdown menu is populated from known windows audio devices. Windows has a default audio communication device. The audio processing system sets Chat as the Windows default audio communication device and all the audio from the Chat audio stream gets routed to the audio selected in the dropdown. In this example, the user does not choose where the audio comes from but chooses where y the audio is redirected to. The game itself outputs audio to the default Windows setting, which may be set by the Windows operating system control panel.

[0193] FIG. 7M-2 shows an example of selecting sound input and sound output options in a device with a Windows® operating system. Other devices may be selected through actuation of the pop-up menu of the output selectors **7633**. In the illustrated example of FIG. 7M, an Arctis 7 Plus headphone system is selected as the output device for the gaming audio stream, the chat audio stream, and the microphone audio stream. Using the output selectors **7633**, the user may route the audio streams to different output devices. FIG. 7M-3 illustrates mixing of different audio levels for the gaming audio stream, the chat audio stream and the player's microphone audio stream. FIG. 7N-1 illustrates

[0194] Thus, the mixer panel **7630** enables separate audio processing for each of the game audio stream, the chat audio stream, and the microphone audio stream. By interacting with the graphical user interface including the mixer panel **7630**, the user can independently control the volume and audio effects of each of these streams.

[0195] FIG. 7N depicts an illustrative embodiment of a parametric equalizer **7635** of the graphical user interface. An equalizer is a tool by which the user can adjust and control the sound to improve audio during gameplay or other situations. The equalizer allows the user to hear more or less of the sound from a specific frequency range. Adjusting each frequency lets the user hear more of what the user wants to hear and less of what does not want to hear.

[0196] The parametric equalizer **7635** provides an interactive but easy to use, professional level 10-band parametric equalizer (EQ) for complete customization on the user's gaming audio stream. The gaming audio stream includes the audio stream of game sounds generated by the gaming system to accompany game visuals. For example, the user can lower the sound of explosions produced by the game and increase the important sounds of footsteps and reloading produced by the game so that the user can better understand activities of opponents in game. The parametric equalizer allows the user to boost or cut the sound over custom frequency bands according to a set of parameters, giving precise control. In particular, for a selected frequency band, the user can select the center frequency, the amplitude, the bandwidth and the filter type. Any tuning will be applied to the user's sound and visible on the graph in real time. In some embodiments, the tuning will affect only the game audio stream. Other audio streams won't be affected.

[0197] The graphical user interface allows the user to interact with the display on a display screen of the user's device using a mouse or other accessory. In the parametric equalizer **7635**, the center frequency can be shifted by the user to any value between 20 Hz and 20 kHz by dragging on the graph one of the available band dots or by selecting a band (from 1 to 10) and then using the frequency slider. Once selected, the amplitude of the band can be tuned up to +12 dB or down to -12 dB directly on the graph or with the gain slider.

[0198] In the graphical user interface illustrated in FIG. 7N, the user may select one of the dots corresponding to a frequency band or filter and adjust the gain applied by a respective filter by dragging the dot up or down vertically on the display screen. A value higher on the screen corresponds to a higher gain for that filter. The continuous line in the graphical user interface shows the actual frequencies that are being applied or curves out between the dots. Selecting the dots allows the user to move the line. Curve smoothing is applied between adjacent dots to create a smooth frequency curve. FIG. 7N-1 illustrates an example of adjusting the parametric equalizer **7635** for the chat audio stream to improve the intelligibility of teammates speech conveyed to the user over the chat audio stream.

[0199] In the illustrated embodiment, the parametric equalizer **7635** includes a parameter selector **7636**, an equalization selector **7637** and a filter type selector **7638**. The parametric equalizer **7635** implements a series of filters for controlling audio processing. Using the features of the parametric equalizer **7635**, the user can independently adjust the gain, frequency, quality (Q) factor and apply audio filters for up to 10 data points or frequency bands to customize the game audio.

[0200] The parameter selector **7636** enables the user to select a parameter to control on the graphical user interface. The user interface is divided into a series of frequency bands of the audible spectrum. In the example of FIG. 7N, the frequency bands are centered at 20 Hz, 50 Hz, 100 Hz, 200 Hz, 500 Hz, 1 kHz, 2 kHz, 5 kHz, 10 kHz, and 20 kHz. These center frequencies are exemplary only. Center frequencies may be selected to emphasize certain regions of the audible spectrum, such as relatively low frequencies. The selection may be based on the particular audio content of the game being played or some aspect of the game being played. For example, in a game involving players walking and

shooting, relatively low frequency sounds carry substantial information that is useful to the gamer to understand unseen action of the game.

[0201] In the illustrated embodiment, the user can separately control the gain of a selected frequency band. Each respective frequency band corresponds to respective filter of the series of filters that form the parametric equalizer **7635**. In the illustrated example, a frequency slider is actuated to select a frequency at 32.03 Hz and a gain slider is selected to adjust the gain at 32.03 Hz to a value of 2.1 dB. Moreover, a Q factor slider may be actuated to adjust the Q factor of each respective filter. The Q factor slider adjusts the Q factor value for each individual filter. The Q factor is related to the breadth of frequencies modified by the filter. A narrow filter bandwidth corresponds to a smaller Q factor value and means the gain is being applied to a smaller range of frequencies. A broader filter bandwidth corresponds to a larger Q value.

[0202] Once a frequency band is selected, the bandwidth of the frequency band can be widened or narrowed with the Q factor slider. Generally, the amplitude is inversely related to the Q Factor. In a gaming context, the Q factor is generally useful for three types of filters, including a peak filter, a notch filter, and a band pass filter. For these filters, the higher the value of Q factor, the narrower the curve or the bandwidth affected by the filter. A peak filter with a low Q factor will be useful to control the timber of the sound, for example, if the user wants to add bass to give a more powerful sound or feel to a car engine. A peak filter with a high Q factor will be useful to highlight or de-emphasize a certain frequency, for example, if the user wants to highlight a bomb sound in a particular game such as CS:Go.

[0203] As noted, the Q factor is a parameter that relates the filter's center frequency to its bandwidth. FIG. 7N-2 illustrates adjustment of a Q factor for a filter with a relatively narrow bandwidth. The Q factor has a value of 10.00. FIG. 7N-3 illustrates adjustment of a Q factor for a filter with a relatively wide bandwidth. The Q factor has a value of 0.5. FIG. 7N-2 and FIG. 7N-3 also illustrate an example in which, as a mouse or other accessory is used to select a filter for adjustment, current values of gain, frequency and Q factor for the selected filter are displayed on the screen near the cursor to simplify adjustment by the user and to increase precision of adjustments by the user.

[0204] The filter type selector **7638** allows the user to select among different available filter types for use in the parametric equalizer. Once selected, for each band, the filter type can be chosen in the in a filter type dropdown menu. In the example, filter types available include bypass, lowpass, high pass, peak, low shelving, high shelving filters. Other filter types may be specified including combinations of filters. In a gaming context, a peak filter with a low Q factor, a low shelf filter, or a high shelf filter may be useful to enhance the entertainment feeling for the user. Further, peak filter with a high Q factor and a notch filter may be useful to give a competitive advantage to a gamer in a gameplay situation. In many applications, a band pass filter is not useful during a game session. However, a band pass filter can be very useful to search for the frequency of a specific sound. For example, a user may want to know what frequency is associated with footsteps in the game audio stream. The band pass filter allows the user to listen to only a small frequency region or band of frequencies.

[0205] Once the parametric equalizer has been used to establish a set of preferences for audio processing for the gaming audio stream, the chat audio stream and the microphone audio stream, the established values may be stored by the user for subsequent use. The user may store data for a single audio stream, or the user may store data for multiple audio streams. The data may be stored on the user's computer or may be communicated over a network for storage at a remote location such as in the cloud. The data may be retrieved for editing or other modification. In some embodiments, the user may share a profile or configuration with another gamer. This may be done in any convenient manner. Further, the profile or the configuration may be specific to a particular game that the user plays among multiple games. Thus, the user may store a first configuration or profile for the Fortnite game and store a second configuration or profile for the World of Warcraft game.

[0206] FIG. 70 illustrates an exemplary embodiment of a graphical user interface in which the user may apply a predetermined configuration or profile to a respective audio stream. The graphical user interface includes a configuration selector **7640**. The configuration selector **7640** includes an audio stream selector **7641** and a configuration pop-up menu **7642**. The user may use the mouse or another accessory to select an audio stream of interest for applying a profile or configuration. In the example embodiment, the user may select the mixer, the gaming audio stream, the chat audio stream, or the microphone audio stream for application of a configuration. Once the audio stream of interest is selected, the user may use a pop-up menu to select a predetermined profile or configuration for the selected audio stream. In the example, for the gaming audio stream, the user has selected a configuration titled "Fortnite footsteps" which may include data defining a set of audio settings for the parametric equalizer that improve the clarity of footsteps heard by the user in the game play. Adjusting the frequency, gain and Q factor of a set of filters may greatly improve the user's ability to discern footsteps during gameplay.

[0207] Thus, using the graphical user interface, the user may create, duplicate, edit and delete configurations for each unique audio stream. This allows the user to change audio stream effects with just one touch to select a predetermined configuration, without having to again select each specific filter setting. In embodiments, the configurations can be set up and applied to each separate audio stream to maximize the personalization for each gamer.

[0208] In some embodiments, the audio processing system will automatically retrieve and implement a user's configurations when a game is launched by the user. For example, the user provides identification information such as logging in to a game application or web site. Based on the identification information or other information the user provides, the audio processing software may select the user's pre-stored configurations for the game. In some embodiments, the process of retrieving the user's pre-stored configurations may be done automatically based on user information or other information. In other embodiments, the user may be prompted to select a set of configurations or a profile. In still other embodiments, the user may have to actively locate and apply the user's pre-stored set of configurations for a game or the user's profile for the game.

[0209] In other examples, a group of pre-set configurations may be made available to users for some games, or to some users of some games. For example, the manufacturer

may design a set of profiles or configuration files that a user may access and select for application to the user's audio processing system. Access to the pre-set configuration files may be on a subscription basis for example. Further, gamers may prepare and store in a common location a library of profiles or configurations for different games and different preferences. The stored profiles may be retrieved, modified, and improved by members of a gamer community to enhance the enjoyment of the group.

[0210] FIG. 7P illustrates an exemplary embodiment of a graphical user interface in which the user may apply basic audio rendering presets for quick and easy audio effects applied to each individual audio stream. The graphical user interface includes a gaming audio render effects selector **7645**. The gaming audio render effects selector **7645** in the example includes an audio stream selector **7646** and an audio adjustment panel **7647**. Other examples may include other adjustments or features.

[0211] The audio stream selector **7646** allows the user to select an audio stream for application of the audio rendering presets. In the example, the user may select between the mixer, the gaming audio stream, the chat audio stream, and the microphone audio stream.

[0212] The audio adjustment panel **7647** includes a set of sliders that may be actuated by a user with a mouse or other accessory. Each of the sliders is associated with an audio adjustment. Each of the sliders is also associated with an on-off selector that may be actuated by the user on the graphical user interface to enable or disable particular audio adjustments. In the example embodiment, the available audio adjustments include bass boost, treble boost, voice clarity, smart volume, general gain, and virtual surround.

[0213] Tuning the bass boost audio adjustment allows the user to boost or cut low frequencies. In an example, the bass frequencies are around 80 Hz. This may have the effect of making more or less audible deep, low frequency sounds in the gaming audio stream such as explosions and engines. The bass boost audio adjustment can be tuned through the slider in a range from +12 dB to -12 dB in the example.

[0214] Tuning the treble boost audio adjustment allows the user to boost or cut high frequencies. In an example, the treble frequencies are around 8000 Hz. This may have the effect of making more or less audible high frequency sounds in the gaming audio stream like footsteps, gunshots or whistles. For example, this can give the user a real advantage in shooting games. The treble boost can be tuned through the slider in a range from -12 dB up to +12 dB in the example.

[0215] The voice clarity audio adjustment identifies and treats the dialogs or voices in the audio stream produced by the game. That means that tuning the voice clarity through the slider enables the user to boost or cut the speech in games, movies, and incoming communications. The voice boost audio adjustment can be tuned through the slider in a range from -12 dB up to +12 dB in the example.

[0216] Some audio streams have very large dynamic range, or the difference between the loudest sounds and the softest sounds. The smart volume audio adjustment automatically makes small adjustments and helps maintain a steady volume in the game for the user. For example, if the user is constantly adjusting the volume on the user's computer while playing games or listening to music, the smart volume will solve address the problem. The user may tune the impact of this audio effect on your sound through the slider in a range from 0 to 1 in the example.

[0217] The general gain audio adjustment simply allows the user to adjust overall sound volume. Such a feature is useful when the audio source volume is low. The general gain audio adjustment can be tuned through the slider through a range from 0 dB up to +12 dB in the example. Also, the general gain audio adjustment does not allow system limits on audio volume or gain to be exceeded.

[0218] The virtual surround sound audio adjustment may be used in arrangements that include a stereo device such as a headset, or notebook computer internal speakers. In one embodiment, a multichannel audio stream such as 5.1 channel audio or 7.1 channel audio is available, switching ON the virtual surround audio adjustment re-creates a multichannel three-dimensional audio listening experience over the user's stereo headphones or speakers. When actuated ON, the virtual surround sound audio adjustment operates to virtualize the multichannel game audio stream from the game engine, or any other media content, and mix the audio stream to retrieve a multichannel listening experience over the user's stereo headphones or speakers. On the other hand, if a multichannel audio stream, such as stereo content, is not available, switching ON the virtual surround sound audio adjustment operates to expand the stereo sound for a wider sound stage.

[0219] In another example, if the user's audio device is a multichannel device such as a home theater system, and if a multichannel audio stream such as 5.1 channel or 7.1 channel audio is available, switching ON the virtual surround sound audio adjustment will virtualize the multichannel audio stream on left and right speakers. In embodiments, only left and right speakers will output sound. In this situation, it may be preferable not to activate the virtual surround sound audio adjustment since a real surround sound effect is available. In this case, if a multichannel audio stream is not available, such as stereo content, switching ON the virtual surround sound audio adjustment operates to expand the stereo sound for a wider sound stage. Only left and right speakers may output sound in this example.

[0220] FIG. 7Q illustrates an exemplary embodiment of a graphical user interface including a chat equalizer 7650. The chat equalizer 7650 in the example is embodied as a ten-band graphic equalizer for the team chat audio stream. The chat equalizer 7650 includes an audio stream selector 7651, a series of audio stream controllers 7652. Other embodiments may include other features in addition to those illustrated in FIG. 7Q.

[0221] The audio stream selector 7651 allows the user to select an audio stream for application of audio equalization. In the example, the user may select between the mixer, the gaming audio stream, the chat audio stream, and the microphone audio stream. The chat audio stream is selected in FIG. 7Q. The chat equalizer allows the user to tailor the audio of the team chat to improve the way in which the user hears teammates or others speaking on the chat audio stream. In a typical embodiment, a game is played on a gaming system with two or more players working together as teammates. Each player has access to a microphone and speakers, such as an integrated headset wearable by the player. The players may use the microphone and speakers to converse with each other or more generally to share speech and non-speech voiced audio. The chat audio stream may include non-speech such as screams or laughter. The audio stream controller 7652 allows the user to adjust gain upward and downward across ten frequency bands.

[0222] In the illustrated embodiment, the chat equalizer 7650 includes a configuration selector 7653. The user may create and store one or more configurations or profiles. For example, after the user adjusts the audio stream controller 7652 to tailor sound processing for the chat audio stream according to the user's preferences, the user may save the data that device the current adjustments. The data may be saved as a profile file or configuration file and may be saved locally or remotely. A saved profile may be shared among users or may be part of a library of available configurations. A profile may be retrieved and edited by the user. The configuration selector 7653 in the example includes a pop-up menu that, when actuated by the user, displays available profiles or configurations. In some examples, a profile or configuration may be designated as a default profile and load automatically when the game is started. A profile may be established for a respective game according to user preferences.

[0223] FIG. 7R illustrates an exemplary embodiment of a graphical user interface in which the user may apply basic audio rendering presets for quick and easy audio effects applied to the chat audio stream. The graphical user interface includes a chat audio render effects selector 7655. The chat audio render effects selector 7655 in the example includes an audio stream selector 7656 and an audio adjustment panel 7657. Other examples may include other adjustments or features.

[0224] The audio stream selector 7656 allows the user to select an audio stream for application of the audio rendering presets. In the example, the user may select between the mixer, the gaming audio stream, the chat audio stream, and the microphone audio stream. In the illustrated example, the chat audio stream is selected.

[0225] The audio adjustment panel 7657 allows the user to select and adjust a variety of aspects of the chat audio stream. In the illustrated example, the user has access to a noise reduction audio adjustment, a volume stabilizer audio adjustment, a noise gate audio adjustment, an automatic noise gate audio adjustment and an impact noise reduction audio adjustment.

[0226] The noise reduction audio adjustment operates to remove static noise from the user's sound by cleaning up spoken words and sentences. This audio adjustment is particularly useful to remove constant noise like personal computer fans. The user may tune the effect of the noise reduction audio adjustment through the slider from 0, appropriate for a quiet environment, to 1, appropriate for a noisy environment. The smart volume audio adjustment automatically makes small

[0227] adjustments and helps maintain a steady volume, particularly when the chat audio stream has a relatively high dynamic range. The smart volume audio adjustment eliminates the need for the user to constantly adjust a computer's volume during gameplay or listening to music. The user may tune the impact of the smart volume audio adjustment on user sound through the slider, over a range from 0 to 1.

[0228] For a given a volume threshold, the noise gate audio adjustment operates to cut off any sound behind it. If sound is above a predetermined noise gate threshold, the audio processing system lets pass all the sound. In other words, the sound is allowed through the gate. On the other hand, if the sound is below the predetermined noise gate threshold, the sound does not pass through the gate. The predetermined noise gate threshold can be changed with the

threshold slider in the range -60 dB -10 dB. This effect can be particularly useful if the user perceives an annoying or distracting noise which is way behind the conversation level.

[0229] For a given volume threshold, the noise gate audio adjustment will cut off any sound behind it. The auto noise gate audio adjustment automatically computes the predetermined noise gate threshold according to detected background noise. In embodiments, the auto noise gate audio adjustment requires the noise gate audio adjustment to be ON, i.e., both audio adjustments must be activated. The auto noise gate audio adjustment operates automatically. However, the user has a layer of control over the auto noise gate audio adjustment with the offset slider which allows the user to apply a gain over a range from -20 dB to $+20$ dB) on the top of the computed threshold. The auto noise gate is useful if the user needs a noise gate, but the user is uncertain about which threshold to use.

[0230] The impact noise reduction audio adjustment operates to detect and suppress impact noises from the user's chat audio stream or input device. The impact noise reduction audio adjustment is useful in reducing the noise coming from sources such as keystroke on a keyboard or noise from mouse clicks.

[0231] FIG. 7S illustrates an exemplary embodiment of a graphical user interface including a microphone equalizer **7660**. The microphone equalizer **7660** in the example is embodied as a ten-band graphic equalizer for equalizing the user's microphone audio stream. The microphone equalizer **7660** includes an audio stream selector **7661** and a series of audio stream controllers **7662**. Other embodiments may include other features in addition to those illustrated in FIG. 7S.

[0232] The audio stream selector **7661** allows the user to select an audio stream for application of audio equalization. In the example, the user may select between the mixer, the gaming audio stream, the chat audio stream, and the microphone audio stream. The microphone audio stream is selected in FIG. 7S. The microphone equalizer allows the user to tailor the audio of the user's microphone provided to the team chat to improve the way in which teammates or others hear the user speaking on the chat audio stream. In a typical embodiment, a game is played with two or more players working together as teammates. Each player has access to a microphone and speakers. The players may use the microphone and speakers to converse with each other in the chat audio which may include non-speech such as screams or laughter. The audio stream controller **7662** allows the user to adjust gain upward and downward across ten frequency bands.

[0233] In the illustrated embodiment, the chat equalizer **7650** includes a configuration selector **7663**. Similar to profiles for the game audio stream and chat audio stream, the user may create and store one or more configurations or profiles. For example, after the user adjusts the audio stream controller **7662** to tailor sound processing for the microphone audio stream according to the user's preferences, the user may save the data that device the current adjustments. The data may be saved as a profile file or configuration file and may be saved locally or remotely. A saved profile may be shared among users or may be part of a library of available configurations. A profile may be retrieved and edited by the user. The configuration selector **7663** in the example includes a pop-up menu that, when actuated by the user, displays available profiles or configurations. In some

examples, a profile or configuration may be designated as a default profile and load automatically when the game is started. A profile may be established for a respective game according to user preferences.

[0234] FIG. 7T illustrates an exemplary embodiment of a graphical user interface in which the user may apply basic audio rendering presets for quick and easy audio effects applied to the microphone audio stream generated by the user's own microphone. The microphone audio stream may be shared with teammates, for example, during game play or at other times. In the illustrated example, the graphical user interface includes a microphone audio render effects selector **7655**. The chat audio render effects selector **7665** in the example includes an audio stream selector **7666** and an audio adjustment panel **7667**. The microphone audio render effects selector **7665** in the exemplary embodiment operates similarly to and includes features like those of chat audio render effects selector **7655** illustrated in FIG. 7R. Other examples may include other adjustments or features.

[0235] The audio stream selector **7656** allows the user to select an audio stream among the audio streams processed by the audio processing system for application of the audio rendering presets. In the example, the user may select between the mixer, the gaming audio stream, the chat audio stream, and the microphone audio stream. In the illustrated example, the microphone audio stream is selected.

[0236] The audio adjustment panel **7667** allows the user to select and adjust a variety of aspects of the microphone audio stream. In the illustrated example, the user has access to a noise reduction audio adjustment, a volume stabilizer audio adjustment, a noise gate audio adjustment, an automatic noise gate audio adjustment and an impact noise reduction audio adjustment.

[0237] The noise reduction audio adjustment operates to remove static noise from the audio stream created by the user's microphone by filtering or cleaning up spoken words and sentences. This audio adjustment is particularly useful to remove constant noise like personal computer fans or other persistent noises in the space with the user that might be picked up by the user's microphone. The user may tune the effect of the noise reduction audio adjustment through the slider from 0, appropriate for a quiet environment, to 1, appropriate for a noisy environment.

[0238] The smart volume audio adjustment automatically makes small adjustments and helps maintain a steady volume, particularly when the microphone audio stream has a relatively high dynamic range. The smart volume audio adjustment eliminates the need for the user to constantly adjust a computer's volume during gameplay or while listening to music. The user may tune the impact of the smart volume audio adjustment on microphone audio through the slider, over a range from 0 to 1.

[0239] For a given a volume threshold, the noise gate audio adjustment operates to cut off any sound behind it. If sound is above a predetermined noise gate threshold, the audio processing system lets all the sound pass. In other words, the sound is allowed through the gate. On the other hand, if the sound is below the predetermined noise gate threshold, the sound does not pass through the gate. The predetermined noise gate threshold can be changed with the threshold slider in the range -60 dB -10 dB. This effect can be particularly useful if the user perceives an annoying or distracting noise which is way behind conversation level.

[0240] For a given volume threshold, the noise gate audio adjustment will cut off any sound behind it. The auto noise gate audio adjustment automatically computes the predetermined noise gate threshold according to detected background noise. In embodiments, the auto noise gate audio adjustment requires the noise gate audio adjustment to be ON, i.e., both audio adjustments must be activated. The auto noise gate audio adjustment operates automatically. However, the user has a layer of control over the auto noise gate audio adjustment with the offset slider which allows the user to apply a gain over a range from -20 dB to +20 dB) on the top of the computed threshold. The auto noise gate is useful if the user needs a noise gate, but the user is uncertain about which threshold to use.

[0241] The impact noise reduction audio adjustment operates to detect and suppress impact noises from the user's chat audio stream or input device. The impact noise reduction audio adjustment is useful in reducing the noise coming from sources such as keystroke on a keyboard or noise from mouse clicks.

[0242] FIG. 7U depicts an illustrative embodiment of a method 7670 utilized in the subject disclosure. The method 7670 may be performed by one or more software routines including data and instructions operating on a processing device such as computing device 206. The method 7670 may be performed on a standalone device of a gamer such as a laptop computer or desktop computer. In other embodiments, the method 7670 may be performed by a device located remotely from the gamer, such as in a cloud network or at a gaming server, and accessible by a gamer device.

[0243] The method 7670 may be used to detect a predetermined audio event, such as the sound of a gunshot or a footstep or other sound, in an audio stream such as a gaming system output audio stream and, if detected, to modify audio processing to affect the audibility the audio event. For example, this may be done to make the audio event more audible to a game player or more easily distinguished from other sounds in the gaming system output stream. In another example, this may be done to deemphasize the audio event such as to remove a distracting sound from the audio stream.

[0244] The method begins at block 7672 when a game is initiated by the gaming system. This may be done in any suitable manner, such as a game player activating an application program on a computing device or accessing a remote location such as a website over a network to initiate the game, or by the game player joining an already-initiated game. In an exemplary embodiment, the game player may play the game as a member of team with teammates and against opponents.

[0245] At block 7674, the method 7670 includes receiving a plurality of game audio input streams from a video game. In the illustrated embodiment, the plurality of audio input streams includes a game output audio stream, a gamer chat audio output stream, and a microphone audio output stream. In other embodiments, fewer, additional, or alternative audio streams may be received. Further, in other embodiments, the received audio streams may originate in a source other than a video game, such as an application program running on the user device or over a network.

[0246] The received audio stream may include multiple audio streams and block 7674 may include a process of separating the received audio stream into constituent audio streams. In this example, the game output audio stream includes game sounds such as explosion sounds, sounds of

footsteps, sounds of gun shots, sounds of opening and closing door, music and other sounds tied to the activity occurring in the video game. The gamer chat audio output stream includes spoken audio shared among a group of gamers participating in the game together such as teammates on a team. The gamer chat may include spoken words of speech in multiple languages as well as non-speech sounds such as laughter, shouts, or screams. The microphone audio output stream includes spoken audio of a local gamer that is to be shared with other gamers, such as part of the gamer chat with teammates. For example, the microphone audio may include spoken words of speech in one or more languages as well as non-speech sounds such as laughter and shouting.

[0247] At block 7676, a predetermined audio event is detected in the audio stream. The predetermined audio event may be any suitable sound or combination of sounds. In an example, the predetermined audio event may be the sound of a footstep in the game being played by the player. In particular, the sound may be a footstep associated with an opponent of the player in the game. In another example, the predetermined audio event may be the sound of a gunshot in the game being played by the player. The gunshot may be associated with an opponent of the player or may be a random sound occurring in the gameplay. The detection of the audio event at block 7676 may be performed any suitable fashion. An example embodiment is discussed in conjunction with FIG. 7U.

[0248] In these examples, the predetermined audio event is detected in the game output audio stream which includes the gamer chat audio output stream and the microphone audio output stream. In other examples, the predetermined audio event may be detected in the gamer chat audio stream or the microphone audio stream or some combination of multiple audio streams.

[0249] At block 7680, a predetermined audio profile is retrieved for processing the detected audio event. In embodiments, the nature or identification of the audio profile is based on the nature or identification of the detected audio event. In one example, at block 7676 the sound of a gunshot is detected, and the audio event is identified as a gunshot. Based on this identification, at block 7680, a predetermined audio profile is retrieved for processing a gunshot sound. In another example, the audio event detected at block 7676 is identified as a car door opening. At block 7680, a profile associated with the sound of a car door opening is retrieved. Further, more detailed, or fine-grained identifications of audio events may be made and used to select a particular profile at block 7680. For example, the step of block 7676 may include distinguishing a single gunshot sound from the sound of an automatic weapon firing multiple rounds. In the first case, the audio profile for a single gunshot is retrieved. In the second case, the audio profile for multiple rounds of an automatic weapon is retrieved.

[0250] The predetermined audio profile may be retrieved at block 7680 from any suitable location. This includes local storage at the computing device on which the gaming system is operating or remote storage such as at a network location or a location in a network cloud accessible over a network location such as the public internet. In some embodiments, the latency for retrieval of the profile should be very short, such as in the range 100 to 500 milliseconds, so the audio event can be processed in near real time, with minor audible delay for the listener.

[0251] The predetermined audio profile may be selected from a group of previously established profiles provided by a third-party source. For example, a game accessory manufacturer may make available and market a set of audio filters for use with game accessories or for use with a particular gaming system. A set of canned profiles may be available for a gamer to select and download over a network. In another example, the gamer may develop custom profiles using trial and error to enhance, suppress or otherwise modify specific sounds in a gaming audio stream. In an example, a gamer may use a graphical user interface to tailor parameters of a parametric equalizer and store the parameters for subsequent access and use. For example, the graphical user interface illustrated in FIGS. 7M through 7T may be used by a gamer to tailor a profile for the user.

[0252] The predetermined audio profile can include any suitable data or information for processing the detected audio event according to requirements or preferences of the listener. In some examples, the audio event may be made more easily audible to the listener. For example, in a game playing situation, the player may obtain an advantage by hearing footsteps, particularly footsteps of an opponent in the game. For a detected sound of a footstep, then, the predetermined audio profile may include filter settings to enhance the gain or volume of frequencies in which a substantial part of the energy of the footstep spectrum exists. Other frequencies may be suppressed by filtering or other suitable method. Also, in the example, the predetermined audio profile for a footstep sound may include filtering in a lowpass filter with a particular roll off frequency and a particular Q value or quality factor.

[0253] In some embodiments, the predetermined audio profile includes data and information for controlling a parametric equalizer. The parametric equalizer may be configured to process an audio stream according to settings established by a user, such as filter frequencies and gain and filter type and filter Q factor. A gamer may set a number of equalizer inputs according to user preferences or other factors. The predetermined audio profile may include at least one equalizer setting for the parametric equalizer, or a set of equalizer settings. The at least one equalizer setting of the profile replaces at least one equalizer input to modify the response of the parametric equalizer.

[0254] At block 7682, the predetermined audio profile is applied to the parametric equalizer. Further, at block 7684, the audio stream is provided to the parametric equalizer. The parametric equalizer, under control of the audio profile, operates, for example, to enhance the gaming system audio stream to improve an ability of the player of the gaming system to hear the predetermined event in the gaming system audio stream. In one example, spatial or temporal aspects of the input audio stream may be modified. For example, an audio processing system may identify a direction of origin of a sound in the input audio stream, such as a footstep. In the example, the gamer controls an avatar in a video game and the direction or origin may be relative to a position of the avatar within the game play. The audio processing system may modify aspects of the input audio stream, such as the audio spectrum of the game audio stream, to emphasize the direction of origin of the footstep sound. This may include, for example, including aspects of echo or reverberation to make clear the location and direction of the footstep. This is particularly useful if the gamer wears a headset, and the audio stream provides a virtual

surround sound experience for the gamer. The output of the parametric equalizer is a modified audio stream.

[0255] In embodiments which do not employ a parametric equalizer, any suitable device or method may be used to modify an input audio stream according to data, parameters, or information of a predetermined audio profile. The modification may be through digital signal processing or through analog signal processing, for example.

[0256] At block 7886, the modified audio stream is provided to audio equipment of the gamer or other user. For example, the gamer may wear a headset with speakers and microphone. The modified audio stream may be provided to the headset of the gamer. In another example, the computing device used by the gamer has stereo speakers and the modified audio is provided to those speakers. In some applications, the modified audio stream may be provided as a virtual surround sound audio stream, such as by processing through a head-related transfer function.

[0257] FIG. 7V illustrates an exemplary embodiment of block 7676 of FIG. 7U for detecting a predetermined audio event in an input audio stream such as an audio stream from a gaming system. At block 7690, the input audio stream is applied to an artificial intelligence engine such as a neural network. The neural network has been trained using training data to identify or detect particular sounds or predetermined audio events. For example, by being provided with training data corresponding to gunshot sounds, the neural network is able to identify gunshot sounds with a particular probability in the input audio stream. Thus, at block 7690, in response to receiving the input audio stream, the neural network returns a probability value in the range 0 to 1. In the example, the larger the probability value, the larger the likelihood or confidence level that the predetermined sound such as a gunshot has been detected.

[0258] At block 7692, the probability value returned by the neural network is compared with a threshold value. The threshold value may be set at any level such as 0.5, 0.8 or 0.95. In some examples, the threshold value may be set dynamically depending on factors such as the state of game play. If the probability value does not exceed the threshold, no audio event is considered to have been detected. Control returns to block 7674, FIG. 7U, and the processing of the input audio continues. If the probability value exceeds the threshold, an audio event is considered to have been detected. Control returns to block 7680 and the predetermined audio profile is retrieved to process the detected audio event.

[0259] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIGS. 7U and 7V, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0260] FIG. 7W and FIG. 7X depict illustrative embodiments of an exemplary audio scene 7700 (FIG. 7W) and audio mix 7702 (FIG. 7X). The audio scene 7700 illustrate an example of audio as presented to a listener 7704. In the example, the listener 7704 employs a multichannel audio arrangement for audio playback. The multichannel audio arrangement includes, in the example, 5 speakers labeled in FIG. 7W as Front-L, Center, Front-R, Surround L and

Surround-R. Further in this example, the audio scene **7700** includes several audio sources, including voice **7706**, footsteps **7708**, a thunderstorm sounds **7710**, vehicle sounds **7712** and gunfire **7714**. In the exemplary audio scene **7700**, the voice **7706** is positioned directly ahead of the listener **7704**, the footsteps **7708** are positioned to the right of the listener **7704**, the thunderstorm sounds **7710** are positioned behind the listener **7704**, the vehicle sounds are positioned to the left of the listener **7704** and the gunfire **7714** is positioned to the left-front of the listener **7704**.

[0261] The audio scene **7700** may be associated with a gaming system audio stream received during a video game, for example as played by the gaming console **206** (FIG. 2). The gaming system audio stream may be generated by a gaming system and may include a wide variety of sounds, including for example, speech, footsteps, sounds of nature such as thunderstorm sounds, and man-made sounds such as vehicle sounds and gunshots. To assist with creating an immersive experience for the listener, the gaming system audio stream may use surround sound capabilities and a surround sound audio system including speakers as shown in FIG. 7W, to cause the sounds to appear from different directions relative to the listener **7704**. In another example, the audio scene **7700** may include music in addition to or in place of sounds of the type illustrated in FIG. 7W. The listener **7704** may use a surround sound arrangement to listen to any type of audio content.

[0262] In FIG. 7X, the audio mix **7702** illustrates example audio channels corresponding to the audio scene **7700** of FIG. 7W. The audio mix **7702** illustrates the five audio channels provided to the loud speakers of FIG. 7W. These include front left (FL) front right (FR) center (C) surround left (SL) and surround right (SR). The vertical axis for each respective channel corresponds to audio volume. The horizontal axis for each respective channel corresponds to time. The respective sounds, including the gunshot, the speech and the footsteps, occur simultaneously in time. Each audio channel receives a portion of each sound.

[0263] Each respective audio channel illustrates relative sound composition for an individual sound, including a gunshot, spoken words and footsteps, in the example. Thus, as illustrated on the left hand side of FIG. 7X, the gunshot, which in the audio scene **7700** (FIG. 7W) appears to originate at the front left of the listener **7704**, includes relatively substantial audio volume or amplitude in the front left (FL) channel, less amplitude in the center channel (C) and even less amplitude in the surround left (SL) channel. Moreover, the front right (FR) channel and the surround right (SR) channel have substantially no audio volume or amplitude for the gunshot.

[0264] Similarly, for the spoken words of the speaker, the front left and front right channels have moderate audio volume or amplitude. Because the speaker appears to stand directly in front of the listener **7704** in the audio scene **7700** of FIG. 7W, the center channel has substantial audio volume or amplitude. The two surround channels surround left and surround right, have substantially no audio volume or amplitude.

[0265] Still further, in the audio scene **7700**, the footsteps appear to come from the right of the listener **7704**. Thus in the audio mix **7702** of FIG. 7X, the front right channel and the surround right channel have substantially all the audio volume or amplitude. Moreover, in this example, because

the gunshot and the speech are relatively loud, those sounds have relatively larger signal amplitudes relative to the softer footsteps.

[0266] On the right-hand side of FIG. 7X, the global mix for each of the five respective audio channels is shown. Each respective channel in the global mix represents a sum of all the different sounds heard from each of the sources, including the gunshot, to speech and the footsteps. Thus, for the time segment illustrated, the front left channel is a combination of the gunshot sound and the speech sound. Similarly, for the illustrated time segment, the front right channel is a combination of the speech sound and the footsteps sound.

[0267] The global mix may correspond to the audio produced by the gaming system during the play of a video game. The gaming system produces a gaming audio stream with audio signals comparable to the global mix of FIG. 7X. Location information is provided by the five channels, including a mix of sound frequencies and amplitudes that drive the respective speakers. However, information about the source of any particular sound in the global mix is not available from the information in the gaming audio stream. As a result, it is generally not possible to recover an individual sound from the global mix, or from the gaming audio stream.

[0268] There is a need to improve the audio listening experience for some users. For example, for some gamers, it is beneficial to be able to hear some sounds or classes of sounds better than is conventionally available in a gaming audio stream. Some individual gamers feel that hearing footsteps better can assist a gamer in identifying individuals around the gamer's location in a gaming environment. If the volume or sound of the footsteps can be increased, or if the footsteps can be made more prominent in the audio stream, the gamer can hear the footsteps better and be more competitive in the game. Similarly, suppressing some sounds may be helpful to the gamer. For example, if the gamer believes there is another person nearby in the game, the gamer may want to accentuate the sound of footsteps in the gaming audio stream. If the gamer can hear footsteps, the gamer can locate potential enemies and be better aware of action in the game. Suppressing or reducing in volume or prominence gunshot sounds in a game may help the gamer to better hear other sounds in the audio stream. This may enable the gamer to manipulate an audio scene such as audio scene **7700** to improve the gamer's competitiveness in the game or enjoyment of the game.

[0269] It is known to use equalization to attempt to separate individual sounds in an audio stream such as a gaming audio stream. An equalizer circuit or digital signal processor configured as an equalizer can receive the gaming audio stream and act globally on the channels of the audio stream for the audio scene **7700**. The equalizer can operate to boost selected frequencies of interest and suppress other selected frequencies. However, equalization operates on all frequencies in a selected band and some individual sounds overlap in frequency. For example, both a footstep and a gunshot have relatively large frequencies that overlap in frequency. Thus, equalization of the gaming audio stream is generally not helpful for distinguishing or separating individual sounds in the audio stream. For a music audio stream, it is known use a neural network on a stereo audio stream to isolate an individual instrument such as a guitar.

[0270] In accordance with some of the aspects of the subject disclosure, a method and apparatus operate to extract

a class of sounds or classes of sounds, such as the sound of footsteps, within a multichannel audio stream. Extraction of the class of sounds is done for example to enable manipulation of sounds of the class of sounds such as to increase the gain of the sounds or to decrease the gain of the sounds of that class, or to apply audio effects to the sounds of that class. Effects may include equalization or compression of sounds only of that class. The manipulation of the sounds of that class is done without changing other sounds not in that class.

[0271] Alternatively, manipulation may include decreasing or removing all sound not in the class of sounds of interest, leaving just sounds in the class of interest. A class of sounds may be defined in any suitable manner. In the example of a gaming audio stream which creates the audio scene 7700, classes of sounds may include gunshots, footsteps, vehicle noises, speech and thunderstorm sounds. In the example of music, a class of sounds may include music produced by a particular instrument, such as drums or piano.

[0272] FIG. 7Y is a block diagram of a system 7720 for extracting individual sounds from an audio stream in accordance with some aspects of the subject disclosure. In an exemplary embodiment, the system 7720 includes a deep neural network 7722 that receives a multichannel input audio stream 7724 and provides a multichannel output audio stream 7726 corresponding to an individual sound or class of sounds of interest. Other embodiments of the system 7720 may include additional or alternative elements for extracting classes of sounds from the input audio stream 7724. For example, in place of a deep neural network, other types of machine learning modules or artificial intelligence modules may be used. In some embodiments, the output audio stream 7726 includes a plurality of audio streams in which, for example, one audio stream includes the sounds of a selected class of sounds such as footsteps or gunshots of the input audio stream 7724 and another audio stream includes the remaining sounds of the input audio stream 7724. The game player may, for example, select which output audio stream to hear during a particular time segment or period of time in order to enhance his gaming experience or performance.

[0273] The deep neural network (DNN) 7722 is an artificial neural network having an input layer and an output layer and having multiple layers between the input layer and the output layer. The layers include artificial neurons which include mathematical functions that receive one or more inputs of data or signals and produce an output according to a transfer function. Layers of artificial neurons are connected by synapses to transmit data or signals between artificial neurons. Artificial neurons and synapses can have a weight that varies as the DNN 7722 learns. The weight of the artificial neuron or synapse will alter the data or signal passed on to a subsequent artificial neuron. The DNN 7722 may be constructed in any suitable form including circuits and other hardware components, software and a combination of the two.

[0274] The system 7720 may be a portion of an audio processing system for processing one or more streams of audio from an audio source. In one example, the one or more audio streams are output audio from a gaming system. Thus, in FIG., 7Y, the input audio stream 7724 is labelled game mix and is received from a gaming system. In embodiments, the input audio stream 7724 may correspond to game audio source 7614 (FIG. 7L) and may be, for example, a gaming system, one or more application programs running on a

device such as computing device 206 and other sources accessed by a user of the gaming system, such as a network source. In the illustrated embodiment, game audio source 7614 may be an application or online source. The input audio stream 7724 from the gaming system may include game sounds associated with video displayed on a video display device of the computing device 206 (FIG. 2). In the case of a war simulation game, for example, the audio input stream 7724 may include sounds of gun shots, footsteps of teammates or opponents, explosions, vehicle noises, speech including spoken words and others.

[0275] The DNN 7722 may be trained in any suitable manner recognize and identify a particular sound or a class of sounds. Training involves selecting appropriate weights for each artificial neuron and each synapse. Training data may include buffers of sound samples from the input audio stream 7724 or another source. The DNN 7722 learns to recognize a sound or class of sounds from the training data. Training may be supervised, meaning the training data is tagged to identify a sound or class of sounds, or unsupervised. In one embodiment, a single DNN 7722 is trained to recognize a single class of sounds. Thus, in the example of FIG. 7Y, the DNN 7722 analyzes five channels of the input audio stream 7724 to identify the sounds of footsteps in the input audio stream 7724. In other embodiments, the single DNN 7722 is trained to recognize multiple classes of sounds. In yet another embodiment, the DNN 7722 includes multiple DNN modules. Each respective module is trained to recognize a class of sounds of interest. The DNN 7722 may include suitable input controls or an interface such as an application programming interface (API) to receive control information defining, for example, what the sound class of interest is, and other control information.

[0276] In one example, audio output from one or more games is obtained. This may be, for example, raw audio from the games without any preprocessing. The raw data may be used to simulate what a game engine can do and compose training data for the DNN 7722. In other embodiments, the training data may be increased by including audio from several or many different game or other sources. The training data should include an intermediate mix, particularly a mix that is intensive for the class of interest. For operating on a musical audio stream, training data may include a final master of the soundtrack as well as an intermediate multitrack recording. The multitrack recording provides spatial information about where in the stereo mix a sound is located. The musical genre of the training data should match the musical genre of the music of interest.

[0277] In an embodiment, the DNN 7722 receives the input audio stream 7724 after being trained to extract one class of sound such as a footstep or a gunshot. The DNN 7722 separates the input audio stream 7724, the game mix in FIG. 7Y, into two components. The first component is the class of interest, such as footsteps. The second component is the rest of the input stream 7724, minus the sound of the class of interest, or minus the sound of the footsteps. The sound of the footsteps has been extracted from the input audio stream and the output audio stream 7726 includes a stream including the sound of the footsteps and a stream including the rest of the sounds from the input audio stream. Thus, the DNN 7722 operates somewhat like a filter, to select and filter out (or extract) the footstep sounds from the rest of the mix.

[0278] The DNN 7722 operates substantially in real time. In this example, substantially in real time means that the output audio stream 7726 is produced at the same rate as the input audio stream 7724 is received, with minimal delay between input audio stream 7724 and the output audio stream 7726. In some embodiments, a buffer may be used to hold a short segment of sounds of the input audio stream 7724 or the output audio stream 7726.

[0279] In the example of FIG. 7Y, both the input audio stream 7724 and the output audio stream 7726 are multichannel audio streams. That is, each respective audio stream includes a plurality of audio channels. In the example, five channel audio streams are used, including a front left channel, a front right channel, a center channel, a surround left channel and a surround right channel. In other examples, the multichannel audio streams could be 5.1 audio streams, 7.1 audio stream, stereo audio streams etc. The channels of the input audio stream 7724 provide spatial information to the listener. In the example, the listener hears the footsteps coming from the right side of the listener. Similarly, the channels of the output audio stream retain the spatial information. In this manner, the footsteps extracted by the DNN 7722 retain spatial information for playback to the listener. The listener hearing just the footsteps hears them from the right-hand side. The extracted front right channel and the extracted surround right channel of the output audio stream 7726 have frequency and amplitude characteristics substantially similar to those of the input audio stream illustrated in FIG. 7X. The rest of the output audio stream 7726 similarly retains spatial information. Each of the channels of the multichannel output audio stream has frequency components and amplitude substantially matching those of the channels of the multichannel input audio stream 7724.

[0280] FIG. 7Z is a block diagram of a system 7730 for extracting individual sounds from an audio stream in accordance with some aspects of the subject disclosure. In exemplary embodiment, the system 7730 includes a deep neural network 7722 that receives an input audio stream 7734 and provides an output audio stream 7736 corresponding to an individual sound or class of sounds of interest. The output audio stream 7736 may include multiple audio streams including, for example, an audio stream with sounds of a sound class of interest, such as footsteps or spoken speech, and an audio stream with the remaining sounds of the input audio stream. The gamer may select which output audio stream to hear or may modify the output audio stream to accentuate features of the first audio stream and the second audio stream. Further, the system 7730 includes a plurality of audio processing circuits 7732 including in the example audio processing circuit 7732a, audio processing circuit 7732b, . . . audio processing circuit 7732n and audio processing circuit 7738. Still further, the system 7730 includes a mixer 7742. Other embodiments of the system 7730 may include additional or alternative elements for extracting classes of sounds from the input audio stream 7724. For example, in place of a deep neural network, other types of machine learning modules or artificial intelligence modules may be used.

[0281] In the embodiment of FIG. 7Z, the DNN 7722 is configured to extract several classes of sounds of interest. The DNN 7722 receives an input audio stream 7734 which may include, for example, a few milliseconds of audio such as the game mix from a gaming system. The game mix may include a plurality of classes of sounds of interest to a

listener. Similar to the example of FIG. 7W, a first class, Class 1, may include footsteps. A second class, Class 2, may include thunderstorm sounds. A third class, class 3 may include vehicle sounds. A fourth class, class 4, may include gunshot sounds. A fifth class, class 5, may include speech or spoken dialogue.

[0282] In the example, the DNN 7722 receives a control signal at input 7740. In response to the control signal, the DNN extracts one or more of the classes of sound of interest from the game mix of the input audio stream 7734. The extracted sounds from the class of interest are provided as outputs from the DNN 7722. This includes sounds of footsteps, sounds of gunshots, sounds of speech, etc. In addition, the rest of the mix that is not extracted is provided as an output labelled "Rest of the mix" in FIG. 7Z.

[0283] In embodiments, each of the outputs of the DNN 7722, including the extracted sounds along with the Rest of the mix output, is time aligned or synchronized with the other outputs. Moreover, in embodiments, the DNN 7722 operates substantially in real time to extract the class of interest from the input audio stream.

[0284] The plurality of audio processing circuits 7732 may be individually controlled to tailor the class of sound provided to each respective audio processing circuit. For example, the audio processing circuits 7732 may provide gain to boost or reduce the level of a class of sound. The audio processing circuits 7732 may provide equalization such as filtering or compression for individual classes of sounds. Thus, each respective class of sound may receive different audio processing.

[0285] In embodiments, the type of audio processing provided by each of the audio processing circuits 7732 may be selected and controlled by a user. Other types of audio processing may be provided as well. A user interface, such as a touchscreen graphical user interface or voice input using voice recognition, may be provided to a user to enable user modification and enhancement of one or more classes of sound associated with the gaming system. Thus if the user desires to hear footsteps and speech more clearly suppress sounds of gunshots and vehicles, the user may actuate the user interface accordingly and the audio processing circuits 7732 will be adjusted to provide the necessary sound shaping and customization.

[0286] In embodiments, each of the classes of sound processed by the DNN 7722 is a multichannel audio stream. Similar to the example of FIG. 7W, the multichannel streams provide spatial information for a listener, for example by providing channels with different frequencies and amplitudes. The DNN 7722 operates on the multiple channels to extract a particular class of sound from the input audio stream 7724. The spatial information is retained in each of the extracted classes of sound so that a gunshot from the listener's front left in the input audio stream 7734 continues to appear to originate from the front left after extraction by the DNN 7722 and audio processing by one of the audio processing circuits 7732.

[0287] The audio processing circuits 7732 may include any suitable processing operation, such as a digital signal processor. The audio processing circuits 7732 may include any combination of hardware and software. The audio processing circuits 7732 may all be substantially identical and modifiable according to control information received, for example, from the game player. In other embodiments, the audio processing circuits 7732 may be selected or

configured for processing specific sounds or classes of sounds. For example, footstep sounds and gunshot sounds have relatively broad frequency bandwidth but spoken speech from the gaming system may have a relatively narrow bandwidth. Different components may be selected for each respective class of sounds according to the respective bandwidth.

[0288] The mixer 7742 combines output signals from the audio processing circuits 7732 and produces the output audio stream 7736. The output audio stream 7736 includes a modified mix of the game mix received with the input audio stream 7734. In embodiments, the mixer 7742 operates under control of the user, such as through a user interface to adjust desired audio characteristics. For example, if the listener or user wants to better hear footsteps, the DNN 7722 is trained to extract footstep sounds from the input audio stream 7734. The DNN 7722 provides as an output two streams, a first a first audio stream including all footstep sounds, for example, and a second audio stream including the rest of the game mix from the input audio stream 7734. In other embodiments, other combinations of output audio streams may be provided. In the example, the audio processing circuit 7732a will increase gain of the footstep sounds in class 1 and the audio processing circuit 7738 will decrease the gain on the rest of the mix, which includes all sounds except the extracted footsteps. The two audio streams are then recombined in the mixer 7742 to produce a modified game mix. In the modified game mix, the sound of the footsteps is enhanced, and the rest of the mix is reduced.

[0289] FIGS. 8-9 illustrate embodiments of a system with a corresponding communication flow diagram for correlating stimulations and gaming action results. In this illustration a user clicks the left button 119 of the gaming controller 115. The gaming controller 115 can include firmware (or circuitry), which creates an event as depicted by event 2 in FIG. 8. The button depression and the event creation are depicted in FIG. 9 as steps 902 and 904. In step 904, the firmware of the gaming controller 115 can, for example, generate an event type “left button #3”, and a unique GUID with a time stamp which is submitted to the AMS application. Referring back to FIG. 8, the AMS application catalogues event 3, and if a substitute stimulation has been predefined, remaps the event according to the substitution. The remapped event is then transmitted to the gaming application at event 4. Event 3 of FIG. 8 is depicted as step 906 in FIG. 9. In this illustration, the AMS application substitutes the left button #3 depression stimulus with a “keyboard ‘F’” depression which can be interpreted by the gaming application as a fire command. The AMS application in this illustration continues to use the same GUID but substitutes the time stamp for another time stamp to identify when the substitution took place.

[0290] Referring back to event 4, the gaming application processes the event and sends back at event 5 a game action result to the AMS application which is processed by the AMS application at event 6. The AMS application then submits the results to the accessory at event 7. Events 4 and 5 are depicted as step 908 in FIG. 9. In this step, the gaming application processes “F” as an action to fire the gamer’s gun, and then determines from the action the result from logistical gaming results generated by the gaming application. In the present illustration, the action of firing resulted in a hit. The gaming application submits to the AMS

application the result type “Hit” with a new time stamp, while utilizing the same GUID for tracking purposes. At step 910, the AMS application correlates the stimulation “left button #3 (and/or the substitute stimulation keyboard “F”) to the game result “Hit” and catalogues them in memory. The AMS application then submits to the accessory (e.g., gaming controller 115) in step 910 the game action results “Hit” with the same GUID, and a new time stamp indicating when the result was received. Upon receiving the message from the AMS application, the accessory in step 912 processes the “Hit” by asserting a red LED on the accessory (e.g., left button 119 illuminates in red or other LED of the gaming controller 115 illuminates in red) to indicate a hit. Other notification notices can be used such as another color for the LED to indicate misses, a specific sound for a hit, or kill, a vibration or other suitable technique for notifying the gamer of the game action result.

[0291] Upon reviewing the aforementioned embodiments, it would be evident to an artisan with ordinary skill in the art that the embodiments of the subject disclosure can be modified, reduced, or enhanced without departing from the scope of the claims described below. For example, the AMS application can be executed from an accessory 115 or computing device 206 to perform the embodiments described in the subject disclosure. The AMS application can also be operated from a remote server (“cloud services”). In yet another embodiment, functions of the AMS application can be distributed between devices. In yet another embodiment, the AMS application can be configured to track the performance of a gamer and adapt a threshold as the gamer improves or declines in performance.

[0292] For instance, as a gamer’s performance improves with a particular gaming action, the threshold associated with the gaming action can be adapted to be less sensitive in detecting an over usage state. Similarly, the sensitivity of the threshold can be increased to promptly identify an over usage state of a gaming action if the gamer’s performance declines as a result of an over usage of the gaming action. Additionally, the AMS application can be adapted to add gaming actions to an exclusion table when the gamer’s performance substantially improves as a result of using the gaming action being excluded. The exclusion table can also be changed by the AMS application by removing a gaming action from the exclusion table responsive to its excessive use causing a decline in a gamer’s performance.

[0293] Other embodiments can be applied to the subject disclosure.

[0294] It should be understood that devices described in the exemplary embodiments can be in communication with each other via various wireless and/or wired methodologies. The methodologies can be links that are described as coupled, connected and so forth, which can include unidirectional and/or bidirectional communication over wireless paths and/or wired paths that utilize one or more of various protocols or methodologies, where the coupling and/or connection can be direct (e.g., no intervening processing device) and/or indirect (e.g., an intermediary processing device such as a router).

[0295] FIG. 10 depicts an exemplary diagrammatic representation of a machine in the form of a computer system 1000 within which a set of instructions, when executed, may cause the machine to perform any one or more of the methods described above. One or more instances of the machine can operate, for example, as an accessory, comput-

ing device or combinations thereof. In some embodiments, the machine may be connected (e.g., using a network **1026**) to other machines. In a networked deployment, the machine may operate in the capacity of a server or a client user machine in a server-client user network environment, or as a peer machine in a peer-to-peer (or distributed) network environment.

[0296] The machine may comprise a server computer, a client user computer, a personal computer (PC), a tablet, a smart phone, a laptop computer, a desktop computer, a control system, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. It will be understood that a communication device of the subject disclosure includes broadly any electronic device that provides voice, video, or data communication. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methods discussed herein.

[0297] The computer system **1000** may include a processor (or controller) **1002** (e.g., a central processing unit (CPU)), a graphics processing unit (GPU, or both), a main memory **1004** and a static memory **1006**, which communicate with each other via a bus **1008**. The computer system **1000** may further include a display unit **1010** (e.g., a liquid crystal display (LCD), a flat panel, or a solid-state display). The computer system **1000** may include an input device **1012** (e.g., a keyboard), a cursor control device **1014** (e.g., a mouse), a disk drive unit **1016**, a signal generation device **1018** (e.g., a speaker or remote control) and a network interface device **1020**. In distributed environments, the embodiments described in the subject disclosure can be adapted to utilize multiple display units **1010** controlled by two or more computer systems **1000**. In this configuration, presentations described by the subject disclosure may in part be shown in a first of the display units **1010**, while the remaining portion is presented in a second of the display units **1010**.

[0298] The disk drive unit **1016** may include a tangible computer-readable storage medium **1022** on which is stored one or more sets of instructions (e.g., software or instructions **1024**) embodying any one or more of the methods or functions described herein, including those methods illustrated above. The instructions **1024** may also reside, completely or at least partially, within the main memory **1004**, the static memory **1006**, and/or within the processor **1002** during execution thereof by the computer system **1000**. The main memory **1004** and the processor **1002** also may constitute tangible computer-readable storage media.

[0299] Dedicated hardware implementations including, but not limited to, application specific integrated circuits, programmable logic arrays and other hardware devices can likewise be constructed to implement the methods described herein. Application specific integrated circuits and programmable logic array can use downloadable instructions for executing state machines and/or circuit configurations to implement embodiments of the subject disclosure. Applications that may include the apparatus and systems of various embodiments broadly include a variety of electronic and computer systems. Some embodiments implement functions in two or more specific interconnected hardware modules or devices with related control and data signals communicated

between and through the modules, or as portions of an application-specific integrated circuit. Thus, the example system is applicable to software, firmware, and hardware implementations.

[0300] In accordance with various embodiments of the subject disclosure, the operations or methods described herein are intended for operation as software programs or instructions running on or executed by a computer processor or other computing device, and which may include other forms of instructions manifested as a state machine implemented with logic components in an application specific integrated circuit or field programmable gate array. Furthermore, software implementations (e.g., software programs, instructions, etc.) including, but not limited to, distributed processing or component/object distributed processing, parallel processing, or virtual machine processing can also be constructed to implement the methods described herein. It is further noted that a computing device such as a processor, a controller, a state machine, or other suitable device for executing instructions to perform operations or methods may perform such operations directly or indirectly by way of one or more intermediate devices directed by the computing device.

[0301] While the tangible computer-readable storage medium **1022** is shown in an example embodiment to be a single medium, the term “tangible computer-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “tangible computer-readable storage medium” shall also be taken to include any non-transitory medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methods of the subject disclosure. The term “non-transitory” as in a non-transitory computer-readable storage includes without limitation memories, drives, devices, and anything tangible but not a signal per se.

[0302] The term “tangible computer-readable storage medium” shall accordingly be taken to include, but not be limited to: solid-state memories such as a memory card or other package that houses one or more read-only (non-volatile) memories, random access memories, or other rewritable (volatile) memories, a magneto-optical or optical medium such as a disk or tape, or other tangible media which can be used to store information. Accordingly, the disclosure is considered to include any one or more of a tangible computer-readable storage medium, as listed herein and including art-recognized equivalents and successor media, in which the software implementations herein are stored.

[0303] Although the present specification describes components and functions implemented in the embodiments with reference to particular standards and protocols, the disclosure is not limited to such standards and protocols. Each of the standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP) represent examples of the state of the art. Such standards are from time-to-time superseded by faster or more efficient equivalents having essentially the same functions. Wireless standards for device detection (e.g., RFID), short-range communications (e.g., Bluetooth®, Wi-Fi, Zigbee®), and long-range communications (e.g., WiMAX, GSM, CDMA, LTE) can be used by computer system **1000**.

[0304] FIG. 13 depicts system 1300 to identify a group of possible trigger clips in accordance with aspects of this disclosure and FIG. 14 depicts a system 1400 to distribute a group of trigger clips to a group of players in accordance with aspects of this disclosure. Referring to both FIGS. 13 and 14, in one or more embodiments, a computer server 1402 can implement a video game (hereafter video game server 1402) involving several different players 1304, 1306, 1308, 1314, 1316, and 1318 in different locations communicating with each other and the video game server 1402 over a communication network (e.g. Internet). Further, some players 1304, 1306, 1314, and 1316 can comprise one team playing the video game playing against another team of players 1308 and 1318. The gaming devices 1404, 1408, 1412, and 1416 are associated with each player 1304, 1306, 1314, and 1316 of one team.

[0305] In one or more embodiments, the video game server 1402 can detect a current trigger during a video game and record a trigger clip. A trigger is an event in the video game that a player may find significant and/or if reviewed, improve player performance in the future. Examples of triggers can include when a player makes a kill, makes a hit, takes a hit, loss of life, fired upon, implements weapon fire, loss of life of another player, change in ammunition, etc. A player may want to review triggers so as to avoid such actions/events (e.g. loss of life, takes a hit, fired upon, etc.) or to continue the actions/events (e.g. makes a kill, makes a hit, etc.) to improve video game performance in the future. A trigger clip is a portion of the video game that is recorded that contains the trigger. As described herein, a video game may have a sliding window of game content that it continuously records portions of the video game. Once the video game server 1402 identifies a trigger, the video game server 1402 can process the recorded sliding window/recorded portion of video game content that contains the trigger as the trigger clip. Further, the trigger clip can be provided and presented to a player for the player's review.

[0306] In one or more embodiments, the trigger clip is a recorded portion of the game content from the perspective or point of view of a player. In some embodiments, a trigger can involve one or more players. Thus, a distributed trigger clip may be requested or provided to all players involved in the trigger. That is, a distributed trigger clip is a group of trigger clips provided to each player involved in the trigger. Each trigger clip provided to each player is a portion of game content from the associated player's perspective or point of view. Referring to FIG. 13, a trigger clip 1302 of a war scenario video game can include a first player 1304 on a first team. The trigger clip 1302 can be from the perspective of the first player 1304. The trigger can be the loss of life of a second player 1306 of the first team. Further, the trigger clip 1302 can include a first player 1308 on a second team that is taking cover behind a building 1310. The first player 1304 of the first player may request to clip the trigger of the loss of life of the second player 1306 of the first team because the first player 1304 on the first team is unsure whether he/she killed the second player 1306 on the first team (i.e. loss of life due to friendly fire), or whether the first player 1308 on the second team killed the second player 1306 on the first team (i.e. loss of life due to enemy fire). The first player 1304 may want to review the trigger clip to review whether he/she committed friendly fire, and if so, avoid doing so in the future.

[0307] In one or more embodiments, the video game server 1402 can identify that the first player 1304 and the second 1306 on the first team are both involved in the trigger. Thus, the video game server 1402 can generate another trigger clip 1302a from the perspective of the second player 1306. Referring to FIG. 14, when requested by the first player 1304, the video game server can provide the distributed trigger clip, which comprises trigger clip 1302 and trigger clip 1302a. Trigger clip 1302 can be sent to gaming device 1404 associated with the first player 1304 over a communication network (wired, wireless, or a combination thereof) to be presented on display 1406 for review by the first player 1304. Further, trigger clip 1302a can be sent to gaming device 1408 associated with the second player 1306 (wired, wireless, or a combination thereof) to be presented on display 1410 for review by the first player 1306.

[0308] Referring to FIG. 13, a third player 1314 and a fourth player 1316 on the first team (same team as first player 1304 and second player 1306) are not involved in the trigger because they are in a different scene/setting 1312 of the video game battling a second player 1318 on the second team in a forest. Referring to FIG. 14, the video game server 1402 would identify that neither the third player 1314 nor the fourth player 1316 would not be provided with the distributed trigger clip because they were not involved with the trigger. Thus, even though the video game server 1402 is communicatively coupled to gaming devices 1412, 1416 and displays 1414 and 1418 associated with players 1314 and 1316 over a communication network (wired, wireless, or a combination thereof), the video game server 1402 would not provide the distributed trigger clip 1302, 1302a to gaming devices 1412 and 1416.

[0309] In one or more embodiments, the video game server 1402 can detect or identify a trigger by receiving user-generated input from the first player 1304, the user-generated input indicating to generate a clip according to the event/trigger and the players involved in the event/trigger. The user-generated input can also include a timestamp. The user-generated input can be provided entering text input, gesture input (using a touchscreen graphical user interface), and/or voice input (involving voice recognition techniques) into the gaming device 1404. Further, the user-generated input provided to the gaming device 1404 is then transmitted to the video game server 1402 for further processing. The video game server 1402 can then record the distributed trigger clip (group of trigger clips) about the timestamp (e.g. 20 seconds before and after the indicated or detected trigger) included in the user-generated input (or detected otherwise) for the trigger/event and for the players indicated in the user-generated input.

[0310] In one or more embodiments, prior to detecting a trigger, the video game server 1402 can receive user-generated input (as described herein) indicating different events that are associated with different trigger types. For example, trigger types can include of kill, hit, weapon fire, loss of life of current player, loss of life of another player, and change in ammunition. The video game server 1402 can associate each event with a trigger type according to the user-generated input.

[0311] In one or more embodiments, the video game server 1402 can detect a trigger by implementing image recognition techniques to identify or otherwise determine that an event within a portion of game content is a trigger.

In other embodiments, the video game server **1402** can access historical game content and identify a previous trigger within the historical game content. That is, the video game server **1402** can analyze historical game content to determine when an event was identified as a trigger according to its trigger type. Based on such a determination, the video game server **1402** can detect a current trigger by determining the current trigger/event is of the same type as the previous trigger.

[0312] In one or more embodiments, the detecting of a current trigger can include accessing training game content, acquiring machine learning techniques to identify different trigger from the different event from the training game content. That is, different events that are identified as different triggers in the training game content can be used to train an artificial intelligence engine on the video game server **1402** to acquire machine learning techniques to identify triggers in portions of game content in the future. Further, the detecting of a current trigger can comprise identifying the current trigger by using, or in response to, implementing the machine learning techniques on the portion of game content.

[0313] In one or more embodiments, the detecting of a trigger can include detecting a particular sound or utterance or word or statement by a player to identify or otherwise determine that an event within a portion of game content is a trigger. This may be done in any suitable manner including using artificial intelligence or machine learning techniques. In an example, a neural network is trained to estimate, within real time, if there is, for example, a laugh or scream from a player. In an example, real time means substantially without delay detectable by a player. In another example, real time means within a time delay of less than one second. Any suitable neural network may be used. Training data may be audio data from other gameplay that is labelled to identify a laugh or a scream or other utterance or statement that is to be designated as a trigger. The neural network is trained to provide as an output an indication that the trigger has been detected or identified. After training, a portion of an audio stream is provided to the neural network and the neural network generates a value that corresponds to a probability that the trigger has been detected. A processing system may compare the value with a predetermined threshold. If the probability exceeds the predetermined threshold, such as 50 percent or 85 percent or 95 percent, the characteristic may be identified as a significant trigger. The training data may include any sort of audio including conversations, laughter, screaming and other sounds as well, including sounds that are representative of typical sounds.

[0314] A system including an audio processing system such as audio processing system **7600** (FIG. 7L) may include a chat audio stream and a microphone audio stream. The chat audio stream may include audio corresponding to conversation and other utterances from gamers such as teammates of a local user. The microphone audio stream may include audio corresponding to conversation and other utterances from the local user, detected by the microphone adjacent to the local user. In a first use case, a neural network, or multiple neural networks, may be trained to detect an occurrence in the chat audio stream or an occurrence in the microphone audio stream and, responsive to the detected occurrence, to cause a subsequent action within the gaming system. In other embodiments, in a second use case, the neural network may operate on a combined audio stream

that includes both the chat audio stream and the microphone audio stream. Similarly, in a third use case, a single audio stream includes a combined audio stream that includes the chat audio stream, the microphone audio stream and a game audio stream that includes sounds generated by the game. The training data for either of these use cases is tailored to the particular use case.

[0315] In a particular embodiment, a trigger is detected by automatically detecting a laugh or a scream in an audio stream. The laugh or scream can originate with any player. A motivation for such automatic triggering is the notion that, if someone is laughing or screaming, something interesting happened in the game and that should be recorded in a trigger clip. Both audio and video are recorded in the trigger clip.

[0316] In some embodiments, a first window of video of a first predetermined time duration is recorded before the occurrence of the trigger and a second window of a second predetermined time duration, which may be different from the first predetermined time duration, is recorded following the occurrence of the trigger. The nature of the detected trigger, such as a laugh or a scream, can be used to select the duration of the trigger clip. For example, a laugh detected as a trigger may cause a 30 second trigger clip having a duration of 30 seconds to be recorded and a scream detected as a trigger may cause a trigger clip having a duration of 60 seconds to be recorded. In some embodiments, any aspect of detecting a trigger, identifying a trigger or the duration of the trigger clip associated with a specific trigger may be user selectable, such as through a graphical user interface such as GUI **7610** (FIG. 7L).

[0317] In some embodiments, one or more players has a webcam or other device collecting video of the player or the player's environment. The video stream from the webcam can be provided to a neural network to identify a video trigger in a manner similar to the identifying an audio trigger in an audio stream. The neural network may be trained with appropriate data that is labelled to identify the trigger of interest. In further embodiments, both an audio stream and a video stream may be monitored in a multimodal approach to identify a trigger event. For example, a video neural network will produce a video probability that a video trigger has been detected and an audio neural network will produce an audio probability that an audio trigger has been detected. The video probability and the audio probability may be combined and evaluated in any suitable manner against one or more probability thresholds. For example, if a player laughs silently, the audio neural network may not detect the sound corresponding to a laugh, but the video neural network may detect the action or movement associated with laughter, or a probability that the motion corresponds to a laugh. Any change of emotion or other response captured in audio or video, may be designated as a trigger.

[0318] In some embodiments, a trigger may be defined according to particular cultural aspects of a player. For example, research may show that European players react to a game situation in a first way, American players react to the same game situation in a second way, south Asian players react to the same game situation in a third way and east Asian players react to the same game situation in a fourth way. Reactions of the different cultures may include utterances and statements made orally or body movements or

postures, or any combination of these. Different models may be directed to geography or language or other characteristics of different gamers.

[0319] In some embodiments, the processing of audio and video streams to identify triggers may occur on a user device such as computing device 206 (FIG. 2). The processing may be part of the game application running on a gamer's device. In other embodiments, the processing of audio and video streams may be done remotely such as at a server hosting the game and accessible over a network such as the public internet.

[0320] In some embodiments, the audio stream may be provided to a speech recognition module. The speech recognition module may detect and identify words stated by a gamer. Any suitable speech recognition module may be used for this purpose. In this manner, the user may be able to provide commands to the gaming system. Moreover, a trigger may be detected based on an identified statement or command of the player.

[0321] In some embodiments, data for a user's audio and video recognition may be stored in a user profile. The user profile may contain any suitable information useful for processing the audio or the video associated with the user. The user profile may be accessed when the user initiates the game or during gameplay, or at any appropriate time. The user profile may be stored locally at a device of the user. In other embodiments, the user profile may be stored remotely at a server or other storage location for access by the local device of the user.

[0322] In some embodiments, a detected trigger may be characterized to determine particular characteristics of the trigger. The detected characteristics may be used to automatically select different characteristics of the trigger clip that is recorded. For example, a sound may be identified as a scream. If the volume of the scream exceeds a certain threshold, or if the duration of the scream exceeds a certain duration, or a combination of these characteristics, the scream may be treated as a more significant trigger. Identification of the more significant trigger may cause the audio processing system or other aspect to record a trigger clip having a particular characteristic, such as a longer duration. Similarly, identification of the more significant trigger may cause the audio processing system or other aspect to distribute the recorded trigger clip to a different audience than the audience for a conventional trigger. The neural network may be trained using appropriate training data to distinguish any characteristics of interest in the audio stream and to generate a value corresponding to a probability that the trigger has been detected. If the probability exceeds a predetermined threshold, such as 85 percent, the characteristic may be identified as a significant trigger. Any suitable statistics in the audio stream and in the neural network can be used to identify one or more triggers.

[0323] In one or more embodiments, the video game server can determine a group of players involved in a current trigger by identifying the group of players using, or in response to, implementing image recognition techniques on a trigger clip. In one or more embodiments, the video game server can instead, or in addition, identify a group of players involved in a current trigger by identifying the group of players using, or in response to, analyzing one or more audio streams such as chat audio streams or microphone audio streams on a trigger clip. In some embodiments, the video game server 1402 can determine the group of players

involved in the current trigger by identifying the group of players by, or in response to, analyzing game action during the same time period each possible trigger clip in the group of trigger clips. That is, when the video game server 1402 identifies a trigger in a portion of game content from one player's perspective, the video game server can then generate a trigger clip for that player (i.e. record a portion of the game content from that player's perspective). However, the video game server 1402 can also record other portions of game content, each portion of game content from a different player's perspective. These group of portions of game content are possible trigger clips to comprise the distributed trigger clip. However, after determining only players involved in the trigger, the video game server 1402 discards portions of game content that are from perspective of players not involved in the trigger and generates the distributed trigger clip to comprise a group of portions of game content (i.e. trigger clips) from the perspective of each player involved in the trigger.

[0324] FIGS. 15A-D depict illustrative embodiments of methods utilized in the subject disclosure. Further, the methods depicted in FIGS. 15A-D can be implemented by a video game server as described herein. Referring to FIG. 15A, the method 1500 can include the video game server, at 1502, associating each event in a group of events with each trigger type from a group of trigger types. Further, the method 1500 can include the video game server, at 1504, detecting a current trigger during a video game. In addition, the method 1500 can include the video game server, at 1506, determining a group of players involved in the current trigger. Also, the method 1500 can include the video game server, at 1508, recording a trigger clip of game content for each player in the group of players resulting in a group of trigger clips. Each trigger clip of the group of trigger clips is a portion of game content that includes the current trigger from a perspective of each player of the group of players. Further, the method 1500 can include the video game server, at 1510, providing the group of trigger clips to a group of gaming devices over a communication network. Each gaming device of the group of gaming devices is associated with one player of the group of players.

[0325] Referring to FIG. 15B, the method 1511 can implement detecting of the current trigger in different embodiments. The method 1511 can include the video game server, at 1512, receiving first user-generated input. The first user-generated input indicates the current trigger and/or to record the group of trigger clips. Further, the method 1511 can include the video game server, at 1514, implementing image recognition techniques on each portion of game content associated with each of trigger clip of the group of trigger clips to determine that an event occurred within each portion of game content such that the event comprises the current trigger. In addition, the method 1511 can include the video game server, at 1516, accessing historical game content. Also, the method 1511 can include the video game server, at 1518, identifying a previous trigger within the historical game content. Further, the method 1511 can include the video game server, at 1520, determining the current trigger is a same type of trigger as the previous trigger. In addition, the method 1511 can include the video game server, at 1522, accessing training game content. Also, the method 1511 can include the video game server, at 1524, acquiring machine learning techniques to identify different triggers from different events from the training game content. Further, the

method 1511 can include the video game server, at 1526, identifying the current trigger in response to implementing the machine learning techniques on the portion of game content. In some embodiments, the current trigger comprises a trigger type, wherein the trigger type comprises one of kill, hit, weapon fire, loss of life of current player, loss of life of another player, and change in ammunition.

[0326] Referring to FIG. 15C, the associating an event with a trigger can be done by the video game server in different embodiments. The method 1531 can include the video game server, at 1530, determining the group of events of the video game. Further, the method 1531 can include the video game server, at 1532, associating each event in the group of events of the video game with each trigger type from the group of trigger types. In addition, the method 1531 can include the video game server, at 1534, receiving second user-generated input. The second user-generated input provides an indication of each event associated with each trigger type.

[0327] Referring to FIG. 15D, the determining the group of players involved in the current trigger can be done by the video game server in different embodiments. The method 1541 can include the video game server, at 1540, implementing image recognition techniques or analyzing audio streams on each trigger clip of the group of trigger clips. In response, the method 1541 can include the video game server, at 1542, identifying the group of players. Further, the method 1541 can include the video game server, at 1544, analyzing game action during a same time period as each trigger clip of the group of trigger clips. In response, the method 1541 can include the video game server, at 1546, identifying the group of players. This analysis can include the video game server determining from game action the location of each player within the video game and determining the group of players involved in the current trigger by determining the location of each player within the game. Further, those players within a certain threshold distance from the event/current trigger are determined to be involved with the current trigger.

[0328] The illustrations of embodiments described herein are intended to provide a general understanding of the structure of various embodiments, and they are not intended to serve as a complete description of all the elements and features of apparatus and systems that might make use of the structures described herein. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. The exemplary embodiments can include combinations of features and/or steps from multiple embodiments. Other embodiments may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. Figures are also merely representational and may not be drawn to scale. Certain proportions thereof may be exaggerated, while others may be minimized. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

[0329] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. In

one or more embodiments, features that are positively recited can also be excluded from the embodiment with or without replacement by another component or step. The steps or functions described with respect to the exemplary processes or methods can be performed in any order. The steps or functions described with respect to the exemplary processes or methods can be performed alone or in combination with other steps or functions (from other embodiments or from other steps that have not been described).

[0330] Less than all of the steps or functions described with respect to the exemplary processes or methods can also be performed in one or more of the exemplary embodiments. Further, the use of numerical terms to describe a device, component, step, or function, such as first, second, third, and so forth, is not intended to describe an order or function unless expressly stated so. The use of the terms first, second, third and so forth, is generally to distinguish between devices, components, steps, or functions unless expressly stated otherwise. Additionally, one or more devices or components described with respect to the exemplary embodiments can facilitate one or more functions, where the facilitating (e.g., facilitating access or facilitating establishing a connection) can include less than every step needed to perform the function or can include all of the steps needed to perform the function.

[0331] In one or more embodiments, a processor (which can include a controller or circuit) has been described that performs various functions. It should be understood that the processor can be multiple processors, which can include distributed processors or parallel processors in a single machine or multiple machines. The processor can be used in supporting a virtual processing environment. The virtual processing environment may support one or more virtual machines representing computers, servers, or other computing devices. In such virtual machines, components such as microprocessors and storage devices may be virtualized or logically represented. The processor can include a state machine, application specific integrated circuit, and/or programmable gate array including a Field PGA. In one or more embodiments, when a processor executes instructions to perform “operations”, this can include the processor performing the operations directly and/or facilitating, directing, or cooperating with another device or component to perform the operations.

[0332] The Abstract of the Disclosure is provided with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

What is claimed is:

1. A device, comprising:

- a processing system including a processor; and
- a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:

obtaining sounds of a selected class of sounds from a plurality of classes of sounds that are extracted from an input audio stream from a gaming system, the input audio stream being a multichannel audio signal and including gaming audio of a video game played by a game player;

providing a plurality of output audio streams including providing a first audio stream including the sounds of the selected class of sounds of the input audio stream and a second audio stream including remaining sounds of the input audio stream;

modifying one or more of the plurality of output audio streams, forming modified output audio, wherein the modifying includes modifying a gain of the sounds of the selected class of sounds according to control information associated with the game player to form the modified output audio; and

providing the modified output audio to the game player.

2. The device of claim 1, wherein the obtaining sounds from the gaming system comprises receiving sounds of the input audio stream having an apparent location, and wherein the providing the plurality of output audio streams comprises providing the sounds of the selected class of sounds with a same apparent location as the sounds of the gaming audio.

3. The device of claim 1, wherein the obtaining of the sounds is from a neural network that performs extracting of the sounds from the input audio stream.

4. The device of claim 1, wherein the obtaining sounds from the gaming system comprises receiving surround sound audio from the gaming system and wherein the providing the first audio stream comprises providing the sounds of the selected class of sounds in surround sound audio.

5. The device of claim 1, wherein the operations further comprise:

receiving footstep sounds having an apparent direction of origin in relation to the game player;

extracting the footstep sounds from the input audio stream; and

providing the first audio stream including the footstep sounds to the game player with same apparent direction of origin in relation to the game player.

6. The device of claim 1, wherein the modified output audio includes footstep sounds.

7. The device of claim 6, wherein the modified output audio includes gunshot sounds.

8. The device of claim 6, wherein the modifying one or more of the plurality of output audio streams comprises:

equalizing sounds of the selected class of sounds according to the control information from the game player to form the modified output audio.

9. The device of claim 6, wherein the modifying one or more of the plurality of output audio streams comprises:

modifying the sounds of the selected class of sounds according to the control information from the game player, forming modified sounds of interest;

modifying sounds of the remaining sounds of the input audio stream according to the control information from the game player, forming modified remaining sounds;

combining the modified sounds of interest and the modified remaining sounds to form the modified output audio, forming combined output audio; and

providing the combined output audio to the game player.

10. The device of claim 9, wherein the operations further comprise:

providing one of the combined output audio and the input audio stream to the game player according to the control information from the game player.

11. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising:

obtaining sounds of a selected class of sounds from a plurality of classes of sounds that are extracted from an input audio stream from a gaming system, the input audio stream being a multichannel audio signal and including gaming audio of a video game played by a game player;

providing a plurality of output audio streams including providing a first audio stream including the sounds of the selected class of sounds of the input audio stream and a second audio stream including remaining sounds of the input audio stream;

modifying one or more of the plurality of output audio streams, forming modified output audio, wherein the modifying includes modifying a gain of the sounds of the selected class of sounds according to control information associated with the game player to form the modified output audio; and

providing the modified output audio to the game player.

12. The non-transitory machine-readable medium of claim 11, wherein the modified output audio include at least one of footsteps or gunshot sounds.

13. The non-transitory machine-readable medium of claim 11, wherein the operations further comprise:

selecting the selected class of sounds in response to audio processing control information from the game player.

14. The non-transitory machine-readable medium of claim 11, wherein the operations further comprise:

receiving a multichannel audio stream as the input audio stream from the gaming system, the multichannel audio stream including location information of sounds of the gaming audio of the video game; and

providing the plurality of output audio streams to the game player with same location information of the sounds of the gaming audio of the video game.

15. The non-transitory machine-readable medium of claim 11, wherein the operations further comprise:

selecting one of footstep sounds and gunshot sounds of the gaming audio of the video game as the selected class of sounds, wherein the selecting is responsive to the audio processing control information from the game player.

16. The non-transitory machine-readable medium of claim 11, wherein the operations further comprise:

providing the input audio stream to a neural network; and receiving, from the neural network, the sounds of the selected class of sounds of the plurality of classes of sounds.

17. A method, comprising:

obtaining, by a processing system including a processor, sounds of a selected class of sounds from a plurality of classes of sounds that are extracted from an input audio stream from a gaming system, the input audio stream being a multichannel audio signal and including gaming audio of a video game played by a game player;

providing, by the processing system, a plurality of output audio streams including providing a first audio stream including the sounds of the selected class of sounds of the input audio stream and a second audio stream including remaining sounds of the input audio stream; modifying, by the processing system, one or more of the plurality of output audio streams, forming modified output audio, wherein the modifying includes enhancing, by the processing system, one or more of the plurality of output audio streams of the gaming audio of the video game, and reducing, by the processing system, the remaining sounds of the gaming audio of the video game; and providing, by the processing system, the modified output audio to the game player.

18. The method of claim **17**, wherein the modified output audio include footstep sounds.

19. The method of claim **17**, wherein the modifying the one or more of the plurality of output audio streams comprises:

modifying, by the processing system, a gain of the one or more of the plurality of output audio streams; equalizing, by the processing system, the one or more of the plurality of output audio streams; or compressing, by the processing system, the one or more of the plurality of output audio streams, or a combination of these.

20. The method of claim **19**, wherein the modified output audio include footstep sounds.

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